# Assessments of the year

**Organisation.** Each deadline has two practice papers for training. They announce the expected methods; they may be given at the start of the unit, even if not all the topics have yet been taught. The table indicates the planned time of distribution. Versions A and B are reserved for the assessment.

**Marking schemes.** Each exercise is worth 5 points; the values are given question by question. For a 1-point question requiring a proof, award up to 0.5 point for a relevant approach and up to 0.5 point for the justification and conclusion. For a proof by induction marked out of 2 points: property and base case 0.5; hypothesis at a fixed term and inductive step calculation 1; quantified conclusion 0.5. For another value, adapt these proportions. Do not penalise a propagated error twice. The solutions write out the expected reasoning and calculations; accept any correct method within the taught scope.

**Duration.** The 2 h tests have four short exercises; the 4 h mock exams have four developed exercises. Adjust only after real feedback from the class. The four 5-point exercises are a choice consistent with the progression and recent papers, without fixing the instructions of the future session.

| Deadline | Administration guide | Practice papers |
|---|---|---|
| Test 1 | S9 · week of 21 September | From S2 |
| Test 2 | S18 · week of 12 October | From S14 |
| Common test | S28 · week of 16 November; slot to be harmonised | From S14 |
| Test 3 | S34 · week of 30 November | From S21 |
| Mock exam 1 | S45 · 14 to 16 January, 4 h slot to be fixed | From S36 |
| Test 4 | S57 · week of 8 February | From S44 |
| Test 5 | S68 · week of 22 March | From S52 |
| Mock exam 2 | S71 · 1st to 3 April, 4 h slot to be fixed | From S59 |
| Final mock exam | S90 · week of 31 May, 4 h slot to be fixed | From S75 |

# Test 1 · Practice 1

### PRACTICE PAPER · TRAINING

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · A discrete evolution — 5 points

Consider the sequence $(u_n)$ defined by $u_0 = 1$ and, for every natural number $n$, $u_{n+1} = \frac{1}{2}u_n + 4$.

1. Calculate $u_1$ and $u_2$. *(0.5 pt)*
2. Prove by induction that, for all $n \in \mathbb{N}$, $u_n = 8 - 7\left(\frac{1}{2}\right)^n$. *(2 pt)*
3. Deduce that, for every natural number $n$, $1 \leq u_n < 8$. *(1 pt)*
4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of $(u_n)$. *(1 pt)*
5. A student claims: "There exists an integer $n$ such that $u_n = 8$." Is this claim true? Justify. *(0.5 pt)*

## Exercise 2 · Induction and bounds — 5 points

Consider the sequence $(v_n)$ defined by $v_0 = 0$ and, for every natural number $n$, $v_{n+1} = \sqrt{72 + v_n}$.

1. Calculate $v_1$ and $v_2$. Give the exact values. *(0.5 pt)*
2. Prove by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 9$. *(1.5 pt)*
3. Prove by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. *(1.5 pt)*
4. Prove that, for all $n \in \mathbb{N}$, $v_n < 9$. *(1 pt)*
5. Consider a property $P(n)$, defined for $n \in \mathbb{N}$. We have proved $P(1)$ and, for all $n \geq 1$, the implication $P(n) \Rightarrow P(n+1)$. Do these two results allow us to conclude that $P(0)$ is true? Justify. *(0.5 pt)*

Test 1 · Practice 1 · continued

## Exercise 3 · Vector calculations in a basis — 5 points

In a basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$ of space, we define the vectors

$$\vec{u} = \vec{\imath} + \vec{k}, \qquad \vec{v} = \vec{\jmath} + \vec{k},$$
$$\vec{w} = 4\vec{\imath} + 5\vec{\jmath} + 9\vec{k}, \quad \vec{z} = 4\vec{\imath} + 5\vec{\jmath} + 10\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. *(1 pt)*
2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. *(0.5 pt)*
3. Are the vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. *(1.5 pt)*
4. Express the vector $5\vec{u} - 4\vec{v}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
5. Choose a point $A$, then define $B$, $C$ and $D$ by $\overrightarrow{AB} = \vec{u}$, $\overrightarrow{AC} = \vec{v}$ and $\overrightarrow{AD} = \vec{w}$. Prove that $A$, $B$, $C$ and $D$ are coplanar. *(1 pt)*

## Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped $ABCDEFGH$ that is not flat. The vertices $E$, $F$, $G$ and $H$ are the respective images of $A$, $B$, $C$ and $D$ under the translation by vector $\overrightarrow{AE}$. Let $\vec{\imath} = \overrightarrow{AB}$, $\vec{\jmath} = \overrightarrow{AD}$ and $\vec{k} = \overrightarrow{AE}$. The points $I$, $J$ and $K$ are defined by

$$\overrightarrow{AI} = \frac{1}{11}\vec{\imath}, \qquad \overrightarrow{AJ} = \frac{1}{11}\vec{\jmath}, \qquad \overrightarrow{AK} = \frac{1}{11}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
2. Prove that the lines $(IJ)$ and $(BD)$ are parallel. *(1 pt)*
3. Express $\overrightarrow{CG}$ in the basis, then determine whether $\overrightarrow{AB}$ and $\overrightarrow{CG}$ are collinear. Justify. *(1 pt)*
4. Let $O$ be the centre of the parallelepiped. Prove that $O$ lies on the line $(AG)$. *(1 pt)*
5. Prove that the points $I$, $J$ and $K$ are not collinear. *(1 pt)*

# Solutions · Test 1 · Practice 1

## Exercise 1 · A discrete evolution

1. *[0.5 pt]* The relation is applied successively for indices 0 and 1:

$$u_1 = \frac{1}{2} \times 1 + 4 = \frac{9}{2}, \qquad u_2 = \frac{1}{2} \times \frac{9}{2} + 4 = \frac{25}{4}.$$

2. *[2 pt]* For $n \in \mathbb{N}$, let $P(n)$ the property $u_n = 8 - 7\left(\frac{1}{2}\right)^n$.

   **Base case.** For $n = 0$, $8 - 7 \times 1 = 1 = u_0$. So $P(0)$ is true.

   **Inductive step.** Fix an integer $n \geq 0$ and assume $P(n)$ true. By the relation defining the sequence, then by the induction hypothesis,

$$\begin{aligned} u_{n+1} &= \frac{1}{2}u_n + 4 \\ &= \frac{1}{2}\left[8 - 7\left(\frac{1}{2}\right)^n\right] + 4 \\ &= 4 + 4 - 7\left(\frac{1}{2}\right)^{n+1} \\ &= 8 - 7\left(\frac{1}{2}\right)^{n+1}. \end{aligned}$$

   This is $P(n+1)$: the property is hereditary.

   **Conclusion.** The principle of induction gives $u_n = 8 - 7\left(\frac{1}{2}\right)^n$ for all $n \in \mathbb{N}$.

3. *[1 pt]* Since $0 < \frac{1}{2} < 1$, for all $n \in \mathbb{N}$ we have $0 < (\frac{1}{2})^n \leq 1$. Multiplying by $-7 < 0$ reverses the inequalities:

$$-7 \leq -7(\frac{1}{2})^n < 0.$$

   Adding 8 and using the previous formula, we obtain $1 \leq u_n < 8$.

4. *[1 pt]* For all $n \in \mathbb{N}$, the explicit formula gives

$$\begin{aligned} u_{n+1} - u_n &= -7(\frac{1}{2})^{n+1} + 7(\frac{1}{2})^n \\ &= 7(1 - \frac{1}{2})(\frac{1}{2})^n = \frac{7}{2}(\frac{1}{2})^n > 0. \end{aligned}$$

   The sequence is therefore strictly increasing.

5. *[0.5 pt]* The claim is false. For each integer $n$, $8 - u_n = 7(\frac{1}{2})^n > 0$. So no term is equal to 8, whatever the number of steps.

## Exercise 2 · Induction and bounds

1. *[0.5 pt]* $v_1 = \sqrt{72 + 0} = \sqrt{72}$ and $v_2 = \sqrt{72 + \sqrt{72}}$.

2. *[1.5 pt]* Let $P(n)$: "$v_n$ is defined and $0 \leq v_n \leq 9$".

   **Base case.** $v_0 = 0$ is defined and belongs to $[0, 9]$; $P(0)$ is true.

   **Inductive step.** Fix $n \in \mathbb{N}$ and assume $P(n)$ true. Then

$$0 \leq 72 \leq 72 + v_n \leq 72 + 9 = 81.$$

The square root is therefore defined and, since this function is increasing on $[0, +\infty[$,

$$0 \leq v_{n+1} = \sqrt{72 + v_n} \leq \sqrt{81} = 9.$$

Thus $P(n+1)$ is true.

**Conclusion.** By induction, all terms are defined and belong to $[0, 9]$.

3. *[1.5 pt]* For $n \in \mathbb{N}$, let $R(n)$: "$v_{n+1} \geq v_n$".

   **Base case.** $v_1 = \sqrt{72} > 0 = v_0$; $R(0)$ is true.

   **Inductive step.** Fix $n \geq 0$ and assume $v_{n+1} \geq v_n$. Adding 72 and then using the fact that the square root is increasing,

   $$v_{n+2} = \sqrt{72 + v_{n+1}} \geq \sqrt{72 + v_n} = v_{n+1}.$$

   This is $R(n+1)$.

   **Conclusion.** By induction, $v_{n+1} \geq v_n$ for all $n \geq 0$: $(v_n)$ is increasing.

4. *[1 pt]* Let $S(n)$: "$v_n < 9$". At index 0, $v_0 = 0 < 9$. Fix $n \geq 0$ and assume $S(n)$ true. The radicands being positive, the strict increase of the square root gives

   $$v_{n+1} = \sqrt{72 + v_n} < \sqrt{72 + 9} = \sqrt{81} = 9.$$

   So $S(n+1)$ is true. The base case and the inductive step establish by induction that $v_n < 9$ for all $n \in \mathbb{N}$.

5. *[0.5 pt]* No. For example $P(n) : n \geq 1$ is true for all integers $n \geq 1$ but false at 0.

## Exercise 3 · Vector calculations in a basis

1. *[1 pt]* In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{\jmath}$ would be $1 = 0$, a contradiction by uniqueness of coordinates in a basis.

2. *[0.5 pt]* Expanding, $4\vec{u} + 5\vec{v} = 4(\vec{\imath} + \vec{k}) + 5(\vec{\jmath} + \vec{k}) = 4\vec{\imath} + 5\vec{\jmath} + 9\vec{k} = \vec{w}$.

3. *[1.5 pt]* Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis requires

   $$\lambda = 4, \qquad \mu = 5, \qquad \lambda + \mu = 10.$$

   The first two equalities give $\lambda + \mu = 9$, contradicting the third. The three vectors are therefore not coplanar.

4. *[1 pt]* $5\vec{u} - 4\vec{v} = 5(\vec{\imath} + \vec{k}) - 4(\vec{\jmath} + \vec{k}) = 5\vec{\imath} - 4\vec{\jmath} + \vec{k}$.

5. *[1 pt]* $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are non-collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

## Exercise 4 · Geometry of a parallelepiped

1. *[1 pt]* By Chasles' relation, $\overrightarrow{AG} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CG} = \vec{\imath} + \vec{\jmath} + \vec{k}$. Similarly, $\overrightarrow{BH} = \overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DH} = -\vec{\imath} + \vec{\jmath} + \vec{k}$.

2. *[1 pt]* By Chasles' relation, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = \frac{\vec{\jmath} - \vec{\imath}}{11} = \frac{\overrightarrow{BD}}{11}$. These vectors are non-zero and collinear: $(IJ)$ and $(BD)$ are parallel.

3. *[1 pt]* We have $\overrightarrow{CG} = \overrightarrow{AE} = \vec{k}$ and $\overrightarrow{AB} = \vec{\imath}$. These two basis vectors are not collinear: an equality $\vec{k} = t\vec{\imath}$ would contradict the uniqueness of the decomposition in this basis.

4. *[1 pt]* The diagonals of a parallelepiped bisect each other. Thus $\overrightarrow{AO} = \frac{1}{2}\overrightarrow{AG}$, which proves $O \in (AG)$.

5. *[1 pt]* The points satisfy $\overrightarrow{IJ} = \frac{-\vec{\imath} + \vec{\jmath}}{11}$ and $\overrightarrow{IK} = \frac{-\vec{\imath} + \vec{k}}{11}$. If $\overrightarrow{IK} = t\,\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would give $\frac{1}{11} = 0$, which is impossible. The vectors are not collinear, so $I, J, K$ are not collinear.

# Test 1 · Practice 2

**PRACTICE PAPER · TRAINING**

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · A discrete evolution — 5 points

Consider the sequence $(u_n)$ defined by $u_0 = 2$ and, for every natural number $n$, $u_{n+1} = \frac{1}{5}u_n + \frac{36}{5}$.

1. Calculate $u_1$ and $u_2$. *(0.5 pt)*
2. Prove by induction that, for all $n \in \mathbb{N}$, $u_n = 9 - 7\left(\frac{1}{5}\right)^n$. *(2 pt)*
3. Deduce that, for every natural number $n$, $2 \leq u_n < 9$. *(1 pt)*
4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of $(u_n)$. *(1 pt)*
5. A student claims: "There exists an integer $n$ such that $u_n = 9$." Is this claim true? Justify. *(0.5 pt)*

## Exercise 2 · Induction and bounds — 5 points

Consider the sequence $(v_n)$ defined by $v_0 = 0$ and, for every natural number $n$, $v_{n+1} = \sqrt{90 + v_n}$.

1. Calculate $v_1$ and $v_2$. Give the exact values. *(0.5 pt)*
2. Prove by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 10$. *(1.5 pt)*
3. Prove by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. *(1.5 pt)*
4. Prove that, for all $n \in \mathbb{N}$, $v_n < 10$. *(1 pt)*
5. Consider a property $P(n)$, defined for $n \in \mathbb{N}$. We have proved $P(1)$ and, for all $n \geq 1$, the implication $P(n) \Rightarrow P(n+1)$. Do these two results allow us to conclude that $P(0)$ is true? Justify. *(0.5 pt)*

Test 1 · Practice 2 · continued

## Exercise 3 · Vector calculations in a basis — 5 points

In a basis $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, we define the vectors

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=3\vec{\imath}+7\vec{\jmath}+10\vec{k}, \quad \vec{z}=3\vec{\imath}+7\vec{\jmath}+11\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. *(1 pt)*
2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. *(0.5 pt)*
3. Are the vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. *(1.5 pt)*
4. Express the vector $7\vec{u}-3\vec{v}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. *(1 pt)*
5. Choose a point $A$, then define $B$, $C$ and $D$ by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Prove that $A$, $B$, $C$ and $D$ are coplanar. *(1 pt)*

## Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped $ABCDEFGH$ that is not flat. The vertices $E$, $F$, $G$ and $H$ are the respective images of $A$, $B$, $C$ and $D$ under the translation by vector $\overrightarrow{AE}$. Let $\vec{\imath}=\overrightarrow{AB}$, $\vec{\jmath}=\overrightarrow{AD}$ and $\vec{k}=\overrightarrow{AE}$. The points $I$, $J$ and $K$ are defined by

$$\overrightarrow{AI}=\frac{1}{18}\vec{\imath}, \qquad \overrightarrow{AJ}=\frac{1}{18}\vec{\jmath}, \qquad \overrightarrow{AK}=\frac{1}{18}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. *(1 pt)*
2. Prove that the lines $(IJ)$ and $(BD)$ are parallel. *(1 pt)*
3. Express $\overrightarrow{CG}$ in the basis, then determine whether $\overrightarrow{AB}$ and $\overrightarrow{CG}$ are collinear. Justify. *(1 pt)*
4. Let $O$ be the centre of the parallelepiped. Prove that $O$ lies on the line $(AG)$. *(1 pt)*
5. Prove that the points $I$, $J$ and $K$ are not collinear. *(1 pt)*

# Solutions · Test 1 · Practice 2

## Exercise 1 · A discrete evolution

1. *[0.5 pt]* The relation is applied successively for indices 0 and 1:

$$u_1 = \frac{1}{5} \times 2 + \frac{36}{5} = \frac{38}{5}, \qquad u_2 = \frac{1}{5} \times \frac{38}{5} + \frac{36}{5} = \frac{218}{25}.$$

2. *[2 pt]* For $n \in \mathbb{N}$, let $P(n)$ the property $u_n = 9 - 7\left(\frac{1}{5}\right)^n$.

   **Base case.** For $n = 0$, $9 - 7 \times 1 = 2 = u_0$. So $P(0)$ is true.

   **Inductive step.** Fix an integer $n \geq 0$ and assume $P(n)$ true. By the relation defining the sequence, then by the induction hypothesis,

$$\begin{aligned} u_{n+1} &= \frac{1}{5}u_n + \frac{36}{5} \\ &= \frac{1}{5}\left[9 - 7\left(\frac{1}{5}\right)^n\right] + \frac{36}{5} \\ &= \frac{9}{5} + \frac{36}{5} - 7\left(\frac{1}{5}\right)^{n+1} \\ &= 9 - 7\left(\frac{1}{5}\right)^{n+1}. \end{aligned}$$

   This is $P(n+1)$: the property is hereditary.

   **Conclusion.** The principle of induction gives $u_n = 9 - 7\left(\frac{1}{5}\right)^n$ for all $n \in \mathbb{N}$.

3. *[1 pt]* Since $0 < \frac{1}{5} < 1$, for all $n \in \mathbb{N}$ we have $0 < (\frac{1}{5})^n \leq 1$. Multiplying by $-7 < 0$ reverses the inequalities:

$$-7 \leq -7(\frac{1}{5})^n < 0.$$

   Adding 9 and using the previous formula, we obtain $2 \leq u_n < 9$.

4. *[1 pt]* For all $n \in \mathbb{N}$, the explicit formula gives

$$\begin{aligned} u_{n+1} - u_n &= -7(\frac{1}{5})^{n+1} + 7(\frac{1}{5})^n \\ &= 7(1 - \frac{1}{5})(\frac{1}{5})^n = \frac{28}{5}(\frac{1}{5})^n > 0. \end{aligned}$$

   The sequence is therefore strictly increasing.

5. *[0.5 pt]* The claim is false. For each integer $n$, $9 - u_n = 7(\frac{1}{5})^n > 0$. So no term is equal to 9, whatever the number of steps.

## Exercise 2 · Induction and bounds

1. *[0.5 pt]* $v_1 = \sqrt{90 + 0} = \sqrt{90}$ and $v_2 = \sqrt{90 + \sqrt{90}}$.

2. *[1.5 pt]* Let $P(n)$: "$v_n$ is defined and $0 \leq v_n \leq 10$".

   **Base case.** $v_0 = 0$ is defined and belongs to $[0, 10]$; $P(0)$ is true.

   **Inductive step.** Fix $n \in \mathbb{N}$ and assume $P(n)$ true. Then

$$0 \leq 90 \leq 90 + v_n \leq 90 + 10 = 100.$$

The square root is therefore defined and, since this function is increasing on $[0, +\infty[$,

$$0 \leq v_{n+1} = \sqrt{90 + v_n} \leq \sqrt{100} = 10.$$

Thus $P(n+1)$ is true.

**Conclusion.** By induction, all terms are defined and belong to $[0, 10]$.

3. *[1.5 pt]* For $n \in \mathbb{N}$, let $R(n)$: "$v_{n+1} \geq v_n$".

   **Base case.** $v_1 = \sqrt{90} > 0 = v_0$; $R(0)$ is true.

   **Inductive step.** Fix $n \geq 0$ and assume $v_{n+1} \geq v_n$. Adding 90 and then using the fact that the square root is increasing,

   $$v_{n+2} = \sqrt{90 + v_{n+1}} \geq \sqrt{90 + v_n} = v_{n+1}.$$

   This is $R(n+1)$.

   **Conclusion.** By induction, $v_{n+1} \geq v_n$ for all $n \geq 0$: $(v_n)$ is increasing.

4. *[1 pt]* Let $S(n)$: "$v_n < 10$". At index 0, $v_0 = 0 < 10$. Fix $n \geq 0$ and assume $S(n)$ true. The radicands being positive, the strict increase of the square root gives

   $$v_{n+1} = \sqrt{90 + v_n} < \sqrt{90 + 10} = \sqrt{100} = 10.$$

   So $S(n+1)$ is true. The base case and the inductive step establish by induction that $v_n < 10$ for all $n \in \mathbb{N}$.

5. *[0.5 pt]* No. For example $P(n) : n \geq 1$ is true for all integers $n \geq 1$ but false at 0.

## Exercise 3 · Vector calculations in a basis

1. *[1 pt]* In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{\jmath}$ would be $1 = 0$, a contradiction by uniqueness of coordinates in a basis.

2. *[0.5 pt]* Expanding, $3\vec{u} + 7\vec{v} = 3(\vec{\imath} + \vec{k}) + 7(\vec{\jmath} + \vec{k}) = 3\vec{\imath} + 7\vec{\jmath} + 10\vec{k} = \vec{w}$.

3. *[1.5 pt]* Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis requires

   $$\lambda = 3, \qquad \mu = 7, \qquad \lambda + \mu = 11.$$

   The first two equalities give $\lambda + \mu = 10$, contradicting the third. The three vectors are therefore not coplanar.

4. *[1 pt]* $7\vec{u} - 3\vec{v} = 7(\vec{\imath} + \vec{k}) - 3(\vec{\jmath} + \vec{k}) = 7\vec{\imath} - 3\vec{\jmath} + 4\vec{k}$.

5. *[1 pt]* $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are non-collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

## Exercise 4 · Geometry of a parallelepiped

1. *[1 pt]* By Chasles' relation, $\overrightarrow{AG} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CG} = \vec{\imath} + \vec{\jmath} + \vec{k}$. Similarly, $\overrightarrow{BH} = \overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DH} = -\vec{\imath} + \vec{\jmath} + \vec{k}$.

2. *[1 pt]* By Chasles' relation, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = \frac{\vec{\jmath} - \vec{\imath}}{18} = \frac{\overrightarrow{BD}}{18}$. These vectors are non-zero and collinear: $(IJ)$ and $(BD)$ are parallel.

3. *[1 pt]* We have $\overrightarrow{CG} = \overrightarrow{AE} = \vec{k}$ and $\overrightarrow{AB} = \vec{\imath}$. These two basis vectors are not collinear: an equality $\vec{k} = t\vec{\imath}$ would contradict the uniqueness of the decomposition in this basis.

4. *[1 pt]* The diagonals of a parallelepiped bisect each other. Thus $\overrightarrow{AO} = \frac{1}{2}\overrightarrow{AG}$, which proves $O \in (AG)$.

5. *[1 pt]* The points satisfy $\overrightarrow{IJ} = \frac{-\vec{\imath} + \vec{\jmath}}{18}$ and $\overrightarrow{IK} = \frac{-\vec{\imath} + \vec{k}}{18}$. If $\overrightarrow{IK} = t\,\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would give $\frac{1}{18} = 0$, which is impossible. The vectors are not collinear, so $I, J, K$ are not collinear.

# Test 1 · A

**PAPER RESERVED FOR ASSESSMENT**

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · A discrete evolution — 5 points

Consider the sequence $(u_n)$ defined by $u_0 = 3$ and, for every natural number $n$, $u_{n+1} = \frac{1}{4}u_n + \frac{33}{4}$.

1. Calculate $u_1$ and $u_2$. *(0.5 pt)*
2. Prove by induction that, for all $n \in \mathbb{N}$, $u_n = 11 - 8\left(\frac{1}{4}\right)^n$. *(2 pt)*
3. Deduce that, for every natural number $n$, $3 \leq u_n < 11$. *(1 pt)*
4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of $(u_n)$. *(1 pt)*
5. A student claims: "There exists an integer $n$ such that $u_n = 11$." Is this claim true? Justify. *(0.5 pt)*

## Exercise 2 · Induction and bounds — 5 points

Consider the sequence $(v_n)$ defined by $v_0 = 0$ and, for every natural number $n$, $v_{n+1} = \sqrt{110 + v_n}$.

1. Calculate $v_1$ and $v_2$. Give the exact values. *(0.5 pt)*
2. Prove by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 11$. *(1.5 pt)*
3. Prove by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. *(1.5 pt)*
4. Prove that, for all $n \in \mathbb{N}$, $v_n < 11$. *(1 pt)*
5. Consider a property $P(n)$, defined for $n \in \mathbb{N}$. We have proved $P(1)$ and, for all $n \geq 1$, the implication $P(n) \Rightarrow P(n+1)$. Do these two results allow us to conclude that $P(0)$ is true? Justify. *(0.5 pt)*

Test 1 · A · continued

## Exercise 3 · Vector calculations in a basis — 5 points

In a basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$ of space, we define the vectors

$$\vec{u} = \vec{\imath} + \vec{k}, \qquad \vec{v} = \vec{\jmath} + \vec{k},$$
$$\vec{w} = 2\vec{\imath} + 4\vec{\jmath} + 6\vec{k}, \quad \vec{z} = 2\vec{\imath} + 4\vec{\jmath} + 7\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. *(1 pt)*
2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. *(0.5 pt)*
3. Are the vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. *(1.5 pt)*
4. Express the vector $4\vec{u} - 2\vec{v}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
5. Choose a point $A$, then define $B$, $C$ and $D$ by $\overrightarrow{AB} = \vec{u}$, $\overrightarrow{AC} = \vec{v}$ and $\overrightarrow{AD} = \vec{w}$. Prove that $A$, $B$, $C$ and $D$ are coplanar. *(1 pt)*

## Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped $ABCDEFGH$ that is not flat. The vertices $E$, $F$, $G$ and $H$ are the respective images of $A$, $B$, $C$ and $D$ under the translation by vector $\overrightarrow{AE}$. Let $\vec{\imath} = \overrightarrow{AB}$, $\vec{\jmath} = \overrightarrow{AD}$ and $\vec{k} = \overrightarrow{AE}$. The points $I$, $J$ and $K$ are defined by

$$\overrightarrow{AI} = \frac{1}{8}\vec{\imath}, \qquad \overrightarrow{AJ} = \frac{1}{8}\vec{\jmath}, \qquad \overrightarrow{AK} = \frac{1}{8}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
2. Prove that the lines $(IJ)$ and $(BD)$ are parallel. *(1 pt)*
3. Express $\overrightarrow{CG}$ in the basis, then determine whether $\overrightarrow{AB}$ and $\overrightarrow{CG}$ are collinear. Justify. *(1 pt)*
4. Let $O$ be the centre of the parallelepiped. Prove that $O$ lies on the line $(AG)$. *(1 pt)*
5. Prove that the points $I$, $J$ and $K$ are not collinear. *(1 pt)*

# Solutions · Test 1 · A

## Exercise 1 · A discrete evolution

1. *[0.5 pt]* The relation is applied successively for indices 0 and 1:

$$u_1 = \frac{1}{4} \times 3 + \frac{33}{4} = 9, \qquad u_2 = \frac{1}{4} \times 9 + \frac{33}{4} = \frac{21}{2}.$$

2. *[2 pt]* For $n \in \mathbb{N}$, let $P(n)$ the property $u_n = 11 - 8\left(\frac{1}{4}\right)^n$.

   **Base case.** For $n = 0$, $11 - 8 \times 1 = 3 = u_0$. So $P(0)$ is true.

   **Inductive step.** Fix an integer $n \geq 0$ and assume $P(n)$ true. By the relation defining the sequence, then by the induction hypothesis,

$$\begin{aligned} u_{n+1} &= \frac{1}{4}u_n + \frac{33}{4} \\ &= \frac{1}{4}\left[11 - 8\left(\frac{1}{4}\right)^n\right] + \frac{33}{4} \\ &= \frac{11}{4} + \frac{33}{4} - 8\left(\frac{1}{4}\right)^{n+1} \\ &= 11 - 8\left(\frac{1}{4}\right)^{n+1}. \end{aligned}$$

   This is $P(n+1)$: the property is hereditary.

   **Conclusion.** The principle of induction gives $u_n = 11 - 8\left(\frac{1}{4}\right)^n$ for all $n \in \mathbb{N}$.

3. *[1 pt]* Since $0 < \frac{1}{4} < 1$, for all $n \in \mathbb{N}$ we have $0 < (\frac{1}{4})^n \leq 1$. Multiplying by $-8 < 0$ reverses the inequalities:

$$-8 \leq -8(\frac{1}{4})^n < 0.$$

   Adding 11 and using the previous formula, we obtain $3 \leq u_n < 11$.

4. *[1 pt]* For all $n \in \mathbb{N}$, the explicit formula gives

$$\begin{aligned} u_{n+1} - u_n &= -8(\frac{1}{4})^{n+1} + 8(\frac{1}{4})^n \\ &= 8(1 - \frac{1}{4})(\frac{1}{4})^n = 6(\frac{1}{4})^n > 0. \end{aligned}$$

   The sequence is therefore strictly increasing.

5. *[0.5 pt]* The claim is false. For each integer $n$, $11 - u_n = 8(\frac{1}{4})^n > 0$. So no term is equal to 11, whatever the number of steps.

## Exercise 2 · Induction and bounds

1. *[0.5 pt]* $v_1 = \sqrt{110 + 0} = \sqrt{110}$ and $v_2 = \sqrt{110 + \sqrt{110}}$.

2. *[1.5 pt]* Let $P(n)$: "$v_n$ is defined and $0 \leq v_n \leq 11$".

   **Base case.** $v_0 = 0$ is defined and belongs to $[0, 11]$; $P(0)$ is true.

   **Inductive step.** Fix $n \in \mathbb{N}$ and assume $P(n)$ true. Then

$$0 \leq 110 \leq 110 + v_n \leq 110 + 11 = 121.$$

The square root is therefore defined and, since this function is increasing on $[0, +\infty[$,

$$0 \leq v_{n+1} = \sqrt{110 + v_n} \leq \sqrt{121} = 11.$$

Thus $P(n+1)$ is true.

**Conclusion.** By induction, all terms are defined and belong to $[0, 11]$.

3. *[1.5 pt]* For $n \in \mathbb{N}$, let $R(n)$: "$v_{n+1} \geq v_n$".

   **Base case.** $v_1 = \sqrt{110} > 0 = v_0$; $R(0)$ is true.

   **Inductive step.** Fix $n \geq 0$ and assume $v_{n+1} \geq v_n$. Adding 110 and then using the fact that the square root is increasing,

   $$v_{n+2} = \sqrt{110 + v_{n+1}} \geq \sqrt{110 + v_n} = v_{n+1}.$$

   This is $R(n+1)$.

   **Conclusion.** By induction, $v_{n+1} \geq v_n$ for all $n \geq 0$: $(v_n)$ is increasing.

4. *[1 pt]* Let $S(n)$: "$v_n < 11$". At index 0, $v_0 = 0 < 11$. Fix $n \geq 0$ and assume $S(n)$ true. The radicands being positive, the strict increase of the square root gives

   $$v_{n+1} = \sqrt{110 + v_n} < \sqrt{110 + 11} = \sqrt{121} = 11.$$

   So $S(n+1)$ is true. The base case and the inductive step establish by induction that $v_n < 11$ for all $n \in \mathbb{N}$.

5. *[0.5 pt]* No. For example $P(n) : n \geq 1$ is true for all integers $n \geq 1$ but false at 0.

## Exercise 3 · Vector calculations in a basis

1. *[1 pt]* In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{\jmath}$ would be $1 = 0$, a contradiction by uniqueness of coordinates in a basis.

2. *[0.5 pt]* Expanding, $2\vec{u} + 4\vec{v} = 2(\vec{\imath} + \vec{k}) + 4(\vec{\jmath} + \vec{k}) = 2\vec{\imath} + 4\vec{\jmath} + 6\vec{k} = \vec{w}$.

3. *[1.5 pt]* Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis requires

   $$\lambda = 2, \qquad \mu = 4, \qquad \lambda + \mu = 7.$$

   The first two equalities give $\lambda + \mu = 6$, contradicting the third. The three vectors are therefore not coplanar.

4. *[1 pt]* $4\vec{u} - 2\vec{v} = 4(\vec{\imath} + \vec{k}) - 2(\vec{\jmath} + \vec{k}) = 4\vec{\imath} - 2\vec{\jmath} + 2\vec{k}$.

5. *[1 pt]* $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are non-collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

## Exercise 4 · Geometry of a parallelepiped

1. *[1 pt]* By Chasles' relation, $\overrightarrow{AG} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CG} = \vec{\imath} + \vec{\jmath} + \vec{k}$. Similarly, $\overrightarrow{BH} = \overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DH} = -\vec{\imath} + \vec{\jmath} + \vec{k}$.

2. *[1 pt]* By Chasles' relation, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = \frac{\vec{\jmath} - \vec{\imath}}{8} = \frac{\overrightarrow{BD}}{8}$. These vectors are non-zero and collinear: $(IJ)$ and $(BD)$ are parallel.

3. *[1 pt]* We have $\overrightarrow{CG} = \overrightarrow{AE} = \vec{k}$ and $\overrightarrow{AB} = \vec{\imath}$. These two basis vectors are not collinear: an equality $\vec{k} = t\vec{\imath}$ would contradict the uniqueness of the decomposition in this basis.

4. *[1 pt]* The diagonals of a parallelepiped bisect each other. Thus $\overrightarrow{AO} = \frac{1}{2}\overrightarrow{AG}$, which proves $O \in (AG)$.

5. *[1 pt]* The points satisfy $\overrightarrow{IJ} = \frac{-\vec{\imath} + \vec{\jmath}}{8}$ and $\overrightarrow{IK} = \frac{-\vec{\imath} + \vec{k}}{8}$. If $\overrightarrow{IK} = t\,\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would give $\frac{1}{8} = 0$, which is impossible. The vectors are not collinear, so $I, J, K$ are not collinear.

# Test 1 · B

**PAPER RESERVED FOR ASSESSMENT**

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · A discrete evolution — 5 points

Consider the sequence $(u_n)$ defined by $u_0 = 1$ and, for every natural number $n$, $u_{n+1} = \frac{1}{3}u_n + \frac{16}{3}$.

1. Calculate $u_1$ and $u_2$. *(0.5 pt)*
2. Prove by induction that, for all $n \in \mathbb{N}$, $u_n = 8 - 7\left(\frac{1}{3}\right)^n$. *(2 pt)*
3. Deduce that, for every natural number $n$, $1 \leq u_n < 8$. *(1 pt)*
4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of $(u_n)$. *(1 pt)*
5. A student claims: "There exists an integer $n$ such that $u_n = 8$." Is this claim true? Justify. *(0.5 pt)*

## Exercise 2 · Induction and bounds — 5 points

Consider the sequence $(v_n)$ defined by $v_0 = 0$ and, for every natural number $n$, $v_{n+1} = \sqrt{42 + v_n}$.

1. Calculate $v_1$ and $v_2$. Give the exact values. *(0.5 pt)*
2. Prove by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 7$. *(1.5 pt)*
3. Prove by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. *(1.5 pt)*
4. Prove that, for all $n \in \mathbb{N}$, $v_n < 7$. *(1 pt)*
5. Consider a property $P(n)$, defined for $n \in \mathbb{N}$. We have proved $P(1)$ and, for all $n \geq 1$, the implication $P(n) \Rightarrow P(n+1)$. Do these two results allow us to conclude that $P(0)$ is true? Justify. *(0.5 pt)*

Test 1 · B · continued

## Exercise 3 · Vector calculations in a basis — 5 points

In a basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$ of space, we define the vectors

$$\vec{u} = \vec{\imath} + \vec{k}, \qquad \vec{v} = \vec{\jmath} + \vec{k},$$
$$\vec{w} = 5\vec{\imath} + 5\vec{\jmath} + 10\vec{k}, \quad \vec{z} = 5\vec{\imath} + 5\vec{\jmath} + 11\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. *(1 pt)*
2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. *(0.5 pt)*
3. Are the vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. *(1.5 pt)*
4. Express the vector $5\vec{u} - 5\vec{v}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
5. Choose a point $A$, then define $B$, $C$ and $D$ by $\overrightarrow{AB} = \vec{u}$, $\overrightarrow{AC} = \vec{v}$ and $\overrightarrow{AD} = \vec{w}$. Prove that $A$, $B$, $C$ and $D$ are coplanar. *(1 pt)*

## Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped $ABCDEFGH$ that is not flat. The vertices $E$, $F$, $G$ and $H$ are the respective images of $A$, $B$, $C$ and $D$ under the translation by vector $\overrightarrow{AE}$. Let $\vec{\imath} = \overrightarrow{AB}$, $\vec{\jmath} = \overrightarrow{AD}$ and $\vec{k} = \overrightarrow{AE}$. The points $I$, $J$ and $K$ are defined by

$$\overrightarrow{AI} = \frac{1}{15}\vec{\imath}, \qquad \overrightarrow{AJ} = \frac{1}{15}\vec{\jmath}, \qquad \overrightarrow{AK} = \frac{1}{15}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
2. Prove that the lines $(IJ)$ and $(BD)$ are parallel. *(1 pt)*
3. Express $\overrightarrow{CG}$ in the basis, then determine whether $\overrightarrow{AB}$ and $\overrightarrow{CG}$ are collinear. Justify. *(1 pt)*
4. Let $O$ be the centre of the parallelepiped. Prove that $O$ lies on the line $(AG)$. *(1 pt)*
5. Prove that the points $I$, $J$ and $K$ are not collinear. *(1 pt)*

# Solutions · Test 1 · B

## Exercise 1 · A discrete evolution

1. *[0.5 pt]* The relation is applied successively for indices 0 and 1:

$$u_1 = \frac{1}{3} \times 1 + \frac{16}{3} = \frac{17}{3}, \qquad u_2 = \frac{1}{3} \times \frac{17}{3} + \frac{16}{3} = \frac{65}{9}.$$

2. *[2 pt]* For $n \in \mathbb{N}$, let $P(n)$ the property $u_n = 8 - 7\left(\frac{1}{3}\right)^n$.

   **Base case.** For $n = 0$, $8 - 7 \times 1 = 1 = u_0$. So $P(0)$ is true.

   **Inductive step.** Fix an integer $n \geq 0$ and assume $P(n)$ true. By the relation defining the sequence, then by the induction hypothesis,

$$\begin{aligned} u_{n+1} &= \frac{1}{3}u_n + \frac{16}{3} \\ &= \frac{1}{3}\left[8 - 7\left(\frac{1}{3}\right)^n\right] + \frac{16}{3} \\ &= \frac{8}{3} + \frac{16}{3} - 7\left(\frac{1}{3}\right)^{n+1} \\ &= 8 - 7\left(\frac{1}{3}\right)^{n+1}. \end{aligned}$$

   This is $P(n+1)$: the property is hereditary.

   **Conclusion.** The principle of induction gives $u_n = 8 - 7\left(\frac{1}{3}\right)^n$ for all $n \in \mathbb{N}$.

3. *[1 pt]* Since $0 < \frac{1}{3} < 1$, for all $n \in \mathbb{N}$ we have $0 < (\frac{1}{3})^n \leq 1$. Multiplying by $-7 < 0$ reverses the inequalities:

$$-7 \leq -7(\frac{1}{3})^n < 0.$$

   Adding 8 and using the previous formula, we obtain $1 \leq u_n < 8$.

4. *[1 pt]* For all $n \in \mathbb{N}$, the explicit formula gives

$$\begin{aligned} u_{n+1} - u_n &= -7(\frac{1}{3})^{n+1} + 7(\frac{1}{3})^n \\ &= 7(1 - \frac{1}{3})(\frac{1}{3})^n = \frac{14}{3}(\frac{1}{3})^n > 0. \end{aligned}$$

   The sequence is therefore strictly increasing.

5. *[0.5 pt]* The claim is false. For each integer $n$, $8 - u_n = 7(\frac{1}{3})^n > 0$. So no term is equal to 8, whatever the number of steps.

## Exercise 2 · Induction and bounds

1. *[0.5 pt]* $v_1 = \sqrt{42 + 0} = \sqrt{42}$ and $v_2 = \sqrt{42 + \sqrt{42}}$.

2. *[1.5 pt]* Let $P(n)$: "$v_n$ is defined and $0 \leq v_n \leq 7$".

   **Base case.** $v_0 = 0$ is defined and belongs to $[0, 7]$; $P(0)$ is true.

   **Inductive step.** Fix $n \in \mathbb{N}$ and assume $P(n)$ true. Then

$$0 \leq 42 \leq 42 + v_n \leq 42 + 7 = 49.$$

The square root is therefore defined and, since this function is increasing on $[0, +\infty[$,

$$0 \leq v_{n+1} = \sqrt{42 + v_n} \leq \sqrt{49} = 7.$$

Thus $P(n+1)$ is true.

**Conclusion.** By induction, all terms are defined and belong to $[0, 7]$.

3. *[1.5 pt]* For $n \in \mathbb{N}$, let $R(n)$: "$v_{n+1} \geq v_n$".

   **Base case.** $v_1 = \sqrt{42} > 0 = v_0$; $R(0)$ is true.

   **Inductive step.** Fix $n \geq 0$ and assume $v_{n+1} \geq v_n$. Adding 42 and then using the fact that the square root is increasing,

   $$v_{n+2} = \sqrt{42 + v_{n+1}} \geq \sqrt{42 + v_n} = v_{n+1}.$$

   This is $R(n+1)$.

   **Conclusion.** By induction, $v_{n+1} \geq v_n$ for all $n \geq 0$: $(v_n)$ is increasing.

4. *[1 pt]* Let $S(n)$: "$v_n < 7$". At index 0, $v_0 = 0 < 7$. Fix $n \geq 0$ and assume $S(n)$ true. The radicands being positive, the strict increase of the square root gives

   $$v_{n+1} = \sqrt{42 + v_n} < \sqrt{42 + 7} = \sqrt{49} = 7.$$

   So $S(n+1)$ is true. The base case and the inductive step establish by induction that $v_n < 7$ for all $n \in \mathbb{N}$.

5. *[0.5 pt]* No. For example $P(n) : n \geq 1$ is true for all integers $n \geq 1$ but false at 0.

## Exercise 3 · Vector calculations in a basis

1. *[1 pt]* In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{j}$ would be $1 = 0$, a contradiction by uniqueness of coordinates in a basis.

2. *[0.5 pt]* Expanding, $5\vec{u} + 5\vec{v} = 5(\vec{\imath} + \vec{k}) + 5(\vec{\jmath} + \vec{k}) = 5\vec{\imath} + 5\vec{\jmath} + 10\vec{k} = \vec{w}$.

3. *[1.5 pt]* Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis requires

   $$\lambda = 5, \qquad \mu = 5, \qquad \lambda + \mu = 11.$$

   The first two equalities give $\lambda + \mu = 10$, contradicting the third. The three vectors are therefore not coplanar.

4. *[1 pt]* $5\vec{u} - 5\vec{v} = 5(\vec{\imath} + \vec{k}) - 5(\vec{\jmath} + \vec{k}) = 5\vec{\imath} - 5\vec{\jmath}$.

5. *[1 pt]* $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are non-collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

## Exercise 4 · Geometry of a parallelepiped

1. *[1 pt]* By Chasles' relation, $\overrightarrow{AG} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CG} = \vec{\imath} + \vec{\jmath} + \vec{k}$. Similarly, $\overrightarrow{BH} = \overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DH} = -\vec{\imath} + \vec{\jmath} + \vec{k}$.

2. *[1 pt]* By Chasles' relation, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = \frac{\vec{\jmath} - \vec{\imath}}{15} = \frac{\overrightarrow{BD}}{15}$. These vectors are non-zero and collinear: $(IJ)$ and $(BD)$ are parallel.

3. *[1 pt]* We have $\overrightarrow{CG} = \overrightarrow{AE} = \vec{k}$ and $\overrightarrow{AB} = \vec{\imath}$. These two basis vectors are not collinear: an equality $\vec{k} = t\vec{\imath}$ would contradict the uniqueness of the decomposition in this basis.

4. *[1 pt]* The diagonals of a parallelepiped bisect each other. Thus $\overrightarrow{AO} = \frac{1}{2}\overrightarrow{AG}$, which proves $O \in (AG)$.

5. *[1 pt]* The points satisfy $\overrightarrow{IJ} = \frac{-\vec{\imath} + \vec{\jmath}}{15}$ and $\overrightarrow{IK} = \frac{-\vec{\imath} + \vec{k}}{15}$. If $\overrightarrow{IK} = t\,\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would give $\frac{1}{15} = 0$, which is impossible. The vectors are not collinear, so $I, J, K$ are not collinear.

# Test 2 · Practice 1

## PRACTICE PAPER · TRAINING

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{3n^2+4n-1}{4n^2+3}$ and $v_n = \sqrt{n^2+4n} - n$.

1. Determine the limit of $(u_n)$, detailing the transformation of its expression. *(1 pt)*
2. Show that, for every integer $n \geq 1$, $v_n = \frac{4}{\sqrt{1+\frac{4}{n}}+1}$. *(1 pt)*
3. Deduce the limit of $(v_n)$. *(1 pt)*
4. We define $w_n = 3n^2 - 4n$ for $n \geq 1$. Determine the limit of $(w_n)$. *(1 pt)*
5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this claim true? Justify. *(1 pt)*

## Exercise 2 · True or false: reasoning about limits — 5 points

For each of the following statements, say whether it is true or false and justify your answer. An unjustified answer earns no marks. Limits of sequences are taken as $n$ tends to $+\infty$.

1. If $u_n \to 2$ and $v_n \to -5$, then $\frac{u_n}{v_n} \to -\frac{2}{5}$. *(1 pt)*
2. The sequence $u_n = 2 + (-1)^n$ converges to 2. *(1 pt)*
3. If $u_n \to 0$ and $v_n \to +\infty$, then $u_n v_n \to 0$. *(1 pt)*
4. The sequence $w_n = \frac{\sqrt{n^2+3n}}{n}$ for $n \geq 1$ converges to 1. *(1 pt)*
5. For $u_n = \frac{2n+5}{n}$, we have $|u_n - 2| < 0.01$ as soon as $n > 500$. *(1 pt)*

Test 2 · Practice 1 · continued

## Exercise 3 · Vector calculations in a basis — 5 points

In a basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$ of space, we define the vectors

$$\vec{u} = \vec{\imath} + \vec{k}, \qquad \vec{v} = \vec{\jmath} + \vec{k},$$
$$\vec{w} = 2\vec{\imath} + 6\vec{\jmath} + 8\vec{k}, \quad \vec{z} = 2\vec{\imath} + 6\vec{\jmath} + 9\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. *(1 pt)*
2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. *(0.5 pt)*
3. Are the vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. *(1.5 pt)*
4. Express the vector $6\vec{u} - 2\vec{v}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
5. Choose a point $A$, then define $B$, $C$ and $D$ by $\overrightarrow{AB} = \vec{u}$, $\overrightarrow{AC} = \vec{v}$ and $\overrightarrow{AD} = \vec{w}$. Prove that $A$, $B$, $C$ and $D$ are coplanar. *(1 pt)*

## Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped $ABCDEFGH$ that is not flat. The vertices $E$, $F$, $G$ and $H$ are the respective images of $A$, $B$, $C$ and $D$ under the translation by vector $\overrightarrow{AE}$. Let $\vec{\imath} = \overrightarrow{AB}$, $\vec{\jmath} = \overrightarrow{AD}$ and $\vec{k} = \overrightarrow{AE}$. The points $I$, $J$ and $K$ are defined by

$$\overrightarrow{AI} = \frac{1}{18}\vec{\imath}, \qquad \overrightarrow{AJ} = \frac{1}{18}\vec{\jmath}, \qquad \overrightarrow{AK} = \frac{1}{18}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
2. Prove that the lines $(IJ)$ and $(BD)$ are parallel. *(1 pt)*
3. Study the relative position of the lines $(AB)$ and $(CG)$: are they parallel, intersecting or non-coplanar? Justify. *(1 pt)*
4. Let $O$ be the centre of the parallelepiped. Does the point $O$ lie in the plane $(ABG)$? Justify. *(1 pt)*
5. Prove that the points $I$, $J$ and $K$ are not collinear. *(1 pt)*

# Solutions · Test 2 · Practice 1

## Exercise 1 · Limits calculated and justified

1. *[1 pt]* For $n \geq 1$, we divide by $n^2 \neq 0$:

$$u_n = \frac{3 + \frac{4}{n} - \frac{1}{n^2}}{4 + \frac{3}{n^2}}.$$

The terms $\frac{1}{n}$ and $\frac{1}{n^2}$ tend to 0. The numerator tends to 3 and the denominator to $4 \neq 0$; by the quotient rule, $u_n \to \frac{3}{4}$.

2. *[1 pt]* The conjugate is positive. For $n \geq 1$, $\sqrt{n^2} = n$, hence

$$\begin{aligned} v_n &= \frac{(n^2 + 4n) - n^2}{\sqrt{n^2 + 4n} + n} \\ &= \frac{4n}{n(\sqrt{1 + \frac{4}{n}} + 1)} = \frac{4}{\sqrt{1 + \frac{4}{n}} + 1}. \end{aligned}$$

3. *[1 pt]* The denominator tends to 2; limit 2.

4. *[1 pt]* $w_n = n^2(3 - \frac{4}{n}) \to +\infty$; the second factor tends to $3 > 0$.

5. *[1 pt]* No: $(2n) - n = n \to +\infty$. The form $\infty - \infty$ is indeterminate: it does not allow a conclusion without transforming the expression.

## Exercise 2 · True or false: reasoning about limits

1. *[1 pt]* True: the denominator has a non-zero limit; the limit of the quotient is $-\frac{2}{5}$.

2. *[1 pt]* False: the even terms equal 3, the odd terms 1. Both values stay at distance 1 from 2.

3. *[1 pt]* False: $u_n = \frac{1}{n}$, $v_n = n$ for $n \geq 1$ give a constant product equal to 1.

4. *[1 pt]* True: $w_n = \sqrt{1 + \frac{3}{n}} \to 1$.

5. *[1 pt]* True: the gap equals $\frac{5}{n}$; $n > 500$ gives $\frac{5}{n} < 0.01$.

## Exercise 3 · Vector calculations in a basis

1. *[1 pt]* In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{j}$ would be $1 = 0$, a contradiction by uniqueness of coordinates in a basis.

2. *[0.5 pt]* Expanding, $2\vec{u} + 6\vec{v} = 2(\vec{\imath} + \vec{k}) + 6(\vec{j} + \vec{k}) = 2\vec{\imath} + 6\vec{j} + 8\vec{k} = \vec{w}$.

3. *[1.5 pt]* Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis requires

$$\lambda = 2, \qquad \mu = 6, \qquad \lambda + \mu = 9.$$

The first two equalities give $\lambda + \mu = 8$, contradicting the third. The three vectors are therefore not coplanar.

4. *[1 pt]* $6\vec{u} - 2\vec{v} = 6(\vec{\imath} + \vec{k}) - 2(\vec{j} + \vec{k}) = 6\vec{\imath} - 2\vec{j} + 4\vec{k}$.

5. *[1 pt]* $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are non-collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

## Exercise 4 · Geometry of a parallelepiped

1. *[1 pt]* By Chasles' relation, $\overrightarrow{AG} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CG} = \vec{\imath} + \vec{\jmath} + \vec{k}$. Similarly, $\overrightarrow{BH} = \overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DH} = -\vec{\imath} + \vec{\jmath} + \vec{k}$.

2. *[1 pt]* By Chasles' relation, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = \frac{\vec{\jmath} - \vec{\imath}}{18} = \frac{\overrightarrow{BD}}{18}$. These vectors are non-zero and collinear: $(IJ)$ and $(BD)$ are parallel.

3. *[1 pt]* Direction vectors $\vec{\imath}$ and $\vec{k}$, not collinear. An intersection would require $t\vec{\imath} = \vec{\imath} + \vec{\jmath} + s\vec{k}$, impossible by the coefficient of $\vec{\jmath}$. They are non-coplanar.

4. *[1 pt]* $\overrightarrow{AO} = \frac{\overrightarrow{AG}}{2}$, so $O \in (AG) \subset (ABG)$.

5. *[1 pt]* The points satisfy $\overrightarrow{IJ} = \frac{-\vec{\imath} + \vec{\jmath}}{18}$ and $\overrightarrow{IK} = \frac{-\vec{\imath} + \vec{k}}{18}$. If $\overrightarrow{IK} = t\,\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would give $\frac{1}{18} = 0$, which is impossible. The vectors are not collinear, so $I, J, K$ are not collinear.

# Test 2 · Practice 2

**PRACTICE PAPER · TRAINING**

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{2n^2+6n-1}{3n^2+1}$ and $v_n = \sqrt{n^2+6n} - n$.

1. Determine the limit of $(u_n)$, detailing the transformation of its expression. *(1 pt)*
2. Show that, for every integer $n \geq 1$, $v_n = \frac{6}{\sqrt{1+\frac{6}{n}}+1}$. *(1 pt)*
3. Deduce the limit of $(v_n)$. *(1 pt)*
4. We define $w_n = 2n^2 - 6n$ for $n \geq 1$. Determine the limit of $(w_n)$. *(1 pt)*
5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this claim true? Justify. *(1 pt)*

## Exercise 2 · True or false: reasoning about limits — 5 points

For each of the following statements, say whether it is true or false and justify your answer. An unjustified answer earns no marks. Limits of sequences are taken as $n$ tends to $+\infty$.

1. If $u_n \to 5$ and $v_n \to -6$, then $\frac{u_n}{v_n} \to -\frac{5}{6}$. *(1 pt)*
2. The sequence $u_n = 5 + (-1)^n$ converges to 5. *(1 pt)*
3. If $u_n \to 0$ and $v_n \to +\infty$, then $u_n v_n \to 0$. *(1 pt)*
4. The sequence $w_n = \frac{\sqrt{n^2+n}}{n}$ for $n \geq 1$ converges to 1. *(1 pt)*
5. For $u_n = \frac{5n+6}{n}$, we have $|u_n - 5| < 0.01$ as soon as $n > 600$. *(1 pt)*

Test 2 · Practice 2 · continued

## Exercise 3 · Vector calculations in a basis — 5 points

In a basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$ of space, we define the vectors

$$\vec{u} = \vec{\imath} + \vec{k}, \qquad \vec{v} = \vec{\jmath} + \vec{k},$$
$$\vec{w} = 4\vec{\imath} + 7\vec{\jmath} + 11\vec{k}, \quad \vec{z} = 4\vec{\imath} + 7\vec{\jmath} + 12\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. *(1 pt)*
2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. *(0.5 pt)*
3. Are the vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. *(1.5 pt)*
4. Express the vector $7\vec{u} - 4\vec{v}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
5. Choose a point $A$, then define $B$, $C$ and $D$ by $\overrightarrow{AB} = \vec{u}$, $\overrightarrow{AC} = \vec{v}$ and $\overrightarrow{AD} = \vec{w}$. Prove that $A$, $B$, $C$ and $D$ are coplanar. *(1 pt)*

## Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped $ABCDEFGH$ that is not flat. The vertices $E$, $F$, $G$ and $H$ are the respective images of $A$, $B$, $C$ and $D$ under the translation by vector $\overrightarrow{AE}$. Let $\vec{\imath} = \overrightarrow{AB}$, $\vec{\jmath} = \overrightarrow{AD}$ and $\vec{k} = \overrightarrow{AE}$. The points $I$, $J$ and $K$ are defined by

$$\overrightarrow{AI} = \frac{1}{8}\vec{\imath}, \qquad \overrightarrow{AJ} = \frac{1}{8}\vec{\jmath}, \qquad \overrightarrow{AK} = \frac{1}{8}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
2. Prove that the lines $(IJ)$ and $(BD)$ are parallel. *(1 pt)*
3. Study the relative position of the lines $(AB)$ and $(CG)$: are they parallel, intersecting or non-coplanar? Justify. *(1 pt)*
4. Let $O$ be the centre of the parallelepiped. Does the point $O$ lie in the plane $(ABG)$? Justify. *(1 pt)*
5. Prove that the points $I$, $J$ and $K$ are not collinear. *(1 pt)*

# Solutions · Test 2 · Practice 2

## Exercise 1 · Limits calculated and justified

1. *[1 pt]* For $n \geq 1$, we divide by $n^2 \neq 0$:

$$u_n = \frac{2 + \frac{6}{n} - \frac{1}{n^2}}{3 + \frac{1}{n^2}}.$$

The terms $\frac{1}{n}$ and $\frac{1}{n^2}$ tend to 0. The numerator tends to 2 and the denominator to $3 \neq 0$; by the quotient rule, $u_n \to \frac{2}{3}$.

2. *[1 pt]* The conjugate is positive. For $n \geq 1$, $\sqrt{n^2} = n$, hence

$$\begin{aligned} v_n &= \frac{(n^2 + 6n) - n^2}{\sqrt{n^2 + 6n} + n} \\ &= \frac{6n}{n(\sqrt{1 + \frac{6}{n}} + 1)} = \frac{6}{\sqrt{1 + \frac{6}{n}} + 1}. \end{aligned}$$

3. *[1 pt]* The denominator tends to 2; limit 3.

4. *[1 pt]* $w_n = n^2(2 - \frac{6}{n}) \to +\infty$; the second factor tends to $2 > 0$.

5. *[1 pt]* No: $(2n) - n = n \to +\infty$. The form $\infty - \infty$ is indeterminate: it does not allow a conclusion without transforming the expression.

## Exercise 2 · True or false: reasoning about limits

1. *[1 pt]* True: the denominator has a non-zero limit; the limit of the quotient is $-\frac{5}{6}$.

2. *[1 pt]* False: the even terms equal 6, the odd terms 4. Both values stay at distance 1 from 5.

3. *[1 pt]* False: $u_n = \frac{1}{n}$, $v_n = n$ for $n \geq 1$ give a constant product equal to 1.

4. *[1 pt]* True: $w_n = \sqrt{1 + \frac{1}{n}} \to 1$.

5. *[1 pt]* True: the gap equals $\frac{6}{n}$; $n > 600$ gives $\frac{6}{n} < 0.01$.

## Exercise 3 · Vector calculations in a basis

1. *[1 pt]* In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{\jmath}$ would be $1 = 0$, a contradiction by uniqueness of coordinates in a basis.

2. *[0.5 pt]* Expanding, $4\vec{u} + 7\vec{v} = 4(\vec{\imath} + \vec{k}) + 7(\vec{\jmath} + \vec{k}) = 4\vec{\imath} + 7\vec{\jmath} + 11\vec{k} = \vec{w}$.

3. *[1.5 pt]* Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis requires

$$\lambda = 4, \qquad \mu = 7, \qquad \lambda + \mu = 12.$$

The first two equalities give $\lambda + \mu = 11$, contradicting the third. The three vectors are therefore not coplanar.

4. *[1 pt]* $7\vec{u} - 4\vec{v} = 7(\vec{\imath} + \vec{k}) - 4(\vec{\jmath} + \vec{k}) = 7\vec{\imath} - 4\vec{\jmath} + 3\vec{k}$.

5. *[1 pt]* $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are non-collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

## Exercise 4 · Geometry of a parallelepiped

1. *[1 pt]* By Chasles' relation, $\overrightarrow{AG} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CG} = \vec{\imath} + \vec{\jmath} + \vec{k}$. Similarly, $\overrightarrow{BH} = \overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DH} = -\vec{\imath} + \vec{\jmath} + \vec{k}$.

2. *[1 pt]* By Chasles' relation, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = \frac{\vec{\jmath} - \vec{\imath}}{8} = \frac{\overrightarrow{BD}}{8}$. These vectors are non-zero and collinear: $(IJ)$ and $(BD)$ are parallel.

3. *[1 pt]* Direction vectors $\vec{\imath}$ and $\vec{k}$, not collinear. An intersection would require $t\vec{\imath} = \vec{\imath} + \vec{\jmath} + s\vec{k}$, impossible by the coefficient of $\vec{\jmath}$. They are non-coplanar.

4. *[1 pt]* $\overrightarrow{AO} = \frac{\overrightarrow{AG}}{2}$, so $O \in (AG) \subset (ABG)$.

5. *[1 pt]* The points satisfy $\overrightarrow{IJ} = \frac{-\vec{\imath} + \vec{\jmath}}{8}$ and $\overrightarrow{IK} = \frac{-\vec{\imath} + \vec{k}}{8}$. If $\overrightarrow{IK} = t\,\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would give $\frac{1}{8} = 0$, which is impossible. The vectors are not collinear, so $I, J, K$ are not collinear.

# Test 2 · A

**PAPER RESERVED FOR ASSESSMENT**

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{5n^2+7n-1}{6n^2+2}$ and $v_n = \sqrt{n^2+7n} - n$.

1. Determine the limit of $(u_n)$, detailing the transformation of its expression. *(1 pt)*
2. Show that, for every integer $n \geq 1$, $v_n = \dfrac{7}{\sqrt{1+\frac{7}{n}}+1}$. *(1 pt)*
3. Deduce the limit of $(v_n)$. *(1 pt)*
4. We define $w_n = 5n^2 - 7n$ for $n \geq 1$. Determine the limit of $(w_n)$. *(1 pt)*
5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this claim true? Justify. *(1 pt)*

## Exercise 2 · True or false: reasoning about limits — 5 points

For each of the following statements, say whether it is true or false and justify your answer. An unjustified answer earns no marks. Limits of sequences are taken as $n$ tends to $+\infty$.

1. If $u_n \to 4$ and $v_n \to -8$, then $\frac{u_n}{v_n} \to -\frac{1}{2}$. *(1 pt)*
2. The sequence $u_n = 4 + (-1)^n$ converges to 4. *(1 pt)*
3. If $u_n \to 0$ and $v_n \to +\infty$, then $u_n v_n \to 0$. *(1 pt)*
4. The sequence $w_n = \frac{\sqrt{n^2+2n}}{n}$ for $n \geq 1$ converges to 1. *(1 pt)*
5. For $u_n = \frac{4n+8}{n}$, we have $|u_n - 4| < 0.01$ as soon as $n > 800$. *(1 pt)*

Test 2 · A · continued

## Exercise 3 · Vector calculations in a basis — 5 points

In a basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$ of space, we define the vectors

$$\vec{u} = \vec{\imath} + \vec{k}, \qquad \vec{v} = \vec{\jmath} + \vec{k},$$
$$\vec{w} = 3\vec{\imath} + 4\vec{\jmath} + 7\vec{k}, \quad \vec{z} = 3\vec{\imath} + 4\vec{\jmath} + 8\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. *(1 pt)*
2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. *(0.5 pt)*
3. Are the vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. *(1.5 pt)*
4. Express the vector $4\vec{u} - 3\vec{v}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
5. Choose a point $A$, then define $B$, $C$ and $D$ by $\overrightarrow{AB} = \vec{u}$, $\overrightarrow{AC} = \vec{v}$ and $\overrightarrow{AD} = \vec{w}$. Prove that $A$, $B$, $C$ and $D$ are coplanar. *(1 pt)*

## Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped $ABCDEFGH$ that is not flat. The vertices $E$, $F$, $G$ and $H$ are the respective images of $A$, $B$, $C$ and $D$ under the translation by vector $\overrightarrow{AE}$. Let $\vec{\imath} = \overrightarrow{AB}$, $\vec{\jmath} = \overrightarrow{AD}$ and $\vec{k} = \overrightarrow{AE}$. The points $I$, $J$ and $K$ are defined by

$$\overrightarrow{AI} = \frac{1}{15}\vec{\imath}, \qquad \overrightarrow{AJ} = \frac{1}{15}\vec{\jmath}, \qquad \overrightarrow{AK} = \frac{1}{15}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
2. Prove that the lines $(IJ)$ and $(BD)$ are parallel. *(1 pt)*
3. Study the relative position of the lines $(AB)$ and $(CG)$: are they parallel, intersecting or non-coplanar? Justify. *(1 pt)*
4. Let $O$ be the centre of the parallelepiped. Does the point $O$ lie in the plane $(ABG)$? Justify. *(1 pt)*
5. Prove that the points $I$, $J$ and $K$ are not collinear. *(1 pt)*

# Solutions · Test 2 · A

## Exercise 1 · Limits calculated and justified

1. *[1 pt]* For $n \geq 1$, we divide by $n^2 \neq 0$:

$$u_n = \frac{5 + \frac{7}{n} - \frac{1}{n^2}}{6 + \frac{2}{n^2}}.$$

The terms $\frac{1}{n}$ and $\frac{1}{n^2}$ tend to 0. The numerator tends to 5 and the denominator to $6 \neq 0$; by the quotient rule, $u_n \to \frac{5}{6}$.

2. *[1 pt]* The conjugate is positive. For $n \geq 1$, $\sqrt{n^2} = n$, hence

$$\begin{aligned} v_n &= \frac{(n^2 + 7n) - n^2}{\sqrt{n^2 + 7n} + n} \\ &= \frac{7n}{n(\sqrt{1 + \frac{7}{n}} + 1)} = \frac{7}{\sqrt{1 + \frac{7}{n}} + 1}. \end{aligned}$$

3. *[1 pt]* The denominator tends to 2; limit $\frac{7}{2}$.

4. *[1 pt]* $w_n = n^2(5 - \frac{7}{n}) \to +\infty$; the second factor tends to $5 > 0$.

5. *[1 pt]* No: $(2n) - n = n \to +\infty$. The form $\infty - \infty$ is indeterminate: it does not allow a conclusion without transforming the expression.

## Exercise 2 · True or false: reasoning about limits

1. *[1 pt]* True: the denominator has a non-zero limit; the limit of the quotient is $-\frac{1}{2}$.

2. *[1 pt]* False: the even terms equal 5, the odd terms 3. Both values stay at distance 1 from 4.

3. *[1 pt]* False: $u_n = \frac{1}{n}$, $v_n = n$ for $n \geq 1$ give a constant product equal to 1.

4. *[1 pt]* True: $w_n = \sqrt{1 + \frac{2}{n}} \to 1$.

5. *[1 pt]* True: the gap equals $\frac{8}{n}$; $n > 800$ gives $\frac{8}{n} < 0.01$.

## Exercise 3 · Vector calculations in a basis

1. *[1 pt]* In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{\jmath}$ would be $1 = 0$, a contradiction by uniqueness of coordinates in a basis.

2. *[0.5 pt]* Expanding, $3\vec{u} + 4\vec{v} = 3(\vec{\imath} + \vec{k}) + 4(\vec{\jmath} + \vec{k}) = 3\vec{\imath} + 4\vec{\jmath} + 7\vec{k} = \vec{w}$.

3. *[1.5 pt]* Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis requires

$$\lambda = 3, \qquad \mu = 4, \qquad \lambda + \mu = 8.$$

The first two equalities give $\lambda + \mu = 7$, contradicting the third. The three vectors are therefore not coplanar.

4. *[1 pt]* $4\vec{u} - 3\vec{v} = 4(\vec{\imath} + \vec{k}) - 3(\vec{\jmath} + \vec{k}) = 4\vec{\imath} - 3\vec{\jmath} + \vec{k}$.

5. *[1 pt]* $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are non-collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

## Exercise 4 · Geometry of a parallelepiped

1. *[1 pt]* By Chasles' relation, $\overrightarrow{AG} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CG} = \vec{\imath} + \vec{\jmath} + \vec{k}$. Similarly, $\overrightarrow{BH} = \overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DH} = -\vec{\imath} + \vec{\jmath} + \vec{k}$.

2. *[1 pt]* By Chasles' relation, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = \frac{\vec{\jmath} - \vec{\imath}}{15} = \frac{\overrightarrow{BD}}{15}$. These vectors are non-zero and collinear: $(IJ)$ and $(BD)$ are parallel.

3. *[1 pt]* Direction vectors $\vec{\imath}$ and $\vec{k}$, not collinear. An intersection would require $t\vec{\imath} = \vec{\imath} + \vec{\jmath} + s\vec{k}$, impossible by the coefficient of $\vec{\jmath}$. They are non-coplanar.

4. *[1 pt]* $\overrightarrow{AO} = \frac{\overrightarrow{AG}}{2}$, so $O \in (AG) \subset (ABG)$.

5. *[1 pt]* The points satisfy $\overrightarrow{IJ} = \frac{-\vec{\imath} + \vec{\jmath}}{15}$ and $\overrightarrow{IK} = \frac{-\vec{\imath} + \vec{k}}{15}$. If $\overrightarrow{IK} = t\,\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would give $\frac{1}{15} = 0$, which is impossible. The vectors are not collinear, so $I, J, K$ are not collinear.

# Test 2 · B

**PAPER RESERVED FOR ASSESSMENT**

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{4n^2+4n-1}{5n^2+3}$ and $v_n = \sqrt{n^2+4n} - n$.

1. Determine the limit of $(u_n)$, detailing the transformation of its expression. *(1 pt)*
2. Show that, for every integer $n \geq 1$, $v_n = \frac{4}{\sqrt{1+\frac{4}{n}}+1}$. *(1 pt)*
3. Deduce the limit of $(v_n)$. *(1 pt)*
4. We define $w_n = 4n^2 - 4n$ for $n \geq 1$. Determine the limit of $(w_n)$. *(1 pt)*
5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this claim true? Justify. *(1 pt)*

## Exercise 2 · True or false: reasoning about limits — 5 points

For each of the following statements, say whether it is true or false and justify your answer. An unjustified answer earns no marks. Limits of sequences are taken as $n$ tends to $+\infty$.

1. If $u_n \to 3$ and $v_n \to -5$, then $\frac{u_n}{v_n} \to -\frac{3}{5}$. *(1 pt)*
2. The sequence $u_n = 3 + (-1)^n$ converges to 3. *(1 pt)*
3. If $u_n \to 0$ and $v_n \to +\infty$, then $u_n v_n \to 0$. *(1 pt)*
4. The sequence $w_n = \frac{\sqrt{n^2+3n}}{n}$ for $n \geq 1$ converges to 1. *(1 pt)*
5. For $u_n = \frac{3n+5}{n}$, we have $|u_n - 3| < 0.01$ as soon as $n > 500$. *(1 pt)*

Test 2 · B · continued

## Exercise 3 · Vector calculations in a basis — 5 points

In a basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$ of space, we define the vectors

$$\vec{u} = \vec{\imath} + \vec{k}, \qquad \vec{v} = \vec{\jmath} + \vec{k},$$
$$\vec{w} = 3\vec{\imath} + 6\vec{\jmath} + 9\vec{k}, \quad \vec{z} = 3\vec{\imath} + 6\vec{\jmath} + 10\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. *(1 pt)*
2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. *(0.5 pt)*
3. Are the vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. *(1.5 pt)*
4. Express the vector $6\vec{u} - 3\vec{v}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
5. Choose a point $A$, then define $B$, $C$ and $D$ by $\overrightarrow{AB} = \vec{u}$, $\overrightarrow{AC} = \vec{v}$ and $\overrightarrow{AD} = \vec{w}$. Prove that $A$, $B$, $C$ and $D$ are coplanar. *(1 pt)*

## Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped $ABCDEFGH$ that is not flat. The vertices $E$, $F$, $G$ and $H$ are the respective images of $A$, $B$, $C$ and $D$ under the translation by vector $\overrightarrow{AE}$. Let $\vec{\imath} = \overrightarrow{AB}$, $\vec{\jmath} = \overrightarrow{AD}$ and $\vec{k} = \overrightarrow{AE}$. The points $I$, $J$ and $K$ are defined by

$$\overrightarrow{AI} = \frac{1}{5}\vec{\imath}, \qquad \overrightarrow{AJ} = \frac{1}{5}\vec{\jmath}, \qquad \overrightarrow{AK} = \frac{1}{5}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
2. Prove that the lines $(IJ)$ and $(BD)$ are parallel. *(1 pt)*
3. Study the relative position of the lines $(AB)$ and $(CG)$: are they parallel, intersecting or non-coplanar? Justify. *(1 pt)*
4. Let $O$ be the centre of the parallelepiped. Does the point $O$ lie in the plane $(ABG)$? Justify. *(1 pt)*
5. Prove that the points $I$, $J$ and $K$ are not collinear. *(1 pt)*

# Solutions · Test 2 · B

## Exercise 1 · Limits calculated and justified

1. *[1 pt]* For $n \geq 1$, we divide by $n^2 \neq 0$:

$$u_n = \frac{4 + \frac{4}{n} - \frac{1}{n^2}}{5 + \frac{3}{n^2}}.$$

The terms $\frac{1}{n}$ and $\frac{1}{n^2}$ tend to 0. The numerator tends to 4 and the denominator to $5 \neq 0$; by the quotient rule, $u_n \to \frac{4}{5}$.

2. *[1 pt]* The conjugate is positive. For $n \geq 1$, $\sqrt{n^2} = n$, hence

$$\begin{aligned} v_n &= \frac{(n^2 + 4n) - n^2}{\sqrt{n^2 + 4n} + n} \\ &= \frac{4n}{n(\sqrt{1 + \frac{4}{n}} + 1)} = \frac{4}{\sqrt{1 + \frac{4}{n}} + 1}. \end{aligned}$$

3. *[1 pt]* The denominator tends to 2; limit 2.

4. *[1 pt]* $w_n = n^2(4 - \frac{4}{n}) \to +\infty$; the second factor tends to $4 > 0$.

5. *[1 pt]* No: $(2n) - n = n \to +\infty$. The form $\infty - \infty$ is indeterminate: it does not allow a conclusion without transforming the expression.

## Exercise 2 · True or false: reasoning about limits

1. *[1 pt]* True: the denominator has a non-zero limit; the limit of the quotient is $-\frac{3}{5}$.

2. *[1 pt]* False: the even terms equal 4, the odd terms 2. Both values stay at distance 1 from 3.

3. *[1 pt]* False: $u_n = \frac{1}{n}$, $v_n = n$ for $n \geq 1$ give a constant product equal to 1.

4. *[1 pt]* True: $w_n = \sqrt{1 + \frac{3}{n}} \to 1$.

5. *[1 pt]* True: the gap equals $\frac{5}{n}$; $n > 500$ gives $\frac{5}{n} < 0.01$.

## Exercise 3 · Vector calculations in a basis

1. *[1 pt]* In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{j}$ would be $1 = 0$, a contradiction by uniqueness of coordinates in a basis.

2. *[0.5 pt]* Expanding, $3\vec{u} + 6\vec{v} = 3(\vec{\imath} + \vec{k}) + 6(\vec{j} + \vec{k}) = 3\vec{\imath} + 6\vec{j} + 9\vec{k} = \vec{w}$.

3. *[1.5 pt]* Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis requires

$$\lambda = 3, \qquad \mu = 6, \qquad \lambda + \mu = 10.$$

The first two equalities give $\lambda + \mu = 9$, contradicting the third. The three vectors are therefore not coplanar.

4. *[1 pt]* $6\vec{u} - 3\vec{v} = 6(\vec{\imath} + \vec{k}) - 3(\vec{j} + \vec{k}) = 6\vec{\imath} - 3\vec{j} + 3\vec{k}$.

5. *[1 pt]* $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are non-collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

## Exercise 4 · Geometry of a parallelepiped

1. *[1 pt]* By Chasles' relation, $\overrightarrow{AG} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CG} = \vec{\imath} + \vec{\jmath} + \vec{k}$. Similarly, $\overrightarrow{BH} = \overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DH} = -\vec{\imath} + \vec{\jmath} + \vec{k}$.

2. *[1 pt]* By Chasles' relation, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = \frac{\vec{\jmath} - \vec{\imath}}{5} = \frac{\overrightarrow{BD}}{5}$. These vectors are non-zero and collinear: $(IJ)$ and $(BD)$ are parallel.

3. *[1 pt]* Direction vectors $\vec{\imath}$ and $\vec{k}$, not collinear. An intersection would require $t\vec{\imath} = \vec{\imath} + \vec{\jmath} + s\vec{k}$, impossible by the coefficient of $\vec{\jmath}$. They are non-coplanar.

4. *[1 pt]* $\overrightarrow{AO} = \frac{\overrightarrow{AG}}{2}$, so $O \in (AG) \subset (ABG)$.

5. *[1 pt]* The points satisfy $\overrightarrow{IJ} = \frac{-\vec{\imath} + \vec{\jmath}}{5}$ and $\overrightarrow{IK} = \frac{-\vec{\imath} + \vec{k}}{5}$. If $\overrightarrow{IK} = t\,\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would give $\frac{1}{5} = 0$, which is impossible. The vectors are not collinear, so $I, J, K$ are not collinear.

# Common test · Practice 1

## PRACTICE PAPER · TRAINING

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · A discrete evolution — 5 points

Consider the sequence $(u_n)$ defined by $u_0 = 2$ and, for every natural number $n$, $u_{n+1} = \frac{1}{4}u_n + 6$.

1. Calculate $u_1$ and $u_2$. *(0.5 pt)*
2. Prove by induction that, for all $n \in \mathbb{N}$, $u_n = 8 - 6\left(\frac{1}{4}\right)^n$. *(2 pt)*
3. Deduce that, for every natural number $n$, $2 \leq u_n < 8$. *(1 pt)*
4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of $(u_n)$. *(1 pt)*
5. A student claims: "There exists an integer $n$ such that $u_n = 8$." Is this claim true? Justify. *(0.5 pt)*

## Exercise 2 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{3n^2+5n-1}{4n^2+2}$ and $v_n = \sqrt{n^2+5n} - n$.

1. Determine the limit of $(u_n)$, detailing the transformation of its expression. *(1 pt)*
2. Show that, for every integer $n \geq 1$, $v_n = \frac{5}{\sqrt{1+\frac{5}{n}}+1}$. *(1 pt)*
3. Deduce the limit of $(v_n)$. *(1 pt)*
4. We define $w_n = 3n^2 - 5n$ for $n \geq 1$. Determine the limit of $(w_n)$. *(1 pt)*
5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this claim true? Justify. *(1 pt)*

Common test · Practice 1 · continued

## Exercise 3 · Limits and asymptotes — 5 points

Consider the function $f$ defined by $f(x) = \frac{2x+6}{x-2}$. Denote by $\mathcal{C}_f$ its graph in a coordinate frame of the plane.

1. Determine the domain of $f$. Find two real numbers $A$ and $B$ such that, for all $x \neq 2$, $f(x) = A + \frac{B}{x-2}$. *(1 pt)*
2. Determine the limits of $f$ at $-\infty$ and at $+\infty$. *(1 pt)*
3. Determine the limits of $f$ at $2^-$ and at $2^+$. *(1 pt)*
4. Give the equations of the asymptotes of $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. *(1 pt)*
5. Independently of the previous questions, determine $\lim\limits_{x \to +\infty} x^2 e^{-2x}$ with justification. *(1 pt)*

## Exercise 4 · Codes and selections — 5 points

A code is an ordered sequence of 4 symbols chosen from an alphabet containing 15 distinct symbols. The rules about repetitions are specified in each question, independently of the other questions.

1. How many codes can be formed if repetitions are allowed? *(1 pt)*
2. How many codes can be formed if each symbol must be used at most once? *(1 pt)*
3. A symbol of the alphabet is fixed in advance. Repetitions being allowed, how many codes contain this symbol at least once? *(1 pt)*
4. The first symbol of the code is imposed and repetitions are forbidden. How many codes can be formed? *(1 pt)*
5. Two distinct symbols of the alphabet are forbidden. The other symbols may be repeated. How many codes can be formed? *(1 pt)*

# Solutions · Common test · Practice 1

## Exercise 1 · A discrete evolution

1. *[0.5 pt]* The relation is applied successively for indices 0 and 1:

$$u_1 = \frac{1}{4} \times 2 + 6 = \frac{13}{2}, \qquad u_2 = \frac{1}{4} \times \frac{13}{2} + 6 = \frac{61}{8}.$$

2. *[2 pt]* For $n \in \mathbb{N}$, let $P(n)$ the property $u_n = 8 - 6\left(\frac{1}{4}\right)^n$.

   **Base case.** For $n = 0$, $8 - 6 \times 1 = 2 = u_0$. So $P(0)$ is true.

   **Inductive step.** Fix an integer $n \geq 0$ and assume $P(n)$ true. By the relation defining the sequence, then by the induction hypothesis,

$$\begin{aligned} u_{n+1} &= \frac{1}{4}u_n + 6 \\ &= \frac{1}{4}\left[8 - 6\left(\frac{1}{4}\right)^n\right] + 6 \\ &= 2 + 6 - 6\left(\frac{1}{4}\right)^{n+1} \\ &= 8 - 6\left(\frac{1}{4}\right)^{n+1}. \end{aligned}$$

   This is $P(n+1)$: the property is hereditary.

   **Conclusion.** The principle of induction gives $u_n = 8 - 6\left(\frac{1}{4}\right)^n$ for all $n \in \mathbb{N}$.

3. *[1 pt]* Since $0 < \frac{1}{4} < 1$, for all $n \in \mathbb{N}$ we have $0 < (\frac{1}{4})^n \leq 1$. Multiplying by $-6 < 0$ reverses the inequalities:

$$-6 \leq -6(\frac{1}{4})^n < 0.$$

   Adding 8 and using the previous formula, we obtain $2 \leq u_n < 8$.

4. *[1 pt]* For all $n \in \mathbb{N}$, the explicit formula gives

$$\begin{aligned} u_{n+1} - u_n &= -6(\frac{1}{4})^{n+1} + 6(\frac{1}{4})^n \\ &= 6(1 - \frac{1}{4})(\frac{1}{4})^n = \frac{9}{2}(\frac{1}{4})^n > 0. \end{aligned}$$

   The sequence is therefore strictly increasing.

5. *[0.5 pt]* The claim is false. For each integer $n$, $8 - u_n = 6(\frac{1}{4})^n > 0$. So no term is equal to 8, whatever the number of steps.

## Exercise 2 · Limits calculated and justified

1. *[1 pt]* For $n \geq 1$, we divide by $n^2 \neq 0$:

$$u_n = \frac{3 + \frac{5}{n} - \frac{1}{n^2}}{4 + \frac{2}{n^2}}.$$

   The terms $\frac{1}{n}$ and $\frac{1}{n^2}$ tend to 0. The numerator tends to 3 and the denominator to $4 \neq 0$; by the quotient rule, $u_n \to \frac{3}{4}$.

2. *[1 pt]* The conjugate is positive. For $n \geq 1$, $\sqrt{n^2} = n$, hence

$$v_n = \frac{(n^2+5n)-n^2}{\sqrt{n^2+5n}+n} = \frac{5n}{n(\sqrt{1+\frac{5}{n}}+1)} = \frac{5}{\sqrt{1+\frac{5}{n}}+1}.$$

3. *[1 pt]* The denominator tends to 2; limit $\frac{5}{2}$.

4. *[1 pt]* $w_n = n^2(3-\frac{5}{n}) \to +\infty$; the second factor tends to $3 > 0$.

5. *[1 pt]* No: $(2n) - n = n \to +\infty$. The sign $\infty - \infty$ is an indeterminate form, not a rule of calculation.

## Exercise 3 · Limits and asymptotes

1. *[1 pt]* The denominator vanishes only at 2: $D_f = \mathbb{R} \setminus \{2\}$. Since $2x+6 = 2(x-2)+10$, we obtain $f(x) = 2 + \dfrac{10}{x-2}$.

2. *[1 pt]* Both limits equal 2.

3. *[1 pt]* As $x \to 2^-$, $x-2 \to 0$ through negative values; since $10 > 0$, $\frac{10}{x-2} \to -\infty$, so $f(x) \to -\infty$. On the right, the denominator tends to 0 through positive values: the quotient, and then $f(x)$, tend to $+\infty$.

4. *[1 pt]* $x = 2$ and $y = 2$; the curve is below for $x < 2$ and above for $x > 2$.

5. *[1 pt]* Let $t = 2x$. Then $t \to +\infty$ and $x^2e^{-2x} = \frac{1}{4}\frac{t^2}{e^t}$. Comparison of growth rates gives $\frac{t^2}{e^t} \to 0$; the required limit is therefore 0.

## Exercise 4 · Codes and selections

1. *[1 pt]* $15^4 = 50625$ by the multiplication principle.

2. *[1 pt]* The successive choices have 15, 14, then one choice fewer at each step, down to 12 possibilities. By multiplication,

$$15 \times 14 \times 13 \times 12 = 32760.$$

   The order of the symbols is part of the code.

3. *[1 pt]* $15^4 - 14^4 = 12209$ by complement.

4. *[1 pt]* The symbol in the first position is fixed and can no longer be used. The 3 following positions give successively $14, \ldots, 12$ choices:

$$14 \times 13 \times 12 = 2184.$$

5. *[1 pt]* $13^4 = 28561$.

# Common test · Practice 2

## PRACTICE PAPER · TRAINING

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · A discrete evolution — 5 points

Consider the sequence $(u_n)$ defined by $u_0 = 3$ and, for every natural number $n$, $u_{n+1} = \frac{1}{3}u_n + \frac{20}{3}$.

1. Calculate $u_1$ and $u_2$. *(0.5 pt)*
2. Prove by induction that, for all $n \in \mathbb{N}$, $u_n = 10 - 7\left(\frac{1}{3}\right)^n$. *(2 pt)*
3. Deduce that, for every natural number $n$, $3 \leq u_n < 10$. *(1 pt)*
4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of $(u_n)$. *(1 pt)*
5. A student claims: "There exists an integer $n$ such that $u_n = 10$." Is this claim true? Justify. *(0.5 pt)*

## Exercise 2 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{2n^2+7n-1}{3n^2+3}$ and $v_n = \sqrt{n^2+7n} - n$.

1. Determine the limit of $(u_n)$, detailing the transformation of its expression. *(1 pt)*
2. Show that, for every integer $n \geq 1$, $v_n = \dfrac{7}{\sqrt{1+\frac{7}{n}}+1}$. *(1 pt)*
3. Deduce the limit of $(v_n)$. *(1 pt)*
4. We define $w_n = 2n^2 - 7n$ for $n \geq 1$. Determine the limit of $(w_n)$. *(1 pt)*
5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this claim true? Justify. *(1 pt)*

Common test · Practice 2 · continued

## Exercise 3 · Limits and asymptotes — 5 points

Consider the function $f$ defined by $f(x) = \frac{5x+7}{x-3}$. Denote by $\mathcal{C}_f$ its graph in a coordinate frame of the plane.

1. Determine the domain of $f$. Find two real numbers $A$ and $B$ such that, for all $x \neq 3$, $f(x) = A + \frac{B}{x-3}$. *(1 pt)*
2. Determine the limits of $f$ at $-\infty$ and at $+\infty$. *(1 pt)*
3. Determine the limits of $f$ at $3^-$ and at $3^+$. *(1 pt)*
4. Give the equations of the asymptotes of $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. *(1 pt)*
5. Independently of the previous questions, determine $\lim\limits_{x \to +\infty} x^5 e^{-3x}$ with justification. *(1 pt)*

## Exercise 4 · Codes and selections — 5 points

A code is an ordered sequence of 5 symbols chosen from an alphabet containing 16 distinct symbols. The rules about repetitions are specified in each question, independently of the other questions.

1. How many codes can be formed if repetitions are allowed? *(1 pt)*
2. How many codes can be formed if each symbol must be used at most once? *(1 pt)*
3. A symbol of the alphabet is fixed in advance. Repetitions being allowed, how many codes contain this symbol at least once? *(1 pt)*
4. The first symbol of the code is imposed and repetitions are forbidden. How many codes can be formed? *(1 pt)*
5. Two distinct symbols of the alphabet are forbidden. The other symbols may be repeated. How many codes can be formed? *(1 pt)*

# Solutions · Common test · Practice 2

## Exercise 1 · A discrete evolution

1. *[0.5 pt]* The relation is applied successively for indices 0 and 1:

$$u_1 = \frac{1}{3} \times 3 + \frac{20}{3} = \frac{23}{3}, \qquad u_2 = \frac{1}{3} \times \frac{23}{3} + \frac{20}{3} = \frac{83}{9}.$$

2. *[2 pt]* For $n \in \mathbb{N}$, let $P(n)$ the property $u_n = 10 - 7\left(\frac{1}{3}\right)^n$.

   **Base case.** For $n = 0$, $10 - 7 \times 1 = 3 = u_0$. So $P(0)$ is true.

   **Inductive step.** Fix an integer $n \geq 0$ and assume $P(n)$ true. By the relation defining the sequence, then by the induction hypothesis,

$$\begin{aligned} u_{n+1} &= \frac{1}{3}u_n + \frac{20}{3} \\ &= \frac{1}{3}\left[10 - 7\left(\frac{1}{3}\right)^n\right] + \frac{20}{3} \\ &= \frac{10}{3} + \frac{20}{3} - 7\left(\frac{1}{3}\right)^{n+1} \\ &= 10 - 7\left(\frac{1}{3}\right)^{n+1}. \end{aligned}$$

   This is $P(n+1)$: the property is hereditary.

   **Conclusion.** The principle of induction gives $u_n = 10 - 7\left(\frac{1}{3}\right)^n$ for all $n \in \mathbb{N}$.

3. *[1 pt]* Since $0 < \frac{1}{3} < 1$, for all $n \in \mathbb{N}$ we have $0 < (\frac{1}{3})^n \leq 1$. Multiplying by $-7 < 0$ reverses the inequalities:

$$-7 \leq -7(\frac{1}{3})^n < 0.$$

   Adding 10 and using the previous formula, we obtain $3 \leq u_n < 10$.

4. *[1 pt]* For all $n \in \mathbb{N}$, the explicit formula gives

$$\begin{aligned} u_{n+1} - u_n &= -7(\frac{1}{3})^{n+1} + 7(\frac{1}{3})^n \\ &= 7(1 - \frac{1}{3})(\frac{1}{3})^n = \frac{14}{3}(\frac{1}{3})^n > 0. \end{aligned}$$

   The sequence is therefore strictly increasing.

5. *[0.5 pt]* The claim is false. For each integer $n$, $10 - u_n = 7(\frac{1}{3})^n > 0$. So no term is equal to 10, whatever the number of steps.

## Exercise 2 · Limits calculated and justified

1. *[1 pt]* For $n \geq 1$, we divide by $n^2 \neq 0$:

$$u_n = \frac{2 + \frac{7}{n} - \frac{1}{n^2}}{3 + \frac{3}{n^2}}.$$

   The terms $\frac{1}{n}$ and $\frac{1}{n^2}$ tend to 0. The numerator tends to 2 and the denominator to $3 \neq 0$; by the quotient rule, $u_n \to \frac{2}{3}$.

2. *[1 pt]* The conjugate is positive. For $n \geq 1$, $\sqrt{n^2} = n$, hence

$$v_n = \frac{(n^2+7n)-n^2}{\sqrt{n^2+7n}+n} = \frac{7n}{n(\sqrt{1+\frac{7}{n}}+1)} = \frac{7}{\sqrt{1+\frac{7}{n}}+1}.$$

3. *[1 pt]* The denominator tends to 2; limit $\frac{7}{2}$.

4. *[1 pt]* $w_n = n^2(2-\frac{7}{n}) \to +\infty$; the second factor tends to $2 > 0$.

5. *[1 pt]* No: $(2n) - n = n \to +\infty$. The sign $\infty - \infty$ is an indeterminate form, not a rule of calculation.

## Exercise 3 · Limits and asymptotes

1. *[1 pt]* The denominator vanishes only at 3: $D_f = \mathbb{R} \setminus \{3\}$. Since $5x+7 = 5(x-3)+22$, we obtain $f(x) = 5 + \dfrac{22}{x-3}$.

2. *[1 pt]* Both limits equal 5.

3. *[1 pt]* As $x \to 3^-$, $x-3 \to 0$ through negative values; since $22 > 0$, $\frac{22}{x-3} \to -\infty$, so $f(x) \to -\infty$. On the right, the denominator tends to 0 through positive values: the quotient, and then $f(x)$, tend to $+\infty$.

4. *[1 pt]* $x = 3$ and $y = 5$; the curve is below for $x < 3$ and above for $x > 3$.

5. *[1 pt]* Let $t = 3x$. Then $t \to +\infty$ and $x^5 e^{-3x} = \frac{1}{243}\frac{t^5}{e^t}$. Comparison of growth rates gives $\frac{t^5}{e^t} \to 0$; the required limit is therefore 0.

## Exercise 4 · Codes and selections

1. *[1 pt]* $16^5 = 1048576$ by the multiplication principle.

2. *[1 pt]* The successive choices have 16, 15, then one choice fewer at each step, down to 12 possibilities. By multiplication,

$$16 \times 15 \times 14 \times 13 \times 12 = 524160.$$

   The order of the symbols is part of the code.

3. *[1 pt]* $16^5 - 15^5 = 289201$ by complement.

4. *[1 pt]* The symbol in the first position is fixed and can no longer be used. The 4 following positions give successively $15, \ldots, 12$ choices:

$$15 \times 14 \times 13 \times 12 = 32760.$$

5. *[1 pt]* $14^5 = 537824$.

# Common test · A

**PAPER RESERVED FOR ASSESSMENT**

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · A discrete evolution — 5 points

Consider the sequence $(u_n)$ defined by $u_0 = 1$ and, for every natural number $n$, $u_{n+1} = \frac{1}{2}u_n + 6$.

1. Calculate $u_1$ and $u_2$. *(0.5 pt)*
2. Prove by induction that, for all $n \in \mathbb{N}$, $u_n = 12 - 11\left(\frac{1}{2}\right)^n$. *(2 pt)*
3. Deduce that, for every natural number $n$, $1 \leq u_n < 12$. *(1 pt)*
4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of $(u_n)$. *(1 pt)*
5. A student claims: "There exists an integer $n$ such that $u_n = 12$." Is this claim true? Justify. *(0.5 pt)*

## Exercise 2 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{5n^2+8n-1}{6n^2+1}$ and $v_n = \sqrt{n^2+8n} - n$.

1. Determine the limit of $(u_n)$, detailing the transformation of its expression. *(1 pt)*
2. Show that, for every integer $n \geq 1$, $v_n = \frac{8}{\sqrt{1+\frac{8}{n}}+1}$. *(1 pt)*
3. Deduce the limit of $(v_n)$. *(1 pt)*
4. We define $w_n = 5n^2 - 8n$ for $n \geq 1$. Determine the limit of $(w_n)$. *(1 pt)*
5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this claim true? Justify. *(1 pt)*

Common test · A · continued

## Exercise 3 · Limits and asymptotes — 5 points

Consider the function $f$ defined by $f(x) = \frac{4x+4}{x-1}$. Denote by $\mathcal{C}_f$ its graph in a coordinate frame of the plane.

1. Determine the domain of $f$. Find two real numbers $A$ and $B$ such that, for all $x \neq 1$, $f(x) = A + \frac{B}{x-1}$. *(1 pt)*
2. Determine the limits of $f$ at $-\infty$ and at $+\infty$. *(1 pt)*
3. Determine the limits of $f$ at $1^-$ and at $1^+$. *(1 pt)*
4. Give the equations of the asymptotes of $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. *(1 pt)*
5. Independently of the previous questions, determine $\lim\limits_{x \to +\infty} x^4 e^{-1x}$ with justification. *(1 pt)*

## Exercise 4 · Codes and selections — 5 points

A code is an ordered sequence of 3 symbols chosen from an alphabet containing 12 distinct symbols. The rules about repetitions are specified in each question, independently of the other questions.

1. How many codes can be formed if repetitions are allowed? *(1 pt)*
2. How many codes can be formed if each symbol must be used at most once? *(1 pt)*
3. A symbol of the alphabet is fixed in advance. Repetitions being allowed, how many codes contain this symbol at least once? *(1 pt)*
4. The first symbol of the code is imposed and repetitions are forbidden. How many codes can be formed? *(1 pt)*
5. Two distinct symbols of the alphabet are forbidden. The other symbols may be repeated. How many codes can be formed? *(1 pt)*

# Solutions · Common test · A

## Exercise 1 · A discrete evolution

1. *[0.5 pt]* The relation is applied successively for indices 0 and 1:

$$u_1 = \frac{1}{2} \times 1 + 6 = \frac{13}{2}, \qquad u_2 = \frac{1}{2} \times \frac{13}{2} + 6 = \frac{37}{4}.$$

2. *[2 pt]* For $n \in \mathbb{N}$, let $P(n)$ the property $u_n = 12 - 11\left(\frac{1}{2}\right)^n$.

   **Base case.** For $n = 0$, $12 - 11 \times 1 = 1 = u_0$. So $P(0)$ is true.

   **Inductive step.** Fix an integer $n \geq 0$ and assume $P(n)$ true. By the relation defining the sequence, then by the induction hypothesis,

$$\begin{aligned} u_{n+1} &= \frac{1}{2}u_n + 6 \\ &= \frac{1}{2}\left[12 - 11\left(\frac{1}{2}\right)^n\right] + 6 \\ &= 6 + 6 - 11\left(\frac{1}{2}\right)^{n+1} \\ &= 12 - 11\left(\frac{1}{2}\right)^{n+1}. \end{aligned}$$

   This is $P(n+1)$: the property is hereditary.

   **Conclusion.** The principle of induction gives $u_n = 12 - 11\left(\frac{1}{2}\right)^n$ for all $n \in \mathbb{N}$.

3. *[1 pt]* Since $0 < \frac{1}{2} < 1$, for all $n \in \mathbb{N}$ we have $0 < (\frac{1}{2})^n \leq 1$. Multiplying by $-11 < 0$ reverses the inequalities:

$$-11 \leq -11(\frac{1}{2})^n < 0.$$

   Adding 12 and using the previous formula, we obtain $1 \leq u_n < 12$.

4. *[1 pt]* For all $n \in \mathbb{N}$, the explicit formula gives

$$\begin{aligned} u_{n+1} - u_n &= -11(\frac{1}{2})^{n+1} + 11(\frac{1}{2})^n \\ &= 11(1 - \frac{1}{2})(\frac{1}{2})^n = \frac{11}{2}(\frac{1}{2})^n > 0. \end{aligned}$$

   The sequence is therefore strictly increasing.

5. *[0.5 pt]* The claim is false. For each integer $n$, $12 - u_n = 11(\frac{1}{2})^n > 0$. So no term is equal to 12, whatever the number of steps.

## Exercise 2 · Limits calculated and justified

1. *[1 pt]* For $n \geq 1$, we divide by $n^2 \neq 0$:

$$u_n = \frac{5 + \frac{8}{n} - \frac{1}{n^2}}{6 + \frac{1}{n^2}}.$$

   The terms $\frac{1}{n}$ and $\frac{1}{n^2}$ tend to 0. The numerator tends to 5 and the denominator to $6 \neq 0$; by the quotient rule, $u_n \to \frac{5}{6}$.

2. *[1 pt]* The conjugate is positive. For $n \geq 1$, $\sqrt{n^2} = n$, hence

$$v_n = \frac{(n^2+8n)-n^2}{\sqrt{n^2+8n}+n} = \frac{8n}{n(\sqrt{1+\frac{8}{n}}+1)} = \frac{8}{\sqrt{1+\frac{8}{n}}+1}.$$

3. *[1 pt]* The denominator tends to 2; limit 4.

4. *[1 pt]* $w_n = n^2(5-\frac{8}{n}) \to +\infty$; the second factor tends to $5 > 0$.

5. *[1 pt]* No: $(2n) - n = n \to +\infty$. The sign $\infty - \infty$ is an indeterminate form, not a rule of calculation.

## Exercise 3 · Limits and asymptotes

1. *[1 pt]* The denominator vanishes only at 1: $D_f = \mathbb{R} \setminus \{1\}$. Since $4x+4 = 4(x-1)+8$, we obtain $f(x) = 4 + \dfrac{8}{x-1}$.

2. *[1 pt]* Both limits equal 4.

3. *[1 pt]* As $x \to 1^-$, $x-1 \to 0$ through negative values; since $8 > 0$, $\frac{8}{x-1} \to -\infty$, so $f(x) \to -\infty$. On the right, the denominator tends to 0 through positive values: the quotient, and then $f(x)$, tend to $+\infty$.

4. *[1 pt]* $x = 1$ and $y = 4$; the curve is below for $x < 1$ and above for $x > 1$.

5. *[1 pt]* Comparison of growth rates between a power and the exponential gives directly $x^4 e^{-x} = \frac{x^4}{e^x} \to 0$ as $x \to +\infty$.

## Exercise 4 · Codes and selections

1. *[1 pt]* $12^3 = 1728$ by the multiplication principle.

2. *[1 pt]* The successive choices have 12, 11, then one choice fewer at each step, down to 10 possibilities. By multiplication,

$$12 \times 11 \times 10 = 1320.$$

   The order of the symbols is part of the code.

3. *[1 pt]* $12^3 - 11^3 = 397$ by complement.

4. *[1 pt]* The symbol in the first position is fixed and can no longer be used. The 2 following positions give successively $11, \ldots, 10$ choices:

$$11 \times 10 = 110.$$

5. *[1 pt]* $10^3 = 1000$.

# Common test · B

**PAPER RESERVED FOR ASSESSMENT**

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · A discrete evolution — 5 points

Consider the sequence $(u_n)$ defined by $u_0 = 2$ and, for every natural number $n$, $u_{n+1} = \frac{1}{5}u_n + \frac{32}{5}$.

1. Calculate $u_1$ and $u_2$. *(0.5 pt)*
2. Prove by induction that, for all $n \in \mathbb{N}$, $u_n = 8 - 6\left(\frac{1}{5}\right)^n$. *(2 pt)*
3. Deduce that, for every natural number $n$, $2 \leq u_n < 8$. *(1 pt)*
4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of $(u_n)$. *(1 pt)*
5. A student claims: "There exists an integer $n$ such that $u_n = 8$." Is this claim true? Justify. *(0.5 pt)*

## Exercise 2 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{4n^2+5n-1}{5n^2+2}$ and $v_n = \sqrt{n^2+5n} - n$.

1. Determine the limit of $(u_n)$, detailing the transformation of its expression. *(1 pt)*
2. Show that, for every integer $n \geq 1$, $v_n = \dfrac{5}{\sqrt{1+\frac{5}{n}}+1}$. *(1 pt)*
3. Deduce the limit of $(v_n)$. *(1 pt)*
4. We define $w_n = 4n^2 - 5n$ for $n \geq 1$. Determine the limit of $(w_n)$. *(1 pt)*
5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this claim true? Justify. *(1 pt)*

Common test · B · continued

## Exercise 3 · Limits and asymptotes — 5 points

Consider the function $f$ defined by $f(x) = \frac{3x+6}{x-2}$. Denote by $\mathcal{C}_f$ its graph in a coordinate frame of the plane.

1. Determine the domain of $f$. Find two real numbers $A$ and $B$ such that, for all $x \neq 2$, $f(x) = A + \frac{B}{x-2}$. *(1 pt)*
2. Determine the limits of $f$ at $-\infty$ and at $+\infty$. *(1 pt)*
3. Determine the limits of $f$ at $2^-$ and at $2^+$. *(1 pt)*
4. Give the equations of the asymptotes of $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. *(1 pt)*
5. Independently of the previous questions, determine $\lim\limits_{x \to +\infty} x^3 e^{-2x}$ with justification. *(1 pt)*

## Exercise 4 · Codes and selections — 5 points

A code is an ordered sequence of 4 symbols chosen from an alphabet containing 13 distinct symbols. The rules about repetitions are specified in each question, independently of the other questions.

1. How many codes can be formed if repetitions are allowed? *(1 pt)*
2. How many codes can be formed if each symbol must be used at most once? *(1 pt)*
3. A symbol of the alphabet is fixed in advance. Repetitions being allowed, how many codes contain this symbol at least once? *(1 pt)*
4. The first symbol of the code is imposed and repetitions are forbidden. How many codes can be formed? *(1 pt)*
5. Two distinct symbols of the alphabet are forbidden. The other symbols may be repeated. How many codes can be formed? *(1 pt)*

# Solutions · Common test · B

## Exercise 1 · A discrete evolution

1. *[0.5 pt]* The relation is applied successively for indices 0 and 1:

$$u_1 = \frac{1}{5} \times 2 + \frac{32}{5} = \frac{34}{5}, \qquad u_2 = \frac{1}{5} \times \frac{34}{5} + \frac{32}{5} = \frac{194}{25}.$$

2. *[2 pt]* For $n \in \mathbb{N}$, let $P(n)$ the property $u_n = 8 - 6\left(\frac{1}{5}\right)^n$.

   **Base case.** For $n = 0$, $8 - 6 \times 1 = 2 = u_0$. So $P(0)$ is true.

   **Inductive step.** Fix an integer $n \geq 0$ and assume $P(n)$ true. By the relation defining the sequence, then by the induction hypothesis,

$$\begin{aligned} u_{n+1} &= \frac{1}{5}u_n + \frac{32}{5} \\ &= \frac{1}{5}\left[8 - 6\left(\frac{1}{5}\right)^n\right] + \frac{32}{5} \\ &= \frac{8}{5} + \frac{32}{5} - 6\left(\frac{1}{5}\right)^{n+1} \\ &= 8 - 6\left(\frac{1}{5}\right)^{n+1}. \end{aligned}$$

   This is $P(n+1)$: the property is hereditary.

   **Conclusion.** The principle of induction gives $u_n = 8 - 6\left(\frac{1}{5}\right)^n$ for all $n \in \mathbb{N}$.

3. *[1 pt]* Since $0 < \frac{1}{5} < 1$, for all $n \in \mathbb{N}$ we have $0 < (\frac{1}{5})^n \leq 1$. Multiplying by $-6 < 0$ reverses the inequalities:

$$-6 \leq -6(\frac{1}{5})^n < 0.$$

   Adding 8 and using the previous formula, we obtain $2 \leq u_n < 8$.

4. *[1 pt]* For all $n \in \mathbb{N}$, the explicit formula gives

$$\begin{aligned} u_{n+1} - u_n &= -6(\frac{1}{5})^{n+1} + 6(\frac{1}{5})^n \\ &= 6(1 - \frac{1}{5})(\frac{1}{5})^n = \frac{24}{5}(\frac{1}{5})^n > 0. \end{aligned}$$

   The sequence is therefore strictly increasing.

5. *[0.5 pt]* The claim is false. For each integer $n$, $8 - u_n = 6(\frac{1}{5})^n > 0$. So no term is equal to 8, whatever the number of steps.

## Exercise 2 · Limits calculated and justified

1. *[1 pt]* For $n \geq 1$, we divide by $n^2 \neq 0$:

$$u_n = \frac{4 + \frac{5}{n} - \frac{1}{n^2}}{5 + \frac{2}{n^2}}.$$

   The terms $\frac{1}{n}$ and $\frac{1}{n^2}$ tend to 0. The numerator tends to 4 and the denominator to $5 \neq 0$; by the quotient rule, $u_n \to \frac{4}{5}$.

2. *[1 pt]* The conjugate is positive. For $n \geq 1$, $\sqrt{n^2} = n$, hence

$$v_n = \frac{(n^2+5n)-n^2}{\sqrt{n^2+5n}+n} = \frac{5n}{n(\sqrt{1+\frac{5}{n}}+1)} = \frac{5}{\sqrt{1+\frac{5}{n}}+1}.$$

3. *[1 pt]* The denominator tends to 2; limit $\frac{5}{2}$.

4. *[1 pt]* $w_n = n^2(4-\frac{5}{n}) \to +\infty$; the second factor tends to $4 > 0$.

5. *[1 pt]* No: $(2n) - n = n \to +\infty$. The sign $\infty - \infty$ is an indeterminate form, not a rule of calculation.

### Exercise 3 · Limits and asymptotes

1. *[1 pt]* The denominator vanishes only at 2: $D_f = \mathbb{R} \setminus \{2\}$. Since $3x+6 = 3(x-2)+12$, we obtain $f(x) = 3 + \dfrac{12}{x-2}$.

2. *[1 pt]* Both limits equal 3.

3. *[1 pt]* As $x \to 2^-$, $x-2 \to 0$ through negative values; since $12 > 0$, $\frac{12}{x-2} \to -\infty$, so $f(x) \to -\infty$. On the right, the denominator tends to 0 through positive values: the quotient, and then $f(x)$, tend to $+\infty$.

4. *[1 pt]* $x = 2$ and $y = 3$; the curve is below for $x < 2$ and above for $x > 2$.

5. *[1 pt]* Let $t = 2x$. Then $t \to +\infty$ and $x^3 e^{-2x} = \frac{1}{8}\frac{t^3}{e^t}$. Comparison of growth rates gives $\frac{t^3}{e^t} \to 0$; the required limit is therefore 0.

### Exercise 4 · Codes and selections

1. *[1 pt]* $13^4 = 28561$ by the multiplication principle.

2. *[1 pt]* The successive choices have 13, 12, then one choice fewer at each step, down to 10 possibilities. By multiplication,

$$13 \times 12 \times 11 \times 10 = 17160.$$

   The order of the symbols is part of the code.

3. *[1 pt]* $13^4 - 12^4 = 7825$ by complement.

4. *[1 pt]* The symbol in the first position is fixed and can no longer be used. The 3 following positions give successively $12, \ldots, 10$ choices:

$$12 \times 11 \times 10 = 1320.$$

5. *[1 pt]* $11^4 = 14641$.

# Test 3 · Practice 1

**PRACTICE PAPER · TRAINING**

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · Limits and asymptotes — 5 points

Consider the function $f$ defined by $f(x) = \frac{5x+4}{x-1}$. Let $\mathcal{C}_f$ denote its graph in a coordinate plane.

1. Determine the domain of $f$. Find two real numbers $A$ and $B$ such that, for all $x \neq 1$, $f(x) = A + \frac{B}{x-1}$. *(1 pt)*
2. Determine the limits of $f$ at $-\infty$ and at $+\infty$. *(1 pt)*
3. Determine the limits of $f$ at $1^-$ and at $1^+$. *(1 pt)*
4. Give the equations of the asymptotes of $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. *(1 pt)*
5. Independently of the previous questions, determine $\lim\limits_{x \to +\infty} x^5 e^{-1x}$ with justification. *(1 pt)*

## Exercise 2 · Variations, convexity and equation — 5 points

Consider the function $f$ defined on $\mathbb{R}$ par $f(x) = x^3 + 4x - 5$. Let $\mathcal{C}_f$ denote its graph.

1. Calculate $f'(x)$, then study the direction of variation of $f$ on $\mathbb{R}$. *(1 pt)*
2. Determine the limits of $f$ at $-\infty$ and at $+\infty$. *(0.5 pt)*
3. Prove that the equation $f(x) = 0$ has a unique real solution $\alpha$. Give its exact value if it is an integer; otherwise, bound it between two consecutive integers. *(1.5 pt)*
4. Study the convexity of $f$, then determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point of abscissa 0, then study the position of the curve relative to this tangent. *(1 pt)*

Test 3 · Practice 1 · continued

## Exercise 3 · An exponential curve — 5 points

Consider the function $f$ defined on $[0, +\infty)$ par $f(x) = (x+3)e^{-x}$. Let $\mathcal{C}_f$ denote its graph.

1. Calculate $f(0)$ and determine the limit of $f$ at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw the variation table of $f$. *(1 pt)*
3. Calculate $f''(x)$, then study the convexity of $f$. *(1 pt)*
4. Prove that the equation $f(x) = \frac{3}{2}$ has a unique solution $\alpha$ in $[0, +\infty)$. *(1.5 pt)*
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point of abscissa 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. *(1 pt)*

## Exercise 4 · Forming a group — 5 points

A group consists of 6 teachers and 10 students, all distinct. A team of four people is chosen; the order of selection does not matter.

1. How many teams of four people can be formed? *(1 pt)*
2. How many of these teams contain exactly two teachers? *(1 pt)*
3. How many of these teams contain at least one teacher and at least one student? *(1 pt)*
4. A team containing exactly two teachers is chosen, then among its members a teacher spokesperson and a student reporter are designated. Count the choices of the team together with these two assignments. *(1 pt)*
5. Explain why the number of ordered lists of four distinct people, divided by 4!, gives the number of teams obtained in question 1. *(1 pt)*

# Solutions · Test 3 · Practice 1

## Exercise 1 · Limits and asymptotes

1. *[1 pt]* The denominator vanishes only at 1: $D_f = \mathbb{R} \setminus \{1\}$. Since $5x + 4 = 5(x-1) + 9$, we obtain $f(x) = 5 + \dfrac{9}{x-1}$.
2. *[1 pt]* Both limits equal 5.
3. *[1 pt]* As $x \to 1^-$, $x - 1 \to 0$ through negative values; since $9 > 0$, $\frac{9}{x-1} \to -\infty$, so $f(x) \to -\infty$. On the right, the denominator tends to 0 through positive values: the quotient, and hence $f(x)$, tend to $+\infty$.
4. *[1 pt]* $x = 1$ and $y = 5$; the curve is below for $x < 1$ and above for $x > 1$.
5. *[1 pt]* The comparison of growth rates of the power and the exponential directly gives $x^5 e^{-x} = \frac{x^5}{e^x} \to 0$ as $x \to +\infty$.

## Exercise 2 · Variations, convexity and equation

1. *[1 pt]* $f$ is differentiable on $\mathbb{R}$ and $f'(x) = 3x^2 + 4$. Since $3x^2 \geq 0$ and $4 > 0$, $f'(x) > 0$ for all $x$; $f$ is strictly increasing on $\mathbb{R}$.
2. *[0.5 pt]* The dominant term is $x^3$: limits $-\infty$ and $+\infty$.
3. *[1.5 pt]* $f$ is a polynomial, hence continuous on $\mathbb{R}$. Its strict increase and its limits $-\infty, +\infty$ allow us to apply the corollary of the intermediate value theorem (IVT): there is a unique zero $\alpha$. $f(1) = 0$, so the solution is exactly 1.
4. *[1 pt]* $f''(x) = 6x$: concave for $x \leq 0$, convex for $x \geq 0$, point of inflection $(0, -5)$.
5. *[1 pt]* Tangent $y = 4x - 5$; difference $x^3$, negative to the left and positive to the right of 0.

## Exercise 3 · An exponential curve

1. *[0.5 pt]* $f(0) = 3$; limit 0 by comparison of growth rates.
2. *[1 pt]* By the product rule,

$$f'(x) = e^{-x} - (x+3)e^{-x} = (-x-2)e^{-x}.$$

The two factors are respectively strictly negative and positive on $[0, +\infty[$; hence $f' < 0$.

| $x$ | $0$ | | $+\infty$ |
|---|---|---|---|
| $f'(x)$ | | $-$ | |
| $f(x)$ | $3$ | $\searrow$ | $0$ |

3. *[1 pt]* Differentiating the previous product,

$$f''(x) = -e^{-x} - (-x-2)e^{-x} = (x+1)e^{-x}.$$

On the domain, $x + 1 \geq 0$ and $e^{-x} > 0$: $f$ is convex. There is no change of convexity in the interior of the domain, hence no point of inflection.

4. *[1.5 pt]* Continuity, strict decrease, $f(0) = 3 > \frac{3}{2}$ and limit $0 < \frac{3}{2}$: exactly one solution.
5. *[1 pt]* The tangent at 0 has equation $y = f(0) + f'(0)x = 3 - 2x$. A convex function lies above each of its tangents, so $(x+3)e^{-x} \geq 3 - 2x$ for all $x \geq 0$.

## Exercise 4 · Forming a group

1. *[1 pt]* Order does not distinguish teams: we must choose a subset of size 4 from 16 people, that is $\binom{16}{4} = 1820$ teams.
2. *[1 pt]* We choose two teachers from 6 and two students from 10. Each pair of choices gives a unique team: $\binom{6}{2}\binom{10}{2} = 675$.
3. *[1 pt]* The excluded teams are those made up only of teachers or only of students; these two categories are disjoint. The required number is $\binom{16}{4} - \binom{6}{4} - \binom{10}{4} = 1595$.
4. *[1 pt]* Each team yields $2 \times 2$ assignments: 2700.
5. *[1 pt]* Each team of four distinct people corresponds to exactly 4! ordered lists. This multiplicity is constant, so the division removes the multiple counting.

# Test 3 · Practice 2

**PRACTICE PAPER · TRAINING**

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · Limits and asymptotes — 5 points

Consider the function $f$ defined by $f(x) = \frac{4x+6}{x-2}$. Let $\mathcal{C}_f$ denote its graph in a coordinate plane.

1. Determine the domain of $f$. Find two real numbers $A$ and $B$ such that, for all $x \neq 2$, $f(x) = A + \frac{B}{x-2}$. *(1 pt)*
2. Determine the limits of $f$ at $-\infty$ and at $+\infty$. *(1 pt)*
3. Determine the limits of $f$ at $2^-$ and at $2^+$. *(1 pt)*
4. Give the equations of the asymptotes of $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. *(1 pt)*
5. Independently of the previous questions, determine $\lim\limits_{x \to +\infty} x^4 e^{-2x}$ with justification. *(1 pt)*

## Exercise 2 · Variations, convexity and equation — 5 points

Consider the function $f$ defined on $\mathbb{R}$ par $f(x) = x^3 + 3x - 7$. Let $\mathcal{C}_f$ denote its graph.

1. Calculate $f'(x)$, then study the direction of variation of $f$ on $\mathbb{R}$. *(1 pt)*
2. Determine the limits of $f$ at $-\infty$ and at $+\infty$. *(0.5 pt)*
3. Prove that the equation $f(x) = 0$ has a unique real solution $\alpha$. Give its exact value if it is an integer; otherwise, bound it between two consecutive integers. *(1.5 pt)*
4. Study the convexity of $f$, then determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point of abscissa $0$, then study the position of the curve relative to this tangent. *(1 pt)*

Test 3 · Practice 2 · continued

## Exercise 3 · An exponential curve — 5 points

Consider the function $f$ defined on $[0, +\infty)$ par $f(x) = (x+2)e^{-x}$. Let $\mathcal{C}_f$ denote its graph.

1. Calculate $f(0)$ and determine the limit of $f$ at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw the variation table of $f$. *(1 pt)*
3. Calculate $f''(x)$, then study the convexity of $f$. *(1 pt)*
4. Prove that the equation $f(x) = 1$ has a unique solution $\alpha$ in $[0, +\infty)$. *(1.5 pt)*
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point of abscissa 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. *(1 pt)*

## Exercise 4 · Forming a group — 5 points

A group consists of 9 teachers and 11 students, all distinct. A team of four people is chosen; the order of selection does not matter.

1. How many teams of four people can be formed? *(1 pt)*
2. How many of these teams contain exactly two teachers? *(1 pt)*
3. How many of these teams contain at least one teacher and at least one student? *(1 pt)*
4. A team containing exactly two teachers is chosen, then among its members a teacher spokesperson and a student reporter are designated. Count the choices of the team together with these two assignments. *(1 pt)*
5. Explain why the number of ordered lists of four distinct people, divided by 4!, gives the number of teams obtained in question 1. *(1 pt)*

# Solutions · Test 3 · Practice 2

## Exercise 1 · Limits and asymptotes

1. *[1 pt]* The denominator vanishes only at 2: $D_f = \mathbb{R} \setminus \{2\}$. Since $4x + 6 = 4(x-2) + 14$, we obtain $f(x) = 4 + \dfrac{14}{x-2}$.
2. *[1 pt]* Both limits equal 4.
3. *[1 pt]* As $x \to 2^-$, $x - 2 \to 0$ through negative values; since $14 > 0$, $\frac{14}{x-2} \to -\infty$, so $f(x) \to -\infty$. On the right, the denominator tends to 0 through positive values: the quotient, and hence $f(x)$, tend to $+\infty$.
4. *[1 pt]* $x = 2$ and $y = 4$; the curve is below for $x < 2$ and above for $x > 2$.
5. *[1 pt]* Let $t = 2x$. Then $t \to +\infty$ and $x^4 e^{-2x} = \frac{1}{16}\frac{t^4}{e^t}$. Comparison of growth rates gives $\frac{t^4}{e^t} \to 0$; the required limit is therefore 0.

## Exercise 2 · Variations, convexity and equation

1. *[1 pt]* $f$ is differentiable on $\mathbb{R}$ and $f'(x) = 3x^2 + 3$. Since $3x^2 \geq 0$ and $3 > 0$, $f'(x) > 0$ for all $x$; $f$ is strictly increasing on $\mathbb{R}$.
2. *[0.5 pt]* The dominant term is $x^3$: limits $-\infty$ and $+\infty$.
3. *[1.5 pt]* $f$ is a polynomial, hence continuous on $\mathbb{R}$. Its strict increase and its limits $-\infty, +\infty$ allow us to apply the corollary of the intermediate value theorem (IVT): there is a unique zero $\alpha$. $f(1) = -3 < 0$ and $f(2) = 7 > 0$; so $1 < \alpha < 2$.
4. *[1 pt]* $f''(x) = 6x$: concave for $x \leq 0$, convex for $x \geq 0$, point of inflection $(0, -7)$.
5. *[1 pt]* Tangent $y = 3x - 7$; difference $x^3$, negative to the left and positive to the right of 0.

## Exercise 3 · An exponential curve

1. *[0.5 pt]* $f(0) = 2$; limit 0 by comparison of growth rates.
2. *[1 pt]* By the product rule,

$$f'(x) = e^{-x} - (x+2)e^{-x} = (-x-1)e^{-x}.$$

The two factors are respectively strictly negative and positive on $[0, +\infty[$; hence $f' < 0$.

| $x$ | $0$ | | $+\infty$ |
|---|---|---|---|
| $f'(x)$ | $-$ | | |
| $f(x)$ | $2$ | $\searrow$ | $0$ |

3. *[1 pt]* Differentiating the previous product,

$$f''(x) = -e^{-x} - (-x-1)e^{-x} = xe^{-x}.$$

On the domain, $x \geq 0$ and $e^{-x} > 0$: $f'' \geq 0$, so $f$ is convex. There is no change of convexity in the interior of the domain, hence no inflection point.

4. *[1.5 pt]* Continuity, strict decrease, $f(0) = 2 > 1$ and limit $0 < 1$: exactly one solution.
5. *[1 pt]* The tangent at 0 has equation $y = f(0) + f'(0)x = 2 - x$. A convex function lies above each of its tangents, so $(x+2)e^{-x} \geq 2 - x$ for all $x \geq 0$.

## Exercise 4 · Forming a group

1. *[1 pt]* Order does not distinguish teams: we must choose a subset of size 4 from 20 people, that is $\binom{20}{4} = 4845$ teams.
2. *[1 pt]* We choose two teachers from 9 and two students from 11. Each pair of choices gives a unique team: $\binom{9}{2}\binom{11}{2} = 1980$.
3. *[1 pt]* The excluded teams are those made up only of teachers or only of students; these two categories are disjoint. The required number is $\binom{20}{4} - \binom{9}{4} - \binom{11}{4} = 4389$.
4. *[1 pt]* Each team yields $2 \times 2$ assignments: 7920.
5. *[1 pt]* Each team of four distinct people corresponds to exactly 4! ordered lists. This multiplicity is constant, so the division removes the multiple counting.

# Test 3 · A

**PAPER RESERVED FOR ASSESSMENT**

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · Limits and asymptotes — 5 points

Consider the function $f$ defined by $f(x) = \frac{3x+8}{x-3}$. Let $\mathcal{C}_f$ denote its graph in a coordinate plane.

1. Determine the domain of $f$. Find two real numbers $A$ and $B$ such that, for all $x \neq 3$, $f(x) = A + \frac{B}{x-3}$. *(1 pt)*
2. Determine the limits of $f$ at $-\infty$ and at $+\infty$. *(1 pt)*
3. Determine the limits of $f$ at $3^-$ and at $3^+$. *(1 pt)*
4. Give the equations of the asymptotes of $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. *(1 pt)*
5. Independently of the previous questions, determine $\lim\limits_{x \to +\infty} x^3 e^{-3x}$ with justification. *(1 pt)*

## Exercise 2 · Variations, convexity and equation — 5 points

Consider the function $f$ defined on $\mathbb{R}$ par $f(x) = x^3 + 2x - 4$. Let $\mathcal{C}_f$ denote its graph.

1. Calculate $f'(x)$, then study the direction of variation of $f$ on $\mathbb{R}$. *(1 pt)*
2. Determine the limits of $f$ at $-\infty$ and at $+\infty$. *(0.5 pt)*
3. Prove that the equation $f(x) = 0$ has a unique real solution $\alpha$. Give its exact value if it is an integer; otherwise, bound it between two consecutive integers. *(1.5 pt)*
4. Study the convexity of $f$, then determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point of abscissa $0$, then study the position of the curve relative to this tangent. *(1 pt)*

Test 3 · A · continued

## Exercise 3 · An exponential curve — 5 points

Consider the function $f$ defined on $[0, +\infty)$ par $f(x) = (x+5)e^{-x}$. Let $\mathcal{C}_f$ denote its graph.

1. Calculate $f(0)$ and determine the limit of $f$ at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw the variation table of $f$. *(1 pt)*
3. Calculate $f''(x)$, then study the convexity of $f$. *(1 pt)*
4. Prove that the equation $f(x) = \frac{5}{2}$ has a unique solution $\alpha$ in $[0, +\infty)$. *(1.5 pt)*
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point of abscissa 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. *(1 pt)*

## Exercise 4 · Forming a group — 5 points

A group consists of 8 teachers and 8 students, all distinct. A team of four people is chosen; the order of selection does not matter.

1. How many teams of four people can be formed? *(1 pt)*
2. How many of these teams contain exactly two teachers? *(1 pt)*
3. How many of these teams contain at least one teacher and at least one student? *(1 pt)*
4. A team containing exactly two teachers is chosen, then among its members a teacher spokesperson and a student reporter are designated. Count the choices of the team together with these two assignments. *(1 pt)*
5. Explain why the number of ordered lists of four distinct people, divided by 4!, gives the number of teams obtained in question 1. *(1 pt)*

# Solutions · Test 3 · A

## Exercise 1 · Limits and asymptotes

1. *[1 pt]* The denominator vanishes only at 3: $D_f = \mathbb{R} \setminus \{3\}$. Since $3x + 8 = 3(x-3) + 17$, we obtain $f(x) = 3 + \dfrac{17}{x-3}$.
2. *[1 pt]* Both limits equal 3.
3. *[1 pt]* As $x \to 3^-$, $x - 3 \to 0$ through negative values; since $17 > 0$, $\frac{17}{x-3} \to -\infty$, so $f(x) \to -\infty$. On the right, the denominator tends to 0 through positive values: the quotient, and hence $f(x)$, tend to $+\infty$.
4. *[1 pt]* $x = 3$ and $y = 3$; the curve is below for $x < 3$ and above for $x > 3$.
5. *[1 pt]* Let $t = 3x$. Then $t \to +\infty$ and $x^3 e^{-3x} = \frac{1}{27}\frac{t^3}{e^t}$. Comparison of growth rates gives $\frac{t^3}{e^t} \to 0$; the required limit is therefore 0.

## Exercise 2 · Variations, convexity and equation

1. *[1 pt]* $f$ is differentiable on $\mathbb{R}$ and $f'(x) = 3x^2 + 2$. Since $3x^2 \geq 0$ and $2 > 0$, $f'(x) > 0$ for all $x$; $f$ is strictly increasing on $\mathbb{R}$.
2. *[0.5 pt]* The dominant term is $x^3$: limits $-\infty$ and $+\infty$.
3. *[1.5 pt]* $f$ is a polynomial, hence continuous on $\mathbb{R}$. Its strict increase and its limits $-\infty, +\infty$ allow us to apply the corollary of the intermediate value theorem (IVT): there is a unique zero $\alpha$. $f(1) = -1 < 0$ and $f(2) = 8 > 0$; so $1 < \alpha < 2$.
4. *[1 pt]* $f''(x) = 6x$: concave for $x \leq 0$, convex for $x \geq 0$, point of inflection $(0, -4)$.
5. *[1 pt]* Tangent $y = 2x - 4$; difference $x^3$, negative to the left and positive to the right of 0.

## Exercise 3 · An exponential curve

1. *[0.5 pt]* $f(0) = 5$; limit 0 by comparison of growth rates.
2. *[1 pt]* By the product rule,

$$f'(x) = e^{-x} - (x+5)e^{-x} = (-x-4)e^{-x}.$$

The two factors are respectively strictly negative and positive on $[0, +\infty[$; hence $f' < 0$.

| $x$ | $0$ | | $+\infty$ |
|---|---|---|---|
| $f'(x)$ | $-$ | | |
| $f(x)$ | $5$ | $\searrow$ | $0$ |

3. *[1 pt]* Differentiating the previous product,

$$f''(x) = -e^{-x} - (-x-4)e^{-x} = (x+3)e^{-x}.$$

On the domain, $x + 3 \geq 0$ and $e^{-x} > 0$: $f$ is convex. There is no change of convexity in the interior of the domain, hence no point of inflection.

4. *[1.5 pt]* Continuity, strict decrease, $f(0) = 5 > \frac{5}{2}$ and limit $0 < \frac{5}{2}$: exactly one solution.
5. *[1 pt]* The tangent at 0 has equation $y = f(0) + f'(0)x = 5 - 4x$. A convex function lies above each of its tangents, so $(x+5)e^{-x} \geq 5 - 4x$ for all $x \geq 0$.

### Exercise 4 · Forming a group

1. *[1 pt]* Order does not distinguish teams: we must choose a subset of size 4 from 16 people, that is $\binom{16}{4} = 1820$ teams.
2. *[1 pt]* We choose two teachers from 8 and two students from 8. Each pair of choices gives a unique team: $\binom{8}{2}\binom{8}{2} = 784$.
3. *[1 pt]* The excluded teams are those made up only of teachers or only of students; these two categories are disjoint. The required number is $\binom{16}{4} - \binom{8}{4} - \binom{8}{4} = 1680$.
4. *[1 pt]* Each team yields $2 \times 2$ assignments: 3136.
5. *[1 pt]* Each team of four distinct people corresponds to exactly 4! ordered lists. This multiplicity is constant, so the division removes the multiple counting.

# Test 3 · B

**PAPER RESERVED FOR ASSESSMENT**

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · Limits and asymptotes — 5 points

Consider the function $f$ defined by $f(x) = \frac{2x+5}{x-1}$. Let $\mathcal{C}_f$ denote its graph in a coordinate plane.

1. Determine the domain of $f$. Find two real numbers $A$ and $B$ such that, for all $x \neq 1$, $f(x) = A + \frac{B}{x-1}$. *(1 pt)*
2. Determine the limits of $f$ at $-\infty$ and at $+\infty$. *(1 pt)*
3. Determine the limits of $f$ at $1^-$ and at $1^+$. *(1 pt)*
4. Give the equations of the asymptotes of $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. *(1 pt)*
5. Independently of the previous questions, determine $\lim\limits_{x \to +\infty} x^2 e^{-1x}$ with justification. *(1 pt)*

## Exercise 2 · Variations, convexity and equation — 5 points

Consider the function $f$ defined on $\mathbb{R}$ par $f(x) = x^3 + 5x - 5$. Let $\mathcal{C}_f$ denote its graph.

1. Calculate $f'(x)$, then study the direction of variation of $f$ on $\mathbb{R}$. *(1 pt)*
2. Determine the limits of $f$ at $-\infty$ and at $+\infty$. *(0.5 pt)*
3. Prove that the equation $f(x) = 0$ has a unique real solution $\alpha$. Give its exact value if it is an integer; otherwise, bound it between two consecutive integers. *(1.5 pt)*
4. Study the convexity of $f$, then determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point of abscissa 0, then study the position of the curve relative to this tangent. *(1 pt)*

Test 3 · B · continued

## Exercise 3 · An exponential curve — 5 points

Consider the function $f$ defined on $[0, +\infty)$ par $f(x) = (x + 4)e^{-x}$. Let $\mathcal{C}_f$ denote its graph.

1. Calculate $f(0)$ and determine the limit of $f$ at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw the variation table of $f$. *(1 pt)*
3. Calculate $f''(x)$, then study the convexity of $f$. *(1 pt)*
4. Prove that the equation $f(x) = 2$ has a unique solution $\alpha$ in $[0, +\infty)$. *(1.5 pt)*
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point of abscissa 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. *(1 pt)*

## Exercise 4 · Forming a group — 5 points

A group consists of 7 teachers and 10 students, all distinct. A team of four people is chosen; the order of selection does not matter.

1. How many teams of four people can be formed? *(1 pt)*
2. How many of these teams contain exactly two teachers? *(1 pt)*
3. How many of these teams contain at least one teacher and at least one student? *(1 pt)*
4. A team containing exactly two teachers is chosen, then among its members a teacher spokesperson and a student reporter are designated. Count the choices of the team together with these two assignments. *(1 pt)*
5. Explain why the number of ordered lists of four distinct people, divided by 4!, gives the number of teams obtained in question 1. *(1 pt)*

# Solutions · Test 3 · B

## Exercise 1 · Limits and asymptotes

1. *[1 pt]* The denominator vanishes only at 1: $D_f = \mathbb{R} \setminus \{1\}$. Since $2x+5 = 2(x-1)+7$, we obtain $f(x) = 2 + \dfrac{7}{x-1}$.
2. *[1 pt]* Both limits equal 2.
3. *[1 pt]* As $x \to 1^-$, $x-1 \to 0$ through negative values; since $7 > 0$, $\frac{7}{x-1} \to -\infty$, so $f(x) \to -\infty$. On the right, the denominator tends to 0 through positive values: the quotient, and hence $f(x)$, tend to $+\infty$.
4. *[1 pt]* $x = 1$ and $y = 2$; the curve is below for $x < 1$ and above for $x > 1$.
5. *[1 pt]* The comparison of growth rates of the power and the exponential directly gives $x^2 e^{-x} = \frac{x^2}{e^x} \to 0$ as $x \to +\infty$.

## Exercise 2 · Variations, convexity and equation

1. *[1 pt]* $f$ is differentiable on $\mathbb{R}$ and $f'(x) = 3x^2 + 5$. Since $3x^2 \geq 0$ and $5 > 0$, $f'(x) > 0$ for all $x$; $f$ is strictly increasing on $\mathbb{R}$.
2. *[0.5 pt]* The dominant term is $x^3$: limits $-\infty$ and $+\infty$.
3. *[1.5 pt]* $f$ is a polynomial, hence continuous on $\mathbb{R}$. Its strict increase and its limits $-\infty, +\infty$ allow us to apply the corollary of the intermediate value theorem (IVT): there is a unique zero $\alpha$. $f(0) = -5 < 0$ and $f(1) = 1 > 0$; so $0 < \alpha < 1$.
4. *[1 pt]* $f''(x) = 6x$: concave for $x \leq 0$, convex for $x \geq 0$, point of inflection $(0, -5)$.
5. *[1 pt]* Tangent $y = 5x - 5$; difference $x^3$, negative to the left and positive to the right of 0.

## Exercise 3 · An exponential curve

1. *[0.5 pt]* $f(0) = 4$; limit 0 by comparison of growth rates.
2. *[1 pt]* By the product rule,

$$f'(x) = e^{-x} - (x+4)e^{-x} = (-x-3)e^{-x}.$$

The two factors are respectively strictly negative and positive on $[0, +\infty[$; hence $f' < 0$.

| $x$ | $0$ | | $+\infty$ |
|---|---|---|---|
| $f'(x)$ | | $-$ | |
| $f(x)$ | $4$ | $\searrow$ | $0$ |

3. *[1 pt]* Differentiating the previous product,

$$f''(x) = -e^{-x} - (-x-3)e^{-x} = (x+2)e^{-x}.$$

On the domain, $x+2 \geq 0$ and $e^{-x} > 0$: $f$ is convex. There is no change of convexity in the interior of the domain, hence no point of inflection.

4. *[1.5 pt]* Continuity, strict decrease, $f(0) = 4 > 2$ and limit $0 < 2$: exactly one solution.
5. *[1 pt]* The tangent at 0 has equation $y = f(0) + f'(0)x = 4 - 3x$. A convex function lies above each of its tangents, so $(x+4)e^{-x} \geq 4 - 3x$ for all $x \geq 0$.

## Exercise 4 · Forming a group

1. *[1 pt]* Order does not distinguish teams: we must choose a subset of size 4 from 17 people, that is $\binom{17}{4} = 2380$ teams.
2. *[1 pt]* We choose two teachers from 7 and two students from 10. Each pair of choices gives a unique team: $\binom{7}{2}\binom{10}{2} = 945$.
3. *[1 pt]* The excluded teams are those made up only of teachers or only of students; these two categories are disjoint. The required number is $\binom{17}{4} - \binom{7}{4} - \binom{10}{4} = 2135$.
4. *[1 pt]* Each team yields $2 \times 2$ assignments: 3780.
5. *[1 pt]* Each team of four distinct people corresponds to exactly 4! ordered lists. This multiplicity is constant, so the division removes the multiple counting.

# Mock 1 · Practice 1

### PRACTICE PAPER · TRAINING

**Duration: 4 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · Induction and bounds — 5 points

Consider the sequence $(v_n)$ defined by $v_0 = 0$ and, for every natural number $n$, $v_{n+1} = \sqrt{42 + v_n}$.

1. Calculate $v_1$ and $v_2$. Give the exact values. *(0.25 pt)*
2. Prove by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 7$. *(0.75 pt)*
3. Prove by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. *(0.75 pt)*
4. Prove that, for all $n \in \mathbb{N}$, $v_n < 7$. *(0.5 pt)*
5. Justify that $(v_n)$ converges, then determine its limit. *(0.75 pt)*
6. Prove that, for all $n \in \mathbb{N}$, $0 \leq 7 - v_{n+1} \leq \frac{7 - v_n}{7}$. *(0.5 pt)*
7. Deduce by induction an upper bound for $7 - v_n$ in terms of $n$. Then determine, by comparing integer powers, an index from which this bound guarantees $7 - v_n \leq 10^{-3}$. *(1 pt)*
8. Write a Python function with no argument that returns the first integer $n$ such that $7 - v_n \leq 10^{-3}$. Run the function and compare the index obtained with the sufficient index from the previous question. Explain any difference. *(0.5 pt)*

## Exercise 2 · An exponential curve — 5 points

Consider the function $f$ defined on $[0, +\infty)$ by $f(x) = (x + 5)e^{-x}$. Let $\mathcal{C}_f$ denote its graph.

1. Calculate $f(0)$ and determine the limit of $f$ at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw the variation table of $f$. *(0.5 pt)*
3. Calculate $f''(x)$, then study the convexity of $f$. *(0.5 pt)*
4. Prove that the equation $f(x) = \frac{5}{2}$ has a unique solution $\alpha$ in $[0, +\infty)$. *(1 pt)*
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point of abscissa 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. *(0.5 pt)*
6. Show that $0 < \alpha < 2$. Write a Python function that applies bisection on $[0, 2]$ and returns a bracketing of $\alpha$ of width at most $10^{-3}$. *(0.5 pt)*
7. Determine the minimum number of bisection steps guaranteeing this width. *(0.5 pt)*
8. A student claims: "Since $f(x)$ tends to 0, the curve crosses the horizontal axis." Refute this claim by studying the sign of $f(x)$. *(1 pt)*

## Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$ of space, define the vectors

$$\vec{u} = \vec{\imath} + \vec{k}, \qquad \vec{v} = \vec{\jmath} + \vec{k},$$
$$\vec{w} = 4\vec{\imath} + 6\vec{\jmath} + 10\vec{k}, \quad \vec{z} = 4\vec{\imath} + 6\vec{\jmath} + 11\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. *(0.5 pt)*
2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. *(0.5 pt)*
3. Are the vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. *(0.5 pt)*
4. Express the vector $6\vec{u} - 4\vec{v}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
5. Choose a point $A$, then define $B$, $C$ and $D$ by $\overrightarrow{AB} = \vec{u}$, $\overrightarrow{AC} = \vec{v}$ and $\overrightarrow{AD} = \vec{w}$. Prove that $A$, $B$, $C$ and $D$ are coplanar. *(0.5 pt)*
6. For questions 6 to 8, assume the basis is orthonormal. Calculate the distances $AB$, $AC$ and $BC$. *(0.5 pt)*
7. Let $I$ be the midpoint of $[BC]$. Express $\overrightarrow{AI}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$, then determine whether the points $A$, $I$ and $D$ are collinear. *(0.5 pt)*
8. Define $E$ by $\overrightarrow{AE} = \vec{z}$. Is the line $(DE)$ contained in the plane $(ABC)$? Justify. *(1 pt)*

## Exercise 4 · Repetitions and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 19 times. The trials are independent and the probability of success is $p = \frac{3}{10}$ on each trial. Let $X$ denote the number of successes obtained.

1. Justify that $X$ follows a binomial distribution and state its parameters. *(0.5 pt)*
2. Calculate the probability of obtaining exactly two successes. *(0.5 pt)*
3. Express using complementary events, then calculate, the probability of obtaining at least two successes. *(0.5 pt)*
4. Calculate the expected value and the variance of $X$. *(1 pt)*
5. The same trial is now repeated $N \geq 1$ times, still independently. Determine the smallest integer $N$ for which the probability of obtaining at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. *(0.5 pt)*
6. For an integer $j \in \{0, \ldots, 19\}$, prove the formula for $P(X = j)$ by counting the possible positions of the successes. *(0.5 pt)*
7. A series of 19 trials is accepted if its number of successes is between 1 and 3 inclusive. Calculate the probability that a series is accepted. *(0.5 pt)*
8. Write a Python function with no argument that returns the smallest integer $j$ such that $P(X \leq j) \geq 0.95$. Run the function and check the minimality of the result. *(1 pt)*

# Solutions · Mock 1 · Practice 1

## Exercise 1 · Induction and bounds

1. *[0.25 pt]* $v_1 = \sqrt{42+0} = \sqrt{42}$ and $v_2 = \sqrt{42+\sqrt{42}}$.

2. *[0.75 pt]* Let $P(n)$ be the statement: "$v_n$ is defined and $0 \leq v_n \leq 7$".

   **Base case.** $v_0 = 0$ is defined and belongs to $[0,7]$; $P(0)$ is true.

   **Inductive step.** Fix $n \in \mathbb{N}$ and assume $P(n)$ is true. Then

   $$0 \leq 42 \leq 42 + v_n \leq 42 + 7 = 49.$$

   The square root is therefore defined and, since this function is increasing on $[0, +\infty[$,

   $$0 \leq v_{n+1} = \sqrt{42+v_n} \leq \sqrt{49} = 7.$$

   Thus $P(n+1)$ is true.

   **Conclusion.** By induction, all terms are defined and belong to $[0,7]$.

3. *[0.75 pt]* For $n \in \mathbb{N}$, let $R(n)$ be the statement: "$v_{n+1} \geq v_n$".

   **Base case.** $v_1 = \sqrt{42} > 0 = v_0$; $R(0)$ is true.

   **Inductive step.** Fix $n \geq 0$ and assume $v_{n+1} \geq v_n$. Adding 42 and then applying the increasing property of the square root,

   $$v_{n+2} = \sqrt{42+v_{n+1}} \geq \sqrt{42+v_n} = v_{n+1}.$$

   This is $R(n+1)$.

   **Conclusion.** By induction, $v_{n+1} \geq v_n$ for all $n \geq 0$: $(v_n)$ is increasing.

4. *[0.5 pt]* Let $S(n)$ be the statement: "$v_n < 7$. At index 0, $v_0 = 0 < 7$. Fix $n \geq 0$ and assume $S(n)$ is true. Since the radicands are positive, the strict increase of the square root gives

   $$v_{n+1} = \sqrt{42+v_n} < \sqrt{42+7} = \sqrt{49} = 7.$$

   So $S(n+1)$ is true. The base case and the inductive step establish by induction that $v_n < 7$ for all $n \in \mathbb{N}$.

5. *[0.75 pt]* The sequence is increasing and bounded above by 7: the monotone convergence theorem gives a real limit $\ell \in [0,7]$. The function $x \mapsto \sqrt{42+x}$ is continuous on this interval. Taking the limit in the relation, $\ell = \sqrt{42+\ell}$. Since $\ell \geq 0$, this is equivalent to

   $$\ell^2 - \ell - 42 = 0 \quad \Longleftrightarrow \quad (\ell - 7)(\ell + 6) = 0.$$

   The root $-6$ is excluded by $\ell \geq 0$; so $\lim v_n = 7$.

6. *[0.5 pt]* The previous result ensures $7 - v_{n+1} \geq 0$. Multiplying by the conjugate expression,

   $$\begin{aligned} 7 - v_{n+1} &= \frac{49 - (42+v_n)}{7 + \sqrt{42+v_n}} \\ &= \frac{7 - v_n}{7 + \sqrt{42+v_n}} \leq \frac{7-v_n}{7}. \end{aligned}$$

   The last inequality uses a positive numerator and a denominator at least equal to $7 > 0$.

7. *[1 pt]* Let us prove by induction $T(n)$: "$0 \leq 7 - v_n \leq 7^{1-n}$". At index 0, $7 - v_0 = 7 = 7^1$. Fix $n \geq 0$ and assume $T(n)$ is true. The previous question gives

$$0 \leq 7 - v_{n+1} \leq \frac{7 - v_n}{7} \leq \frac{7^{1-n}}{7} = 7^{1-(n+1)}.$$

This is $T(n+1)$. By induction, the bounds hold at every index. To have $7^{1-n} \leq 10^{-3}$, it suffices that $7^{n-1} \geq 1000$. Now $7^3 = 343 < 1000$ and $7^4 = 2401 \geq 1000$. Index 5 is therefore the first one guaranteed by this bound, without using the logarithm.

8. *[0.5 pt]*

```python
from math import sqrt

def premier_rang():
    n, v = 0, 0.0
    while 7-v > 1e-3:
        v = sqrt(42+v)
        n += 1
    return n
```

Convergence to 7 ensures termination. The loop examines the indices in order: it returns 4, the first index satisfying the criterion for the computed values. Here, the gaps at indices 3 and 4 are approximately 0.00265684 and 0.00018978. They lie clearly on either side of $10^{-3}$. The guaranteed index 5 may be larger; it comes from an upper bound, not from the exact error.

## Exercise 2 · An exponential curve

1. *[0.5 pt]* $f(0) = 5$; limit 0 by comparison of growth rates.

2. *[0.5 pt]* By the product rule,

$$f'(x) = e^{-x} - (x+5)e^{-x} = (-x-4)e^{-x}.$$

The two factors are respectively strictly negative and positive on $[0, +\infty[$; hence $f' < 0$.

| $x$ | $0$ | | $+\infty$ |
|---|---|---|---|
| $f'(x)$ | | $-$ | |
| $f(x)$ | $5$ | $\searrow$ | $0$ |

3. *[0.5 pt]* Differentiating the previous product,

$$f''(x) = -e^{-x} - (-x-4)e^{-x} = (x+3)e^{-x}.$$

On the domain, $x + 3 \geq 0$ and $e^{-x} > 0$: $f$ is convex. There is no change of convexity in the interior of the domain, hence no point of inflection.

4. *[1 pt]* Continuity, strict decrease, $f(0) = 5 > \frac{5}{2}$ and limit $0 < \frac{5}{2}$: exactly one solution.

5. *[0.5 pt]* The tangent at 0 has equation $y = f(0) + f'(0)x = 5 - 4x$. A convex function lies above each of its tangents, so $(x+5)e^{-x} \geq 5 - 4x$ for all $x \geq 0$.

6. *[0.5 pt]* We know $f(0) = 5 > \frac{5}{2}$. Moreover, $f(2) = 7e^{-2} \simeq 0.9473 < \frac{5}{2}$. The zero of $f - \frac{5}{2}$ is therefore in $]0, 2[$. A program is:

```
from math import exp

def encadrement():
    a, b = 0.0, 2.0
    while b-a > 1e-3:
        m = (a+b)/2
        if (m+5)*exp(-m) > 5/2:
            a = m
        else:
            b = m
    return a, b
```

   Since $f$ is decreasing, the bracketing is preserved at each step. The result has length at most $10^{-3}$; its midpoint has an error of at most $5 \times 10^{-4}$.

7. *[0.5 pt]* $\frac{2}{2^n} \leq 10^{-3}$: $n = 11$ steps suffice.

8. *[1 pt]* For $x \geq 0$, $x + 5 > 0$ and $e^{-x} > 0$, so $f(x) > 0$. A limit is not necessarily an attained value.

## Exercise 3 · Vector calculation in a basis

1. *[0.5 pt]* In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{\jmath}$ would be $1 = 0$, a contradiction by uniqueness of coordinates in a basis.

2. *[0.5 pt]* Expanding, $4\vec{u} + 6\vec{v} = 4(\vec{\imath} + \vec{k}) + 6(\vec{\jmath} + \vec{k}) = 4\vec{\imath} + 6\vec{\jmath} + 10\vec{k} = \vec{w}$.

3. *[0.5 pt]* Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis requires

$$\lambda = 4, \qquad \mu = 6, \qquad \lambda + \mu = 11.$$

   The first two equalities give $\lambda + \mu = 10$, contradicting the third. The three vectors are therefore not coplanar.

4. *[1 pt]* $6\vec{u} - 4\vec{v} = 6(\vec{\imath} + \vec{k}) - 4(\vec{\jmath} + \vec{k}) = 6\vec{\imath} - 4\vec{\jmath} + 2\vec{k}$.

5. *[0.5 pt]* $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are not collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

6. *[0.5 pt]* $AB = AC = \sqrt{2}$; $\overrightarrow{BC} = \vec{v} - \vec{u} = (-1, 1, 0)$, so $BC = \sqrt{2}$.

7. *[0.5 pt]* The midpoint formula gives $\overrightarrow{AI} = \frac{1}{2}(\vec{u} + \vec{v})$. Since the vectors $\vec{u}, \vec{v}$ are not collinear,

$$\overrightarrow{AD} = \lambda\overrightarrow{AI} \quad \Longleftrightarrow \quad 4 = \frac{\lambda}{2} \text{ and } 6 = \frac{\lambda}{2}.$$

   Here $4 \neq 6$: there is no such $\lambda$, so $A, I, D$ are not collinear.

8. *[1 pt]* No: $D$ belongs to the plane but $E$ does not, by the non-coplanarity shown above. The intersection is exactly $D$.

## Exercise 4 · Repetitions and estimation

1. *[0.5 pt]* Repetition of a fixed number 19 of identical independent trials with two outcomes; $X \sim \mathcal{B}(19, \frac{3}{10})$.

2. *[0.5 pt]* $P(X = 2) = \binom{19}{2}(\frac{3}{10})^2(\frac{7}{10})^{17} \simeq 0.035802$.

3. *[0.5 pt]* $1 - P(X = 0) - P(X = 1) = 1 - (\frac{7}{10})^{19} - 19(\frac{3}{10})(\frac{7}{10})^{18} \simeq 0.989578$.

4. *[1 pt]* $E(X) = \frac{57}{10}$, $V(X) = \frac{399}{100}$.

5. *[0.5 pt]* For $N$ independent trials, the probability of no success is $(1-p)^N$. The required probability is therefore $1 - (\frac{7}{10})^N$. It is strictly increasing in $N$, since $0 < \frac{7}{10} < 1$. Calculation gives

$$P_8 \simeq 0.942352 < 0.95, \qquad P_9 \simeq 0.959646 \geq 0.95.$$

   Thus 9 is the smallest suitable number of trials.

6. *[0.5 pt]* Let $n = 19$. For $j \in \{0, \ldots, n\}$, an ordered outcome containing $j$ successes and $n - j$ failures has, by independence, probability $p^j(1-p)^{n-j}$. There are $\binom{n}{j}$ choices for the positions of the successes. These outcomes are mutually exclusive, so

$$P(X = j) = \binom{n}{j} p^j (1-p)^{n-j}, \qquad p = \frac{3}{10}.$$

7. *[0.5 pt]* The events $X = 1$, $X = 2$, $X = 3$ are mutually exclusive, so

$$P(1 \leq X \leq 3) = \sum_{j=1}^{3} \binom{19}{j} (\frac{3}{10})^j (\frac{7}{10})^{19-j} \simeq 0.132031.$$

8. *[1 pt]*

```python
from math import comb

def quantile():
    n, p = 19, 3/10
    j, cumul = 0, (1-p)**n
    while cumul < 0.95:
        j += 1
        cumul += comb(n, j)*p**j*(1-p)**(n-j)
    return j
```

   The probabilities added are positive and the final cumulative sum equals 1, so the loop terminates. It returns 9. The cumulative sums at indices 8 and 9 are approximately 0.916085 and 0.967447: this confirms minimality.

# Mock 1 · Practice 2

**PRACTICE PAPER · TRAINING**

**Duration: 4 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · Induction and bounds — 5 points

Consider the sequence $(v_n)$ defined by $v_0 = 0$ and, for every natural number $n$, $v_{n+1} = \sqrt{110 + v_n}$.

1. Calculate $v_1$ and $v_2$. Give the exact values. *(0.25 pt)*
2. Prove by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 11$. *(0.75 pt)*
3. Prove by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. *(0.75 pt)*
4. Prove that, for all $n \in \mathbb{N}$, $v_n < 11$. *(0.5 pt)*
5. Justify that $(v_n)$ converges, then determine its limit. *(0.75 pt)*
6. Prove that, for all $n \in \mathbb{N}$, $0 \leq 11 - v_{n+1} \leq \frac{11 - v_n}{11}$. *(0.5 pt)*
7. Deduce by induction an upper bound for $11 - v_n$ in terms of $n$. Then determine, by comparing integer powers, an index from which this bound guarantees $11 - v_n \leq 10^{-3}$. *(1 pt)*
8. Write a Python function with no argument that returns the first integer $n$ such that $11 - v_n \leq 10^{-3}$. Run the function and compare the index obtained with the sufficient index from the previous question. Explain any difference. *(0.5 pt)*

## Exercise 2 · An exponential curve — 5 points

Consider the function $f$ defined on $[0, +\infty)$ by $f(x) = (x + 4)e^{-x}$. Let $\mathcal{C}_f$ denote its graph.

1. Calculate $f(0)$ and determine the limit of $f$ at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw the variation table of $f$. *(0.5 pt)*
3. Calculate $f''(x)$, then study the convexity of $f$. *(0.5 pt)*
4. Prove that the equation $f(x) = 2$ has a unique solution $\alpha$ in $[0, +\infty)$. *(1 pt)*
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point of abscissa 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. *(0.5 pt)*
6. Show that $0 < \alpha < 2$. Write a Python function that applies bisection on $[0, 2]$ and returns a bracketing of $\alpha$ of width at most $10^{-3}$. *(0.5 pt)*
7. Determine the minimum number of bisection steps guaranteeing this width. *(0.5 pt)*
8. A student claims: "Since $f(x)$ tends to 0, the curve crosses the horizontal axis." Refute this claim by studying the sign of $f(x)$. *(1 pt)*

Mock 1 · Practice 2 · continued

## Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$ of space, define the vectors

$$\vec{u} = \vec{\imath} + \vec{k}, \qquad \vec{v} = \vec{\jmath} + \vec{k},$$
$$\vec{w} = 3\vec{\imath} + 8\vec{\jmath} + 11\vec{k}, \quad \vec{z} = 3\vec{\imath} + 8\vec{\jmath} + 12\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. *(0.5 pt)*
2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. *(0.5 pt)*
3. Are the vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. *(0.5 pt)*
4. Express the vector $8\vec{u} - 3\vec{v}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
5. Choose a point $A$, then define $B$, $C$ and $D$ by $\overrightarrow{AB} = \vec{u}$, $\overrightarrow{AC} = \vec{v}$ and $\overrightarrow{AD} = \vec{w}$. Prove that $A$, $B$, $C$ and $D$ are coplanar. *(0.5 pt)*
6. For questions 6 to 8, assume the basis is orthonormal. Calculate the distances $AB$, $AC$ and $BC$. *(0.5 pt)*
7. Let $I$ be the midpoint of $[BC]$. Express $\overrightarrow{AI}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$, then determine whether the points $A$, $I$ and $D$ are collinear. *(0.5 pt)*
8. Define $E$ by $\overrightarrow{AE} = \vec{z}$. Is the line $(DE)$ contained in the plane $(ABC)$? Justify. *(1 pt)*

## Exercise 4 · Repetitions and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 16 times. The trials are independent and the probability of success is $p = \frac{1}{10}$ on each trial. Let $X$ denote the number of successes obtained.

1. Justify that $X$ follows a binomial distribution and state its parameters. *(0.5 pt)*
2. Calculate the probability of obtaining exactly two successes. *(0.5 pt)*
3. Express using complementary events, then calculate, the probability of obtaining at least two successes. *(0.5 pt)*
4. Calculate the expected value and the variance of $X$. *(1 pt)*
5. The same trial is now repeated $N \geq 1$ times, still independently. Determine the smallest integer $N$ for which the probability of obtaining at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. *(0.5 pt)*
6. For an integer $j \in \{0, \ldots, 16\}$, prove the formula for $P(X = j)$ by counting the possible positions of the successes. *(0.5 pt)*
7. A series of 16 trials is accepted if its number of successes is between 1 and 3 inclusive. Calculate the probability that a series is accepted. *(0.5 pt)*
8. Write a Python function with no argument that returns the smallest integer $j$ such that $P(X \leq j) \geq 0.95$. Run the function and check the minimality of the result. *(1 pt)*

# Solutions · Mock 1 · Practice 2

## Exercise 1 · Induction and bounds

1. *[0.25 pt]* $v_1 = \sqrt{110+0} = \sqrt{110}$ and $v_2 = \sqrt{110+\sqrt{110}}$.

2. *[0.75 pt]* Let $P(n)$ be the statement: "$v_n$ is defined and $0 \leq v_n \leq 11$".

   **Base case.** $v_0 = 0$ is defined and belongs to $[0, 11]$; $P(0)$ is true.

   **Inductive step.** Fix $n \in \mathbb{N}$ and assume $P(n)$ is true. Then

   $$0 \leq 110 \leq 110 + v_n \leq 110 + 11 = 121.$$

   The square root is therefore defined and, since this function is increasing on $[0, +\infty[$,

   $$0 \leq v_{n+1} = \sqrt{110+v_n} \leq \sqrt{121} = 11.$$

   Thus $P(n+1)$ is true.

   **Conclusion.** By induction, all terms are defined and belong to $[0, 11]$.

3. *[0.75 pt]* For $n \in \mathbb{N}$, let $R(n)$ be the statement: "$v_{n+1} \geq v_n$".

   **Base case.** $v_1 = \sqrt{110} > 0 = v_0$; $R(0)$ is true.

   **Inductive step.** Fix $n \geq 0$ and assume $v_{n+1} \geq v_n$. Adding 110 and then applying the increasing property of the square root,

   $$v_{n+2} = \sqrt{110+v_{n+1}} \geq \sqrt{110+v_n} = v_{n+1}.$$

   This is $R(n+1)$.

   **Conclusion.** By induction, $v_{n+1} \geq v_n$ for all $n \geq 0$: $(v_n)$ is increasing.

4. *[0.5 pt]* Let $S(n)$ be the statement: "$v_n < 11$. At index 0, $v_0 = 0 < 11$. Fix $n \geq 0$ and assume $S(n)$ is true. Since the radicands are positive, the strict increase of the square root gives

   $$v_{n+1} = \sqrt{110+v_n} < \sqrt{110+11} = \sqrt{121} = 11.$$

   So $S(n+1)$ is true. The base case and the inductive step establish by induction that $v_n < 11$ for all $n \in \mathbb{N}$.

5. *[0.75 pt]* The sequence is increasing and bounded above by 11: the monotone convergence theorem gives a real limit $\ell \in [0, 11]$. The function $x \mapsto \sqrt{110+x}$ is continuous on this interval. Taking the limit in the relation, $\ell = \sqrt{110+\ell}$. Since $\ell \geq 0$, this is equivalent to

   $$\ell^2 - \ell - 110 = 0 \quad \Longleftrightarrow \quad (\ell - 11)(\ell + 10) = 0.$$

   The root $-10$ is excluded by $\ell \geq 0$; so $\lim v_n = 11$.

6. *[0.5 pt]* The previous result ensures $11 - v_{n+1} \geq 0$. Multiplying by the conjugate expression,

   $$\begin{aligned} 11 - v_{n+1} &= \frac{121 - (110 + v_n)}{11 + \sqrt{110+v_n}} \\ &= \frac{11 - v_n}{11 + \sqrt{110+v_n}} \leq \frac{11 - v_n}{11}. \end{aligned}$$

   The last inequality uses a positive numerator and a denominator at least equal to $11 > 0$.

7. *[1 pt]* Let us prove by induction $T(n)$: "$0 \leq 11 - v_n \leq 11^{1-n}$". At index 0, $11 - v_0 = 11 = 11^1$. Fix $n \geq 0$ and assume $T(n)$ is true. The previous question gives

$$0 \leq 11 - v_{n+1} \leq \frac{11 - v_n}{11} \leq \frac{11^{1-n}}{11} = 11^{1-(n+1)}.$$

This is $T(n+1)$. By induction, the bounds hold at every index. To have $11^{1-n} \leq 10^{-3}$, it suffices that $11^{n-1} \geq 1000$. Now $11^2 = 121 < 1000$ and $11^3 = 1331 \geq 1000$. Index 4 is therefore the first one guaranteed by this bound, without using the logarithm.

8. *[0.5 pt]*

```
from math import sqrt

def premier_rang():
    n, v = 0, 0.0
    while 11-v > 1e-3:
        v = sqrt(110+v)
        n += 1
    return n
```

Convergence to 11 ensures termination. The loop examines the indices in order: it returns 4, the first index satisfying the criterion for the computed values. Here, the gaps at indices 3 and 4 are approximately 0.00105884 and 0.00004813. They lie clearly on either side of $10^{-3}$. The guaranteed index 4 may be larger; it comes from an upper bound, not from the exact error.

## Exercise 2 · An exponential curve

1. *[0.5 pt]* $f(0) = 4$; limit 0 by comparison of growth rates.

2. *[0.5 pt]* By the product rule,

$$f'(x) = e^{-x} - (x+4)e^{-x} = (-x-3)e^{-x}.$$

The two factors are respectively strictly negative and positive on $[0, +\infty[$; hence $f' < 0$.

| $x$ | $0$ | | $+\infty$ |
|---|---|---|---|
| $f'(x)$ | | $-$ | |
| $f(x)$ | $4$ | $\searrow$ | $0$ |

3. *[0.5 pt]* Differentiating the previous product,

$$f''(x) = -e^{-x} - (-x-3)e^{-x} = (x+2)e^{-x}.$$

On the domain, $x + 2 \geq 0$ and $e^{-x} > 0$: $f$ is convex. There is no change of convexity in the interior of the domain, hence no point of inflection.

4. *[1 pt]* Continuity, strict decrease, $f(0) = 4 > 2$ and limit $0 < 2$: exactly one solution.

5. *[0.5 pt]* The tangent at 0 has equation $y = f(0) + f'(0)x = 4 - 3x$. A convex function lies above each of its tangents, so $(x+4)e^{-x} \geq 4 - 3x$ for all $x \geq 0$.

6. *[0.5 pt]* We know $f(0) = 4 > 2$. Moreover, $f(2) = 6e^{-2} \simeq 0.8120 < 2$. The zero of $f - 2$ is therefore in $]0, 2[$. A program is:

```
from math import exp

def encadrement():
    a, b = 0.0, 2.0
    while b-a > 1e-3:
        m = (a+b)/2
        if (m+4)*exp(-m) > 4/2:
            a = m
        else:
            b = m
    return a, b
```

   Since $f$ is decreasing, the bracketing is preserved at each step. The result has length at most $10^{-3}$; its midpoint has an error of at most $5 \times 10^{-4}$.

7. *[0.5 pt]* $\frac{2}{2^n} \leq 10^{-3}$: $n = 11$ steps suffice.

8. *[1 pt]* For $x \geq 0$, $x + 4 > 0$ and $e^{-x} > 0$, so $f(x) > 0$. A limit is not necessarily an attained value.

## Exercise 3 · Vector calculation in a basis

1. *[0.5 pt]* In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{\jmath}$ would be $1 = 0$, a contradiction by uniqueness of coordinates in a basis.

2. *[0.5 pt]* Expanding, $3\vec{u} + 8\vec{v} = 3(\vec{\imath} + \vec{k}) + 8(\vec{\jmath} + \vec{k}) = 3\vec{\imath} + 8\vec{\jmath} + 11\vec{k} = \vec{w}$.

3. *[0.5 pt]* Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis requires

$$\lambda = 3, \qquad \mu = 8, \qquad \lambda + \mu = 12.$$

   The first two equalities give $\lambda + \mu = 11$, contradicting the third. The three vectors are therefore not coplanar.

4. *[1 pt]* $8\vec{u} - 3\vec{v} = 8(\vec{\imath} + \vec{k}) - 3(\vec{\jmath} + \vec{k}) = 8\vec{\imath} - 3\vec{\jmath} + 5\vec{k}$.

5. *[0.5 pt]* $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are not collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

6. *[0.5 pt]* $AB = AC = \sqrt{2}$; $\overrightarrow{BC} = \vec{v} - \vec{u} = (-1, 1, 0)$, so $BC = \sqrt{2}$.

7. *[0.5 pt]* The midpoint formula gives $\overrightarrow{AI} = \frac{1}{2}(\vec{u} + \vec{v})$. Since the vectors $\vec{u}, \vec{v}$ are not collinear,

$$\overrightarrow{AD} = \lambda\overrightarrow{AI} \quad \Longleftrightarrow \quad 3 = \frac{\lambda}{2} \text{ and } 8 = \frac{\lambda}{2}.$$

   Here $3 \neq 8$: there is no such $\lambda$, so $A, I, D$ are not collinear.

8. *[1 pt]* No: $D$ belongs to the plane but $E$ does not, by the non-coplanarity shown above. The intersection is exactly $D$.

## Exercise 4 · Repetitions and estimation

1. *[0.5 pt]* Repetition of a fixed number 16 of identical independent trials with two outcomes; $X \sim \mathcal{B}(16, \frac{1}{10})$.

2. *[0.5 pt]* $P(X = 2) = \binom{16}{2}(\frac{1}{10})^2(\frac{9}{10})^{14} \simeq 0.274522$.

3. *[0.5 pt]* $1 - P(X = 0) - P(X = 1) = 1 - (\frac{9}{10})^{16} - 16(\frac{1}{10})(\frac{9}{10})^{15} \simeq 0.485272$.

4. *[1 pt]* $E(X) = \frac{8}{5}$, $V(X) = \frac{36}{25}$.

5. *[0.5 pt]* For $N$ independent trials, the probability of no success is $(1-p)^N$. The required probability is therefore $1 - (\frac{9}{10})^N$. It is strictly increasing in $N$, since $0 < \frac{9}{10} < 1$. Calculation gives

$$P_{28} \simeq 0.947665 < 0.95, \qquad P_{29} \simeq 0.952899 \geq 0.95.$$

   Thus 29 is the smallest suitable number of trials.

6. *[0.5 pt]* Let $n = 16$. For $j \in \{0, \ldots, n\}$, an ordered outcome containing $j$ successes and $n - j$ failures has, by independence, probability $p^j(1-p)^{n-j}$. There are $\binom{n}{j}$ choices for the positions of the successes. These outcomes are mutually exclusive, so

$$P(X = j) = \binom{n}{j} p^j (1-p)^{n-j}, \qquad p = \frac{1}{10}.$$

7. *[0.5 pt]* The events $X = 1$, $X = 2$, $X = 3$ are mutually exclusive, so

$$P(1 \leq X \leq 3) = \sum_{j=1}^{3} \binom{16}{j} (\frac{1}{10})^j (\frac{9}{10})^{16-j} \simeq 0.746292.$$

8. *[1 pt]*

```python
from math import comb

def quantile():
    n, p = 16, 1/10
    j, cumul = 0, (1-p)**n
    while cumul < 0.95:
        j += 1
        cumul += comb(n, j)*p**j*(1-p)**(n-j)
    return j
```

   The probabilities added are positive and the final cumulative sum equals 1, so the loop terminates. It returns 4. The cumulative sums at indices 3 and 4 are approximately 0.931594 and 0.982996: this confirms minimality.

# Mock 1 · A

**PAPER RESERVED FOR ASSESSMENT**

**Duration: 4 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · Induction and bounds — 5 points

Consider the sequence $(v_n)$ defined by $v_0 = 0$ and, for every natural number $n$, $v_{n+1} = \sqrt{132 + v_n}$.

1. Calculate $v_1$ and $v_2$. Give the exact values. *(0.25 pt)*
2. Prove by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 12$. *(0.75 pt)*
3. Prove by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. *(0.75 pt)*
4. Prove that, for all $n \in \mathbb{N}$, $v_n < 12$. *(0.5 pt)*
5. Justify that $(v_n)$ converges, then determine its limit. *(0.75 pt)*
6. Prove that, for all $n \in \mathbb{N}$, $0 \leq 12 - v_{n+1} \leq \frac{12 - v_n}{12}$. *(0.5 pt)*
7. Deduce by induction an upper bound for $12 - v_n$ in terms of $n$. Then determine, by comparing integer powers, an index from which this bound guarantees $12 - v_n \leq 10^{-3}$. *(1 pt)*
8. Write a Python function with no argument that returns the first integer $n$ such that $12 - v_n \leq 10^{-3}$. Run the function and compare the index obtained with the sufficient index from the previous question. Explain any difference. *(0.5 pt)*

## Exercise 2 · An exponential curve — 5 points

Consider the function $f$ defined on $[0, +\infty)$ by $f(x) = (x + 3)e^{-x}$. Let $\mathcal{C}_f$ denote its graph.

1. Calculate $f(0)$ and determine the limit of $f$ at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw the variation table of $f$. *(0.5 pt)*
3. Calculate $f''(x)$, then study the convexity of $f$. *(0.5 pt)*
4. Prove that the equation $f(x) = \frac{3}{2}$ has a unique solution $\alpha$ in $[0, +\infty)$. *(1 pt)*
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point of abscissa 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. *(0.5 pt)*
6. Show that $0 < \alpha < 2$. Write a Python function that applies bisection on $[0, 2]$ and returns a bracketing of $\alpha$ of width at most $10^{-3}$. *(0.5 pt)*
7. Determine the minimum number of bisection steps guaranteeing this width. *(0.5 pt)*
8. A student claims: "Since $f(x)$ tends to 0, the curve crosses the horizontal axis." Refute this claim by studying the sign of $f(x)$. *(1 pt)*

Mock 1 · A · continued

## Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$ of space, define the vectors

$$\vec{u} = \vec{\imath} + \vec{k}, \qquad \vec{v} = \vec{\jmath} + \vec{k},$$
$$\vec{w} = 2\vec{\imath} + 5\vec{\jmath} + 7\vec{k}, \quad \vec{z} = 2\vec{\imath} + 5\vec{\jmath} + 8\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. *(0.5 pt)*
2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. *(0.5 pt)*
3. Are the vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. *(0.5 pt)*
4. Express the vector $5\vec{u} - 2\vec{v}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
5. Choose a point $A$, then define $B$, $C$ and $D$ by $\overrightarrow{AB} = \vec{u}$, $\overrightarrow{AC} = \vec{v}$ and $\overrightarrow{AD} = \vec{w}$. Prove that $A$, $B$, $C$ and $D$ are coplanar. *(0.5 pt)*
6. For questions 6 to 8, assume the basis is orthonormal. Calculate the distances $AB$, $AC$ and $BC$. *(0.5 pt)*
7. Let $I$ be the midpoint of $[BC]$. Express $\overrightarrow{AI}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$, then determine whether the points $A$, $I$ and $D$ are collinear. *(0.5 pt)*
8. Define $E$ by $\overrightarrow{AE} = \vec{z}$. Is the line $(DE)$ contained in the plane $(ABC)$? Justify. *(1 pt)*

## Exercise 4 · Repetitions and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 17 times. The trials are independent and the probability of success is $p = \frac{1}{5}$ on each trial. Let $X$ denote the number of successes obtained.

1. Justify that $X$ follows a binomial distribution and state its parameters. *(0.5 pt)*
2. Calculate the probability of obtaining exactly two successes. *(0.5 pt)*
3. Express using complementary events, then calculate, the probability of obtaining at least two successes. *(0.5 pt)*
4. Calculate the expected value and the variance of $X$. *(1 pt)*
5. The same trial is now repeated $N \geq 1$ times, still independently. Determine the smallest integer $N$ for which the probability of obtaining at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. *(0.5 pt)*
6. For an integer $j \in \{0, \ldots, 17\}$, prove the formula for $P(X = j)$ by counting the possible positions of the successes. *(0.5 pt)*
7. A series of 17 trials is accepted if its number of successes is between 1 and 3 inclusive. Calculate the probability that a series is accepted. *(0.5 pt)*
8. Write a Python function with no argument that returns the smallest integer $j$ such that $P(X \leq j) \geq 0.95$. Run the function and check the minimality of the result. *(1 pt)*

# Solutions · Mock 1 · A

## Exercise 1 · Induction and bounds

1. *[0.25 pt]* $v_1 = \sqrt{132+0} = \sqrt{132}$ and $v_2 = \sqrt{132+\sqrt{132}}$.

2. *[0.75 pt]* Let $P(n)$ be the statement: "$v_n$ is defined and $0 \leq v_n \leq 12$".

   **Base case.** $v_0 = 0$ is defined and belongs to $[0,12]$; $P(0)$ is true.

   **Inductive step.** Fix $n \in \mathbb{N}$ and assume $P(n)$ is true. Then

   $$0 \leq 132 \leq 132 + v_n \leq 132 + 12 = 144.$$

   The square root is therefore defined and, since this function is increasing on $[0,+\infty[$,

   $$0 \leq v_{n+1} = \sqrt{132+v_n} \leq \sqrt{144} = 12.$$

   Thus $P(n+1)$ is true.

   **Conclusion.** By induction, all terms are defined and belong to $[0,12]$.

3. *[0.75 pt]* For $n \in \mathbb{N}$, let $R(n)$ be the statement: "$v_{n+1} \geq v_n$".

   **Base case.** $v_1 = \sqrt{132} > 0 = v_0$; $R(0)$ is true.

   **Inductive step.** Fix $n \geq 0$ and assume $v_{n+1} \geq v_n$. Adding 132 and then applying the increasing property of the square root,

   $$v_{n+2} = \sqrt{132+v_{n+1}} \geq \sqrt{132+v_n} = v_{n+1}.$$

   This is $R(n+1)$.

   **Conclusion.** By induction, $v_{n+1} \geq v_n$ for all $n \geq 0$: $(v_n)$ is increasing.

4. *[0.5 pt]* Let $S(n)$ be the statement: "$v_n < 12$. At index 0, $v_0 = 0 < 12$. Fix $n \geq 0$ and assume $S(n)$ is true. Since the radicands are positive, the strict increase of the square root gives

   $$v_{n+1} = \sqrt{132+v_n} < \sqrt{132+12} = \sqrt{144} = 12.$$

   So $S(n+1)$ is true. The base case and the inductive step establish by induction that $v_n < 12$ for all $n \in \mathbb{N}$.

5. *[0.75 pt]* The sequence is increasing and bounded above by 12: the monotone convergence theorem gives a real limit $\ell \in [0,12]$. The function $x \mapsto \sqrt{132+x}$ is continuous on this interval. Taking the limit in the relation, $\ell = \sqrt{132+\ell}$. Since $\ell \geq 0$, this is equivalent to

   $$\ell^2 - \ell - 132 = 0 \quad \Longleftrightarrow \quad (\ell - 12)(\ell + 11) = 0.$$

   The root $-11$ is excluded by $\ell \geq 0$; so $\lim v_n = 12$.

6. *[0.5 pt]* The previous result ensures $12 - v_{n+1} \geq 0$. Multiplying by the conjugate expression,

   $$\begin{aligned} 12 - v_{n+1} &= \frac{144 - (132+v_n)}{12 + \sqrt{132+v_n}} \\ &= \frac{12 - v_n}{12 + \sqrt{132+v_n}} \leq \frac{12-v_n}{12}. \end{aligned}$$

   The last inequality uses a positive numerator and a denominator at least equal to $12 > 0$.

7. *[1 pt]* Let us prove by induction $T(n)$: "$0 \leq 12 - v_n \leq 12^{1-n}$". At index 0, $12 - v_0 = 12 = 12^1$. Fix $n \geq 0$ and assume $T(n)$ is true. The previous question gives

$$0 \leq 12 - v_{n+1} \leq \frac{12 - v_n}{12} \leq \frac{12^{1-n}}{12} = 12^{1-(n+1)}.$$

This is $T(n+1)$. By induction, the bounds hold at every index. To have $12^{1-n} \leq 10^{-3}$, it suffices that $12^{n-1} \geq 1000$. Now $12^2 = 144 < 1000$ and $12^3 = 1728 \geq 1000$. Index 4 is therefore the first one guaranteed by this bound, without using the logarithm.

8. *[0.5 pt]*

```python
from math import sqrt

def premier_rang():
    n, v = 0, 0.0
    while 12-v > 1e-3:
        v = sqrt(132+v)
        n += 1
    return n
```

Convergence to 12 ensures termination. The loop examines the indices in order: it returns 3, the first index satisfying the criterion for the computed values. Here, the gaps at indices 2 and 3 are approximately 0.02130536 and 0.00088776. They lie clearly on either side of $10^{-3}$. The guaranteed index 4 may be larger; it comes from an upper bound, not from the exact error.

## Exercise 2 · An exponential curve

1. *[0.5 pt]* $f(0) = 3$; limit 0 by comparison of growth rates.

2. *[0.5 pt]* By the product rule,

$$f'(x) = e^{-x} - (x+3)e^{-x} = (-x-2)e^{-x}.$$

The two factors are respectively strictly negative and positive on $[0, +\infty[$; hence $f' < 0$.

| $x$ | $0$ | | $+\infty$ |
|---|---|---|---|
| $f'(x)$ | | $-$ | |
| $f(x)$ | $3$ | $\searrow$ | $0$ |

3. *[0.5 pt]* Differentiating the previous product,

$$f''(x) = -e^{-x} - (-x-2)e^{-x} = (x+1)e^{-x}.$$

On the domain, $x + 1 \geq 0$ and $e^{-x} > 0$: $f$ is convex. There is no change of convexity in the interior of the domain, hence no point of inflection.

4. *[1 pt]* Continuity, strict decrease, $f(0) = 3 > \frac{3}{2}$ and limit $0 < \frac{3}{2}$: exactly one solution.

5. *[0.5 pt]* The tangent at 0 has equation $y = f(0) + f'(0)x = 3 - 2x$. A convex function lies above each of its tangents, so $(x+3)e^{-x} \geq 3 - 2x$ for all $x \geq 0$.

6. *[0.5 pt]* We know $f(0) = 3 > \frac{3}{2}$. Moreover, $f(2) = 5e^{-2} \simeq 0.6767 < \frac{3}{2}$. The zero of $f - \frac{3}{2}$ is therefore in $]0, 2[$. A program is:

```
from math import exp

def encadrement():
    a, b = 0.0, 2.0
    while b-a > 1e-3:
        m = (a+b)/2
        if (m+3)*exp(-m) > 3/2:
            a = m
        else:
            b = m
    return a, b
```

   Since $f$ is decreasing, the bracketing is preserved at each step. The result has length at most $10^{-3}$; its midpoint has an error of at most $5 \times 10^{-4}$.

7. *[0.5 pt]* $\frac{2}{2^n} \leq 10^{-3}$: $n = 11$ steps suffice.

8. *[1 pt]* For $x \geq 0$, $x + 3 > 0$ and $e^{-x} > 0$, so $f(x) > 0$. A limit is not necessarily an attained value.

## Exercise 3 · Vector calculation in a basis

1. *[0.5 pt]* In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{\jmath}$ would be $1 = 0$, a contradiction by uniqueness of coordinates in a basis.

2. *[0.5 pt]* Expanding, $2\vec{u} + 5\vec{v} = 2(\vec{\imath} + \vec{k}) + 5(\vec{\jmath} + \vec{k}) = 2\vec{\imath} + 5\vec{\jmath} + 7\vec{k} = \vec{w}$.

3. *[0.5 pt]* Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis requires

$$\lambda = 2, \qquad \mu = 5, \qquad \lambda + \mu = 8.$$

   The first two equalities give $\lambda + \mu = 7$, contradicting the third. The three vectors are therefore not coplanar.

4. *[1 pt]* $5\vec{u} - 2\vec{v} = 5(\vec{\imath} + \vec{k}) - 2(\vec{\jmath} + \vec{k}) = 5\vec{\imath} - 2\vec{\jmath} + 3\vec{k}$.

5. *[0.5 pt]* $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are not collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

6. *[0.5 pt]* $AB = AC = \sqrt{2}$; $\overrightarrow{BC} = \vec{v} - \vec{u} = (-1, 1, 0)$, so $BC = \sqrt{2}$.

7. *[0.5 pt]* The midpoint formula gives $\overrightarrow{AI} = \frac{1}{2}(\vec{u} + \vec{v})$. Since the vectors $\vec{u}, \vec{v}$ are not collinear,

$$\overrightarrow{AD} = \lambda\overrightarrow{AI} \quad \Longleftrightarrow \quad 2 = \frac{\lambda}{2} \text{ and } 5 = \frac{\lambda}{2}.$$

   Here $2 \neq 5$: there is no such $\lambda$, so $A, I, D$ are not collinear.

8. *[1 pt]* No: $D$ belongs to the plane but $E$ does not, by the non-coplanarity shown above. The intersection is exactly $D$.

## Exercise 4 · Repetitions and estimation

1. *[0.5 pt]* Repetition of a fixed number 17 of identical independent trials with two outcomes; $X \sim \mathcal{B}(17, \frac{1}{5})$.

2. *[0.5 pt]* $P(X = 2) = \binom{17}{2}(\frac{1}{5})^2(\frac{4}{5})^{15} \simeq 0.191403$.

3. *[0.5 pt]* $1 - P(X = 0) - P(X = 1) = 1 - (\frac{4}{5})^{17} - 17(\frac{1}{5})(\frac{4}{5})^{16} \simeq 0.881781$.

4. *[1 pt]* $E(X) = \frac{17}{5}$, $V(X) = \frac{68}{25}$.

5. *[0.5 pt]* For $N$ independent trials, the probability of no success is $(1-p)^N$. The required probability is therefore $1 - (\frac{4}{5})^N$. It is strictly increasing in $N$, since $0 < \frac{4}{5} < 1$. Calculation gives

$$P_{13} \simeq 0.945024 < 0.95, \qquad P_{14} \simeq 0.956020 \geq 0.95.$$

   Thus 14 is the smallest suitable number of trials.

6. *[0.5 pt]* Let $n = 17$. For $j \in \{0, \ldots, n\}$, an ordered outcome containing $j$ successes and $n - j$ failures has, by independence, probability $p^j(1-p)^{n-j}$. There are $\binom{n}{j}$ choices for the positions of the successes. These outcomes are mutually exclusive, so

$$P(X = j) = \binom{n}{j} p^j (1-p)^{n-j}, \qquad p = \frac{1}{5}.$$

7. *[0.5 pt]* The events $X = 1$, $X = 2$, $X = 3$ are mutually exclusive, so

$$P(1 \leq X \leq 3) = \sum_{j=1}^{3} \binom{17}{j} (\frac{1}{5})^j (\frac{4}{5})^{17-j} \simeq 0.526358.$$

8. *[1 pt]*

```python
from math import comb

def quantile():
    n, p = 17, 2/10
    j, cumul = 0, (1-p)**n
    while cumul < 0.95:
        j += 1
        cumul += comb(n, j)*p**j*(1-p)**(n-j)
    return j
```

   The probabilities added are positive and the final cumulative sum equals 1, so the loop terminates. It returns 6. The cumulative sums at indices 5 and 6 are approximately 0.894299 and 0.962337: this confirms minimality.

# Mock 1 · B

**PAPER RESERVED FOR ASSESSMENT**

**Duration: 4 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · Induction and bounds — 5 points

Consider the sequence $(v_n)$ defined by $v_0 = 0$ and, for every natural number $n$, $v_{n+1} = \sqrt{56 + v_n}$.

1. Calculate $v_1$ and $v_2$. Give the exact values. *(0.25 pt)*
2. Prove by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 8$. *(0.75 pt)*
3. Prove by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. *(0.75 pt)*
4. Prove that, for all $n \in \mathbb{N}$, $v_n < 8$. *(0.5 pt)*
5. Justify that $(v_n)$ converges, then determine its limit. *(0.75 pt)*
6. Prove that, for all $n \in \mathbb{N}$, $0 \leq 8 - v_{n+1} \leq \frac{8 - v_n}{8}$. *(0.5 pt)*
7. Deduce by induction an upper bound for $8 - v_n$ in terms of $n$. Then determine, by comparing integer powers, an index from which this bound guarantees $8 - v_n \leq 10^{-3}$. *(1 pt)*
8. Write a Python function with no argument that returns the first integer $n$ such that $8 - v_n \leq 10^{-3}$. Run the function and compare the index obtained with the sufficient index from the previous question. Explain any difference. *(0.5 pt)*

## Exercise 2 · An exponential curve — 5 points

Consider the function $f$ defined on $[0, +\infty)$ by $f(x) = (x + 2)e^{-x}$. Let $\mathcal{C}_f$ denote its graph.

1. Calculate $f(0)$ and determine the limit of $f$ at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw the variation table of $f$. *(0.5 pt)*
3. Calculate $f''(x)$, then study the convexity of $f$. *(0.5 pt)*
4. Prove that the equation $f(x) = 1$ has a unique solution $\alpha$ in $[0, +\infty)$. *(1 pt)*
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point of abscissa 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. *(0.5 pt)*
6. Show that $0 < \alpha < 2$. Write a Python function that applies bisection on $[0, 2]$ and returns a bracketing of $\alpha$ of width at most $10^{-3}$. *(0.5 pt)*
7. Determine the minimum number of bisection steps guaranteeing this width. *(0.5 pt)*
8. A student claims: “Since $f(x)$ tends to 0, the curve crosses the horizontal axis.” Refute this claim by studying the sign of $f(x)$. *(1 pt)*

Mock 1 · B · continued

## Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$ of space, define the vectors

$$\vec{u} = \vec{\imath} + \vec{k}, \qquad \vec{v} = \vec{\jmath} + \vec{k},$$
$$\vec{w} = 5\vec{\imath} + 6\vec{\jmath} + 11\vec{k}, \quad \vec{z} = 5\vec{\imath} + 6\vec{\jmath} + 12\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. *(0.5 pt)*
2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. *(0.5 pt)*
3. Are the vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. *(0.5 pt)*
4. Express the vector $6\vec{u} - 5\vec{v}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
5. Choose a point $A$, then define $B$, $C$ and $D$ by $\overrightarrow{AB} = \vec{u}$, $\overrightarrow{AC} = \vec{v}$ and $\overrightarrow{AD} = \vec{w}$. Prove that $A$, $B$, $C$ and $D$ are coplanar. *(0.5 pt)*
6. For questions 6 to 8, assume the basis is orthonormal. Calculate the distances $AB$, $AC$ and $BC$. *(0.5 pt)*
7. Let $I$ be the midpoint of $[BC]$. Express $\overrightarrow{AI}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$, then determine whether the points $A$, $I$ and $D$ are collinear. *(0.5 pt)*
8. Define $E$ by $\overrightarrow{AE} = \vec{z}$. Is the line $(DE)$ contained in the plane $(ABC)$? Justify. *(1 pt)*

## Exercise 4 · Repetitions and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 19 times. The trials are independent and the probability of success is $p = \frac{1}{10}$ on each trial. Let $X$ denote the number of successes obtained.

1. Justify that $X$ follows a binomial distribution and state its parameters. *(0.5 pt)*
2. Calculate the probability of obtaining exactly two successes. *(0.5 pt)*
3. Express using complementary events, then calculate, the probability of obtaining at least two successes. *(0.5 pt)*
4. Calculate the expected value and the variance of $X$. *(1 pt)*
5. The same trial is now repeated $N \geq 1$ times, still independently. Determine the smallest integer $N$ for which the probability of obtaining at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. *(0.5 pt)*
6. For an integer $j \in \{0, \ldots, 19\}$, prove the formula for $P(X = j)$ by counting the possible positions of the successes. *(0.5 pt)*
7. A series of 19 trials is accepted if its number of successes is between 1 and 3 inclusive. Calculate the probability that a series is accepted. *(0.5 pt)*
8. Write a Python function with no argument that returns the smallest integer $j$ such that $P(X \leq j) \geq 0.95$. Run the function and check the minimality of the result. *(1 pt)*

# Solutions · Mock 1 · B

## Exercise 1 · Induction and bounds

1. *[0.25 pt]* $v_1 = \sqrt{56+0} = \sqrt{56}$ and $v_2 = \sqrt{56+\sqrt{56}}$.

2. *[0.75 pt]* Let $P(n)$ be the statement: "$v_n$ is defined and $0 \leq v_n \leq 8$".

   **Base case.** $v_0 = 0$ is defined and belongs to $[0,8]$; $P(0)$ is true.

   **Inductive step.** Fix $n \in \mathbb{N}$ and assume $P(n)$ is true. Then

   $$0 \leq 56 \leq 56 + v_n \leq 56 + 8 = 64.$$

   The square root is therefore defined and, since this function is increasing on $[0, +\infty[$,

   $$0 \leq v_{n+1} = \sqrt{56+v_n} \leq \sqrt{64} = 8.$$

   Thus $P(n+1)$ is true.

   **Conclusion.** By induction, all terms are defined and belong to $[0,8]$.

3. *[0.75 pt]* For $n \in \mathbb{N}$, let $R(n)$ be the statement: "$v_{n+1} \geq v_n$".

   **Base case.** $v_1 = \sqrt{56} > 0 = v_0$; $R(0)$ is true.

   **Inductive step.** Fix $n \geq 0$ and assume $v_{n+1} \geq v_n$. Adding 56 and then applying the increasing property of the square root,

   $$v_{n+2} = \sqrt{56+v_{n+1}} \geq \sqrt{56+v_n} = v_{n+1}.$$

   This is $R(n+1)$.

   **Conclusion.** By induction, $v_{n+1} \geq v_n$ for all $n \geq 0$: $(v_n)$ is increasing.

4. *[0.5 pt]* Let $S(n)$ be the statement: "$v_n < 8$. At index 0, $v_0 = 0 < 8$. Fix $n \geq 0$ and assume $S(n)$ is true. Since the radicands are positive, the strict increase of the square root gives

   $$v_{n+1} = \sqrt{56+v_n} < \sqrt{56+8} = \sqrt{64} = 8.$$

   So $S(n+1)$ is true. The base case and the inductive step establish by induction that $v_n < 8$ for all $n \in \mathbb{N}$.

5. *[0.75 pt]* The sequence is increasing and bounded above by 8: the monotone convergence theorem gives a real limit $\ell \in [0,8]$. The function $x \mapsto \sqrt{56+x}$ is continuous on this interval. Taking the limit in the relation, $\ell = \sqrt{56+\ell}$. Since $\ell \geq 0$, this is equivalent to

   $$\ell^2 - \ell - 56 = 0 \quad \Longleftrightarrow \quad (\ell - 8)(\ell + 7) = 0.$$

   The root $-7$ is excluded by $\ell \geq 0$; so $\lim v_n = 8$.

6. *[0.5 pt]* The previous result ensures $8 - v_{n+1} \geq 0$. Multiplying by the conjugate expression,

   $$\begin{aligned} 8 - v_{n+1} &= \frac{64 - (56+v_n)}{8 + \sqrt{56+v_n}} \\ &= \frac{8 - v_n}{8 + \sqrt{56+v_n}} \leq \frac{8-v_n}{8}. \end{aligned}$$

   The last inequality uses a positive numerator and a denominator at least equal to $8 > 0$.

7. *[1 pt]* Let us prove by induction $T(n)$: "$0 \leq 8 - v_n \leq 8^{1-n}$". At index 0, $8 - v_0 = 8 = 8^1$. Fix $n \geq 0$ and assume $T(n)$ is true. The previous question gives

$$0 \leq 8 - v_{n+1} \leq \frac{8 - v_n}{8} \leq \frac{8^{1-n}}{8} = 8^{1-(n+1)}.$$

This is $T(n+1)$. By induction, the bounds hold at every index. To have $8^{1-n} \leq 10^{-3}$, it suffices that $8^{n-1} \geq 1000$. Now $8^3 = 512 < 1000$ and $8^4 = 4096 \geq 1000$. Index 5 is therefore the first one guaranteed by this bound, without using the logarithm.

8. *[0.5 pt]*

```python
from math import sqrt

def premier_rang():
    n, v = 0, 0.0
    while 8-v > 1e-3:
        v = sqrt(56+v)
        n += 1
    return n
```

Convergence to 8 ensures termination. The loop examines the indices in order: it returns 4, the first index satisfying the criterion for the computed values. Here, the gaps at indices 3 and 4 are approximately 0.00202265 and 0.00012642. They lie clearly on either side of $10^{-3}$. The guaranteed index 5 may be larger; it comes from an upper bound, not from the exact error.

## Exercise 2 · An exponential curve

1. *[0.5 pt]* $f(0) = 2$; limit 0 by comparison of growth rates.

2. *[0.5 pt]* By the product rule,

$$f'(x) = e^{-x} - (x+2)e^{-x} = (-x-1)e^{-x}.$$

The two factors are respectively strictly negative and positive on $[0, +\infty[$; hence $f' < 0$.

| $x$ | $0$ | | $+\infty$ |
|---|---|---|---|
| $f'(x)$ | | $-$ | |
| $f(x)$ | $2$ | $\searrow$ | $0$ |

3. *[0.5 pt]* Differentiating the previous product,

$$f''(x) = -e^{-x} - (-x-1)e^{-x} = xe^{-x}.$$

On the domain, $x + 0 \geq 0$ and $e^{-x} > 0$: $f$ is convex. There is no change of convexity in the interior of the domain, hence no point of inflection.

4. *[1 pt]* Continuity, strict decrease, $f(0) = 2 > 1$ and limit $0 < 1$: exactly one solution.

5. *[0.5 pt]* The tangent at 0 has equation $y = f(0) + f'(0)x = 2 - x$. A convex function lies above each of its tangents, so $(x+2)e^{-x} \geq 2 - x$ for all $x \geq 0$.

6. *[0.5 pt]* We know $f(0) = 2 > 1$. Moreover, $f(2) = 4e^{-2} \simeq 0.5413 < 1$. The zero of $f - 1$ is therefore in $]0, 2[$. A program is:

```
from math import exp

def encadrement():
    a, b = 0.0, 2.0
    while b-a > 1e-3:
        m = (a+b)/2
        if (m+2)*exp(-m) > 2/2:
            a = m
        else:
            b = m
    return a, b
```

   Since $f$ is decreasing, the bracketing is preserved at each step. The result has length at most $10^{-3}$; its midpoint has an error of at most $5 \times 10^{-4}$.

7. *[0.5 pt]* $\frac{2}{2^n} \leq 10^{-3}$: $n = 11$ steps suffice.

8. *[1 pt]* For $x \geq 0$, $x + 2 > 0$ and $e^{-x} > 0$, so $f(x) > 0$. A limit is not necessarily an attained value.

## Exercise 3 · Vector calculation in a basis

1. *[0.5 pt]* In an equality $\vec{v} = t\vec{u}$, the coefficient of $\vec{\jmath}$ would be $1 = 0$, a contradiction by uniqueness of coordinates in a basis.

2. *[0.5 pt]* Expanding, $5\vec{u} + 6\vec{v} = 5(\vec{\imath} + \vec{k}) + 6(\vec{\jmath} + \vec{k}) = 5\vec{\imath} + 6\vec{\jmath} + 11\vec{k} = \vec{w}$.

3. *[0.5 pt]* Since $\vec{u}, \vec{v}$ are not collinear, coplanarity would be equivalent to $\vec{z} = \lambda\vec{u} + \mu\vec{v}$. Uniqueness of the decomposition in the basis requires

$$\lambda = 5, \qquad \mu = 6, \qquad \lambda + \mu = 12.$$

   The first two equalities give $\lambda + \mu = 11$, contradicting the third. The three vectors are therefore not coplanar.

4. *[1 pt]* $6\vec{u} - 5\vec{v} = 6(\vec{\imath} + \vec{k}) - 5(\vec{\jmath} + \vec{k}) = 6\vec{\imath} - 5\vec{\jmath} + \vec{k}$.

5. *[0.5 pt]* $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are not collinear and $\overrightarrow{AD}$ is their linear combination: $D \in (ABC)$.

6. *[0.5 pt]* $AB = AC = \sqrt{2}$; $\overrightarrow{BC} = \vec{v} - \vec{u} = (-1, 1, 0)$, so $BC = \sqrt{2}$.

7. *[0.5 pt]* The midpoint formula gives $\overrightarrow{AI} = \frac{1}{2}(\vec{u} + \vec{v})$. Since the vectors $\vec{u}, \vec{v}$ are not collinear,

$$\overrightarrow{AD} = \lambda\overrightarrow{AI} \quad \Longleftrightarrow \quad 5 = \frac{\lambda}{2} \text{ and } 6 = \frac{\lambda}{2}.$$

   Here $5 \neq 6$: there is no such $\lambda$, so $A, I, D$ are not collinear.

8. *[1 pt]* No: $D$ belongs to the plane but $E$ does not, by the non-coplanarity shown above. The intersection is exactly $D$.

## Exercise 4 · Repetitions and estimation

1. *[0.5 pt]* Repetition of a fixed number 19 of identical independent trials with two outcomes; $X \sim \mathcal{B}(19, \frac{1}{10})$.

2. *[0.5 pt]* $P(X = 2) = \binom{19}{2}(\frac{1}{10})^2(\frac{9}{10})^{17} \simeq 0.285180$.

3. *[0.5 pt]* $1 - P(X = 0) - P(X = 1) = 1 - (\frac{9}{10})^{19} - 19(\frac{1}{10})(\frac{9}{10})^{18} \simeq 0.579735$.

4. *[1 pt]* $E(X) = \frac{19}{10}$, $V(X) = \frac{171}{100}$.

5. *[0.5 pt]* For $N$ independent trials, the probability of no success is $(1-p)^N$. The required probability is therefore $1 - (\frac{9}{10})^N$. It is strictly increasing in $N$, since $0 < \frac{9}{10} < 1$. Calculation gives

$$P_{28} \simeq 0.947665 < 0.95, \qquad P_{29} \simeq 0.952899 \geq 0.95.$$

   Thus 29 is the smallest suitable number of trials.

6. *[0.5 pt]* Let $n = 19$. For $j \in \{0, \ldots, n\}$, an ordered outcome containing $j$ successes and $n - j$ failures has, by independence, probability $p^j(1-p)^{n-j}$. There are $\binom{n}{j}$ choices for the positions of the successes. These outcomes are mutually exclusive, so

$$P(X = j) = \binom{n}{j} p^j (1-p)^{n-j}, \qquad p = \frac{1}{10}.$$

7. *[0.5 pt]* The events $X = 1$, $X = 2$, $X = 3$ are mutually exclusive, so

$$P(1 \leq X \leq 3) = \sum_{j=1}^{3} \binom{19}{j} (\frac{1}{10})^j (\frac{9}{10})^{19-j} \simeq 0.749917.$$

8. *[1 pt]*

```python
from math import comb

def quantile():
    n, p = 19, 1/10
    j, cumul = 0, (1-p)**n
    while cumul < 0.95:
        j += 1
        cumul += comb(n, j)*p**j*(1-p)**(n-j)
    return j
```

   The probabilities added are positive and the final cumulative sum equals 1, so the loop terminates. It returns 4. The cumulative sums at indices 3 and 4 are approximately 0.885002 and 0.964806: this confirms minimality.

# Test 4 · Practice 1

**PRACTICE PAPER · TRAINING**

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete attempts will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · A logarithmic function — 5 points

Consider the function $f$ defined on $(0, +\infty)$ by $f(x) = \frac{3+\ln x}{x}$. We denote by $\mathcal{C}_f$ its graph.

1. Determine the limits of $f$ at $0^+$ and at $+\infty$. *(1 pt)*
2. Calculate $f'(x)$, then draw the table of variations of $f$. *(1 pt)*
3. State the value of the maximum of $f$, then solve $f(x) = 0$. *(1 pt)*
4. Study the convexity of $f$ and determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Prove that the equation $f(x) = \frac{1}{2}e^2$ has exactly two solutions in $(0, +\infty)$. *(1 pt)*

## Exercise 2 · Antiderivatives with a given condition — 5 points

The five questions are independent. For each antiderivative calculation, state the expression of the antiderivatives requested and check the result by differentiation.

1. Determine all antiderivatives on $\mathbb{R}$ of $x \mapsto 2x^2 - 6x + 2$. *(1 pt)*
2. Determine the antiderivative $F$ of $x \mapsto e^{2x}$ on $\mathbb{R}$ such that $F(0) = 6$. *(1 pt)*
3. Determine all antiderivatives on $\mathbb{R}$ of $x \mapsto \frac{2x}{x^2+6}$. *(1 pt)*
4. Determine an antiderivative on $\mathbb{R}$ of $x \mapsto 2x(x^2+2)^2$. *(1 pt)*
5. Let $F$ and $G$ be two antiderivatives of the same function on an interval $I$. Prove that there exists a real constant $C$ such that $F(x) - G(x) = C$ for all $x \in I$. *(1 pt)*

Test 4 · Practice 1 · continued

## Exercise 3 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{5}y + \frac{8}{5}$ on $[0, +\infty)$. In questions 2 to 5, denote by $y$ the solution satisfying $y(0) = 13$.

1. Determine the constant solution of $(E)$. *(1 pt)*
2. Determine the expression of the solution satisfying the given initial condition. *(1 pt)*
3. Determine the limit of $y(t)$ as $t$ tends to $+\infty$. *(1 pt)*
4. Study the variations of $y$, then determine the smallest real number $t \geq 0$ such that $y(t) \leq 9$. *(1 pt)*
5. Apply Euler's method with step size $h > 0$ and denote by $Y_k$ the approximation of $y(kh)$. Give the value of $Y_0$ and express $Y_{k+1}$ in terms of $Y_k$ and $h$. *(1 pt)*

## Exercise 4 · Projection and volume — 5 points

In an orthonormal coordinate system with origin $O$, consider $A(11, 0, 0)$, $B(0, 11, 0)$ and $C(0, 0, 11)$. We are given $\vec{n} = (1, 1, 1)$. Lengths, areas and volumes will be expressed in units of length, area and volume respectively.

1. Prove that $\vec{n}$ is normal to the plane $(ABC)$. *(1 pt)*
2. Let $H$ be the orthogonal projection of $O$ onto $(ABC)$. Determine the coordinates of $H$ using $\overrightarrow{AH} \cdot \vec{n} = 0$. *(1 pt)*
3. Prove that $OH$ is the distance from $O$ to the plane $(ABC)$: for every $P \in (ABC)$, prove that $OP^2 \geq OH^2$. *(1 pt)*
4. Calculate the area of triangle $ABC$. *(1 pt)*
5. Calculate the volume of the tetrahedron $OABC$ in two ways, by choosing two different bases. *(1 pt)*

# Solutions · Test 4 · Practice 1

## Exercise 1 · A logarithmic function

1. *[1 pt]* At $0^+$, the numerator tends to $-\infty$ and the denominator to $0^+$: limit $-\infty$; at $+\infty$, limit 0 by comparison of growth rates.

2. *[1 pt]* By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (3 + \ln x)}{x^2} = \frac{-2 - \ln x}{x^2}.$$

The denominator is positive. The numerator vanishes at $s = e^{-2}$, is positive for $0 < x < s$ and negative for $x > s$. Hence $f$ is strictly increasing on $]0, s]$, then strictly decreasing on $[s, +\infty[$.

3. *[1 pt]* The maximum is attained only at $s = e^{-2}$ and equals

$$f(s) = \frac{3-2}{e^{-2}} = e^2.$$

Since $x > 0$, $f(x) = 0$ is equivalent to $3 + \ln x = 0$, that is $x = e^{-3}$.

4. *[1 pt]* We differentiate $(-2 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-2 - \ln x)x^{-3} = \frac{2\ln x + 3}{x^3}.$$

The numerator vanishes at $t = e^{-\frac{3}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at $t$ and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. *[1 pt]* Let $m = \frac{e^2}{2}$ and $s = e^{-2}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at 0 and value $2m$ at $s$: the corollary of the intermediate value theorem (IVT) gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from $2m$ to $0 < m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

## Exercise 2 · Antiderivatives with a given condition

1. *[1 pt]* By linearity, the antiderivatives are $F(x) = \frac{2}{3}x^3 - 3x^2 + 2x + C$, $C \in \mathbb{R}$. Indeed, $F'(x) = 2x^2 - 6x + 2$. The derivative does not depend on $C$, and all antiderivatives differ by a constant on $\mathbb{R}$.

2. *[1 pt]* The antiderivatives are $F(x) = \frac{e^{2x}}{2} + C$. The condition $F(0) = 6$ requires $\frac{1}{2} + C = 6$, hence $C = \frac{11}{2}$. The derivative of $\frac{e^{2x}}{2}$ is $e^{2x}$: the function obtained does satisfy both conditions.

3. *[1 pt]* Let $u(x) = x^2 + 6 > 0$ on $\mathbb{R}$; $u'(x) = 2x$. The antiderivatives are therefore $\ln(x^2 + 6) + C$, since their derivative equals $\frac{u'}{u} = \frac{2x}{x^2+6}$.

4. *[1 pt]* An antiderivative is $F(x) = \dfrac{(x^2+2)^3}{3}$. Indeed, by differentiating a composite function,

$$F'(x) = \frac{3}{3}\,2x(x^2+2)^2 = 2x(x^2+2)^2.$$

5. *[1 pt]* The derivative of their difference is zero on the interval; the difference is therefore constant. The fact that the domain is an interval is used in this result.

## Exercise 3 · Differential equation and evolution

1. *[1 pt]* $0 = -\frac{1}{5}y + \frac{8}{5}$ gives $y = 8$.
2. *[1 pt]* The constant solution is 8. Let $z = y - 8$; the equation becomes $z' = -\frac{z}{5}$. Its solutions are $z(t) = Ce^{-\frac{t}{5}}$, so all the solutions of the original equation are $y(t) = 8 + Ce^{-\frac{t}{5}}$. The condition $y(0) = 13$ gives $C = 5$. For the function obtained,

$$y'(t) = -e^{-\frac{t}{5}} = -\frac{y(t)}{5} + \frac{8}{5}, \qquad y(0) = 13.$$

   The solution of the initial value problem is therefore $y(t) = 8 + 5e^{-\frac{t}{5}}$.
3. *[1 pt]* $y' = -e^{-\frac{t}{5}} < 0$; limit 8.
4. *[1 pt]* Since $e^{-\frac{t}{5}} > 0$, the derivative $y'(t) = -e^{-\frac{t}{5}}$ is negative: $y$ is strictly decreasing. We solve

$$\begin{aligned} y(t) \leq 9 &\iff 5e^{-\frac{t}{5}} \leq 1 \\ &\iff -\frac{t}{5} \leq -\ln 5 \\ &\iff t \geq 5\ln 5. \end{aligned}$$

   The last step reverses the inequality because we multiply by $-5 < 0$. The first time is $5\ln 5 \simeq 8.0472$.
5. *[1 pt]* $Y_0 = 13$, $Y_{j+1} = Y_j + h(-\frac{Y_j}{5} + \frac{8}{5})$, $t_j = jh$.

## Exercise 4 · Projection and volume

1. *[1 pt]* $\overrightarrow{AB} = (-11, 11, 0)$ and $\overrightarrow{AC} = (-11, 0, 11)$ are non-collinear and their dot products with $\vec{n}$ are zero.
2. *[1 pt]* The line normal to the plane passing through $O$ has direction vector $\vec{n} = (1, 1, 1)$. Its point $H$ is therefore written $(t, t, t)$. For $H$ to lie in $(ABC)$, it is necessary and sufficient that

$$\overrightarrow{AH} \cdot \vec{n} = (t - 11) + t + t = 3t - 11 = 0.$$

   Hence $t = \frac{11}{3}$ and $H = (\frac{11}{3}, \frac{11}{3}, \frac{11}{3})$. This point lies in the plane and $\overrightarrow{OH}$ is collinear with the normal vector; it is indeed the required orthogonal projection.
3. *[1 pt]* $OH = \frac{11}{\sqrt{3}}$; for every $P$ in the plane, $OP^2 = OH^2 + HP^2 \geq OH^2$, with equality only for $P = H$.
4. *[1 pt]* The coordinate differences give $AB = AC = BC = 11\sqrt{2}$. The triangle is equilateral. Its height splits a side into two equal parts and equals

$$\sqrt{(11\sqrt{2})^2 - (\frac{11\sqrt{2}}{2})^2} = \frac{11\sqrt{6}}{2}.$$

   Its area is therefore $\frac{1}{2} \times 11\sqrt{2} \times \frac{11\sqrt{6}}{2} = \frac{121}{2}\sqrt{3}$ square units.
5. *[1 pt]* Base $ABC$: volume $\frac{(\frac{121}{2}\sqrt{3})(\frac{11}{\sqrt{3}})}{3} = \frac{1331}{6}$. Base $OAB$: area $\frac{121}{2}$ and height 11, the same value.

# Test 4 · Practice 2

**PRACTICE PAPER · TRAINING**

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete attempts will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · A logarithmic function — 5 points

Consider the function $f$ defined on $(0, +\infty)$ by $f(x) = \frac{2+\ln x}{x}$. We denote by $\mathcal{C}_f$ its graph.

1. Determine the limits of $f$ at $0^+$ and at $+\infty$. *(1 pt)*
2. Calculate $f'(x)$, then draw the table of variations of $f$. *(1 pt)*
3. State the value of the maximum of $f$, then solve $f(x) = 0$. *(1 pt)*
4. Study the convexity of $f$ and determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Prove that the equation $f(x) = \frac{1}{2}e^1$ has exactly two solutions in $(0, +\infty)$. *(1 pt)*

## Exercise 2 · Antiderivatives with a given condition — 5 points

The five questions are independent. For each antiderivative calculation, state the expression of the antiderivatives requested and check the result by differentiation.

1. Determine all antiderivatives on $\mathbb{R}$ of $x \mapsto 5x^2 - 7x + 3$. *(1 pt)*
2. Determine the antiderivative $F$ of $x \mapsto e^{5x}$ on $\mathbb{R}$ such that $F(0) = 7$. *(1 pt)*
3. Determine all antiderivatives on $\mathbb{R}$ of $x \mapsto \frac{2x}{x^2+7}$. *(1 pt)*
4. Determine an antiderivative on $\mathbb{R}$ of $x \mapsto 2x(x^2+3)^5$. *(1 pt)*
5. Let $F$ and $G$ be two antiderivatives of the same function on an interval $I$. Prove that there exists a real constant $C$ such that $F(x) - G(x) = C$ for all $x \in I$. *(1 pt)*

Test 4 · Practice 2 · continued

## Exercise 3 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{4}y + \frac{5}{2}$ on $[0, +\infty)$. In questions 2 to 5, denote by $y$ the solution satisfying $y(0) = 16$.

1. Determine the constant solution of $(E)$. *(1 pt)*
2. Determine the expression of the solution satisfying the given initial condition. *(1 pt)*
3. Determine the limit of $y(t)$ as $t$ tends to $+\infty$. *(1 pt)*
4. Study the variations of $y$, then determine the smallest real number $t \geq 0$ such that $y(t) \leq 11$. *(1 pt)*
5. Apply Euler's method with step size $h > 0$ and denote by $Y_k$ the approximation of $y(kh)$. Give the value of $Y_0$ and express $Y_{k+1}$ in terms of $Y_k$ and $h$. *(1 pt)*

## Exercise 4 · Projection and volume — 5 points

In an orthonormal coordinate system with origin $O$, consider $A(7,0,0)$, $B(0,7,0)$ and $C(0,0,7)$. We are given $\vec{n} = (1,1,1)$. Lengths, areas and volumes will be expressed in units of length, area and volume respectively.

1. Prove that $\vec{n}$ is normal to the plane $(ABC)$. *(1 pt)*
2. Let $H$ be the orthogonal projection of $O$ onto $(ABC)$. Determine the coordinates of $H$ using $\overrightarrow{AH} \cdot \vec{n} = 0$. *(1 pt)*
3. Prove that $OH$ is the distance from $O$ to the plane $(ABC)$: for every $P \in (ABC)$, prove that $OP^2 \geq OH^2$. *(1 pt)*
4. Calculate the area of triangle $ABC$. *(1 pt)*
5. Calculate the volume of the tetrahedron $OABC$ in two ways, by choosing two different bases. *(1 pt)*

# Solutions · Test 4 · Practice 2

## Exercise 1 · A logarithmic function

1. *[1 pt]* At $0^+$, the numerator tends to $-\infty$ and the denominator to $0^+$: limit $-\infty$; at $+\infty$, limit 0 by comparison of growth rates.

2. *[1 pt]* By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (2 + \ln x)}{x^2} = \frac{-1 - \ln x}{x^2}.$$

The denominator is positive. The numerator vanishes at $s = e^{-1}$, is positive for $0 < x < s$ and negative for $x > s$. Hence $f$ is strictly increasing on $]0, s]$, then strictly decreasing on $[s, +\infty[$.

3. *[1 pt]* The maximum is attained only at $s = e^{-1}$ and equals

$$f(s) = \frac{2 - 1}{e^{-1}} = e^1.$$

Since $x > 0$, $f(x) = 0$ is equivalent to $2 + \ln x = 0$, that is $x = e^{-2}$.

4. *[1 pt]* We differentiate $(-1 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-1 - \ln x)x^{-3} = \frac{2\ln x + 1}{x^3}.$$

The numerator vanishes at $t = e^{-\frac{1}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at $t$ and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. *[1 pt]* Let $m = \frac{e^1}{2}$ and $s = e^{-1}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at 0 and value $2m$ at $s$: the corollary of the intermediate value theorem (IVT) gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from $2m$ to $0 < m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

## Exercise 2 · Antiderivatives with a given condition

1. *[1 pt]* By linearity, the antiderivatives are $F(x) = \frac{5}{3}x^3 - \frac{7}{2}x^2 + 3x + C$, $C \in \mathbb{R}$. Indeed, $F'(x) = 5x^2 - 7x + 3$. The derivative does not depend on $C$, and all antiderivatives differ by a constant on $\mathbb{R}$.

2. *[1 pt]* The antiderivatives are $F(x) = \frac{e^{5x}}{5} + C$. The condition $F(0) = 7$ requires $\frac{1}{5} + C = 7$, hence $C = \frac{34}{5}$. The derivative of $\frac{e^{5x}}{5}$ is $e^{5x}$: the function obtained does satisfy both conditions.

3. *[1 pt]* Let $u(x) = x^2 + 7 > 0$ on $\mathbb{R}$; $u'(x) = 2x$. The antiderivatives are therefore $\ln(x^2 + 7) + C$, since their derivative equals $\frac{u'}{u} = \frac{2x}{x^2+7}$.

4. *[1 pt]* An antiderivative is $F(x) = \frac{(x^2 + 3)^6}{6}$. Indeed, by differentiating a composite function,

$$F'(x) = \frac{6}{6}\,2x(x^2 + 3)^5 = 2x(x^2 + 3)^5.$$

5. *[1 pt]* The derivative of their difference is zero on the interval; the difference is therefore constant. The fact that the domain is an interval is used in this result.

## Exercise 3 · Differential equation and evolution

1. *[1 pt]* $0 = -\frac{1}{4}y + \frac{5}{2}$ gives $y = 10$.

2. *[1 pt]* The constant solution is 10. Let $z = y - 10$; the equation becomes $z' = -\frac{z}{4}$. Its solutions are $z(t) = Ce^{-\frac{t}{4}}$, so all the solutions of the original equation are $y(t) = 10 + Ce^{-\frac{t}{4}}$. The condition $y(0) = 16$ gives $C = 6$. For the function obtained,

$$y'(t) = -\frac{3}{2}e^{-\frac{t}{4}} = -\frac{y(t)}{4} + \frac{5}{2}, \qquad y(0) = 16.$$

   The solution of the initial value problem is therefore $y(t) = 10 + 6e^{-\frac{t}{4}}$.

3. *[1 pt]* $y' = -\frac{3}{2}e^{-\frac{t}{4}} < 0$; limit 10.

4. *[1 pt]* Since $e^{-\frac{t}{4}} > 0$, the derivative $y'(t) = -\frac{3}{2}e^{-\frac{t}{4}}$ is negative: $y$ is strictly decreasing. We solve

$$\begin{aligned} y(t) \leq 11 &\iff 6e^{-\frac{t}{4}} \leq 1 \\ &\iff -\frac{t}{4} \leq -\ln 6 \\ &\iff t \geq 4\ln 6. \end{aligned}$$

   The last step reverses the inequality because we multiply by $-4 < 0$. The first time is $4\ln 6 \simeq 7.1670$.

5. *[1 pt]* $Y_0 = 16$, $Y_{j+1} = Y_j + h(-\frac{Y_j}{4} + \frac{5}{2})$, $t_j = jh$.

## Exercise 4 · Projection and volume

1. *[1 pt]* $\overrightarrow{AB} = (-7, 7, 0)$ and $\overrightarrow{AC} = (-7, 0, 7)$ are non-collinear and their dot products with $\vec{n}$ are zero.

2. *[1 pt]* The line normal to the plane passing through $O$ has direction vector $\vec{n} = (1, 1, 1)$. Its point $H$ is therefore written $(t, t, t)$. For $H$ to lie in $(ABC)$, it is necessary and sufficient that

$$\overrightarrow{AH} \cdot \vec{n} = (t - 7) + t + t = 3t - 7 = 0.$$

   Hence $t = \frac{7}{3}$ and $H = (\frac{7}{3}, \frac{7}{3}, \frac{7}{3})$. This point lies in the plane and $\overrightarrow{OH}$ is collinear with the normal vector; it is indeed the required orthogonal projection.

3. *[1 pt]* $OH = \frac{7}{\sqrt{3}}$; for every $P$ in the plane, $OP^2 = OH^2 + HP^2 \geq OH^2$, with equality only for $P = H$.

4. *[1 pt]* The coordinate differences give $AB = AC = BC = 7\sqrt{2}$. The triangle is equilateral. Its height splits a side into two equal parts and equals

$$\sqrt{(7\sqrt{2})^2 - (\frac{7\sqrt{2}}{2})^2} = \frac{7\sqrt{6}}{2}.$$

   Its area is therefore $\frac{1}{2} \times 7\sqrt{2} \times \frac{7\sqrt{6}}{2} = \frac{49}{2}\sqrt{3}$ square units.

5. *[1 pt]* Base $ABC$: volume $\frac{(\frac{49}{2}\sqrt{3})(\frac{7}{\sqrt{3}})}{3} = \frac{343}{6}$. Base $OAB$: area $\frac{49}{2}$ and height 7, the same value.

# Test 4 · A

**PAPER RESERVED FOR ASSESSMENT**

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete attempts will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · A logarithmic function — 5 points

Consider the function $f$ defined on $(0, +\infty)$ by $f(x) = \frac{5+\ln x}{x}$. We denote by $\mathcal{C}_f$ its graph.

1. Determine the limits of $f$ at $0^+$ and at $+\infty$. *(1 pt)*
2. Calculate $f'(x)$, then draw the table of variations of $f$. *(1 pt)*
3. State the value of the maximum of $f$, then solve $f(x) = 0$. *(1 pt)*
4. Study the convexity of $f$ and determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Prove that the equation $f(x) = \frac{1}{2}e^4$ has exactly two solutions in $(0, +\infty)$. *(1 pt)*

## Exercise 2 · Antiderivatives with a given condition — 5 points

The five questions are independent. For each antiderivative calculation, state the expression of the antiderivatives requested and check the result by differentiation.

1. Determine all antiderivatives on $\mathbb{R}$ of $x \mapsto 4x^2 - 4x + 1$. *(1 pt)*
2. Determine the antiderivative $F$ of $x \mapsto e^{4x}$ on $\mathbb{R}$ such that $F(0) = 4$. *(1 pt)*
3. Determine all antiderivatives on $\mathbb{R}$ of $x \mapsto \frac{2x}{x^2+4}$. *(1 pt)*
4. Determine an antiderivative on $\mathbb{R}$ of $x \mapsto 2x(x^2+1)^4$. *(1 pt)*
5. Let $F$ and $G$ be two antiderivatives of the same function on an interval $I$. Prove that there exists a real constant $C$ such that $F(x) - G(x) = C$ for all $x \in I$. *(1 pt)*

Test 4 · A · continued

## Exercise 3 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{3}y + \frac{7}{3}$ on $[0, +\infty)$. In questions 2 to 5, denote by $y$ the solution satisfying $y(0) = 11$.

1. Determine the constant solution of $(E)$. *(1 pt)*
2. Determine the expression of the solution satisfying the given initial condition. *(1 pt)*
3. Determine the limit of $y(t)$ as $t$ tends to $+\infty$. *(1 pt)*
4. Study the variations of $y$, then determine the smallest real number $t \geq 0$ such that $y(t) \leq 8$. *(1 pt)*
5. Apply Euler's method with step size $h > 0$ and denote by $Y_k$ the approximation of $y(kh)$. Give the value of $Y_0$ and express $Y_{k+1}$ in terms of $Y_k$ and $h$. *(1 pt)*

## Exercise 4 · Projection and volume — 5 points

In an orthonormal coordinate system with origin $O$, consider $A(8,0,0)$, $B(0,8,0)$ and $C(0,0,8)$. We are given $\vec{n} = (1,1,1)$. Lengths, areas and volumes will be expressed in units of length, area and volume respectively.

1. Prove that $\vec{n}$ is normal to the plane $(ABC)$. *(1 pt)*
2. Let $H$ be the orthogonal projection of $O$ onto $(ABC)$. Determine the coordinates of $H$ using $\overrightarrow{AH} \cdot \vec{n} = 0$. *(1 pt)*
3. Prove that $OH$ is the distance from $O$ to the plane $(ABC)$: for every $P \in (ABC)$, prove that $OP^2 \geq OH^2$. *(1 pt)*
4. Calculate the area of triangle $ABC$. *(1 pt)*
5. Calculate the volume of the tetrahedron $OABC$ in two ways, by choosing two different bases. *(1 pt)*

# Solutions · Test 4 · A

## Exercise 1 · A logarithmic function

1. *[1 pt]* At $0^+$, the numerator tends to $-\infty$ and the denominator to $0^+$: limit $-\infty$; at $+\infty$, limit 0 by comparison of growth rates.

2. *[1 pt]* By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (5 + \ln x)}{x^2} = \frac{-4 - \ln x}{x^2}.$$

The denominator is positive. The numerator vanishes at $s = e^{-4}$, is positive for $0 < x < s$ and negative for $x > s$. Hence $f$ is strictly increasing on $]0, s]$, then strictly decreasing on $[s, +\infty[$.

3. *[1 pt]* The maximum is attained only at $s = e^{-4}$ and equals

$$f(s) = \frac{5 - 4}{e^{-4}} = e^4.$$

Since $x > 0$, $f(x) = 0$ is equivalent to $5 + \ln x = 0$, that is $x = e^{-5}$.

4. *[1 pt]* We differentiate $(-4 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-4 - \ln x)x^{-3} = \frac{2\ln x + 7}{x^3}.$$

The numerator vanishes at $t = e^{-\frac{7}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at $t$ and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. *[1 pt]* Let $m = \frac{e^4}{2}$ and $s = e^{-4}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at 0 and value $2m$ at $s$: the corollary of the intermediate value theorem (IVT) gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from $2m$ to $0 < m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

## Exercise 2 · Antiderivatives with a given condition

1. *[1 pt]* By linearity, the antiderivatives are $F(x) = \frac{4}{3}x^3 - 2x^2 + x + C$, $C \in \mathbb{R}$. Indeed, $F'(x) = 4x^2 - 4x + 1$. The derivative does not depend on $C$, and all antiderivatives differ by a constant on $\mathbb{R}$.

2. *[1 pt]* The antiderivatives are $F(x) = \frac{e^{4x}}{4} + C$. The condition $F(0) = 4$ requires $\frac{1}{4} + C = 4$, hence $C = \frac{15}{4}$. The derivative of $\frac{e^{4x}}{4}$ is $e^{4x}$: the function obtained does satisfy both conditions.

3. *[1 pt]* Let $u(x) = x^2 + 4 > 0$ on $\mathbb{R}$; $u'(x) = 2x$. The antiderivatives are therefore $\ln(x^2 + 4) + C$, since their derivative equals $\frac{u'}{u} = \frac{2x}{x^2+4}$.

4. *[1 pt]* An antiderivative is $F(x) = \dfrac{(x^2 + 1)^5}{5}$. Indeed, by differentiating a composite function,

$$F'(x) = \frac{5}{5}\,2x(x^2 + 1)^4 = 2x(x^2 + 1)^4.$$

5. *[1 pt]* The derivative of their difference is zero on the interval; the difference is therefore constant. The fact that the domain is an interval is used in this result.

## Exercise 3 · Differential equation and evolution

1. *[1 pt]* $0 = -\frac{1}{3}y + \frac{7}{3}$ gives $y = 7$.

2. *[1 pt]* The constant solution is 7. Let $z = y - 7$; the equation becomes $z' = -\frac{z}{3}$. Its solutions are $z(t) = Ce^{-\frac{t}{3}}$, so all the solutions of the original equation are $y(t) = 7 + Ce^{-\frac{t}{3}}$. The condition $y(0) = 11$ gives $C = 4$. For the function obtained,

$$y'(t) = -\frac{4}{3}e^{-\frac{t}{3}} = -\frac{y(t)}{3} + \frac{7}{3}, \qquad y(0) = 11.$$

   The solution of the initial value problem is therefore $y(t) = 7 + 4e^{-\frac{t}{3}}$.

3. *[1 pt]* $y' = -\frac{4}{3}e^{-\frac{t}{3}} < 0$; limit 7.

4. *[1 pt]* Since $e^{-\frac{t}{3}} > 0$, the derivative $y'(t) = -\frac{4}{3}e^{-\frac{t}{3}}$ is negative: $y$ is strictly decreasing. We solve

$$\begin{aligned} y(t) \leq 8 &\iff 4e^{-\frac{t}{3}} \leq 1 \\ &\iff -\frac{t}{3} \leq -\ln 4 \\ &\iff t \geq 3\ln 4. \end{aligned}$$

   The last step reverses the inequality because we multiply by $-3 < 0$. The first time is $3\ln 4 \simeq 4.1589$.

5. *[1 pt]* $Y_0 = 11$, $Y_{j+1} = Y_j + h(-\frac{Y_j}{3} + \frac{7}{3})$, $t_j = jh$.

## Exercise 4 · Projection and volume

1. *[1 pt]* $\overrightarrow{AB} = (-8, 8, 0)$ and $\overrightarrow{AC} = (-8, 0, 8)$ are non-collinear and their dot products with $\vec{n}$ are zero.

2. *[1 pt]* The line normal to the plane passing through $O$ has direction vector $\vec{n} = (1, 1, 1)$. Its point $H$ is therefore written $(t, t, t)$. For $H$ to lie in $(ABC)$, it is necessary and sufficient that

$$\overrightarrow{AH} \cdot \vec{n} = (t - 8) + t + t = 3t - 8 = 0.$$

   Hence $t = \frac{8}{3}$ and $H = (\frac{8}{3}, \frac{8}{3}, \frac{8}{3})$. This point lies in the plane and $\overrightarrow{OH}$ is collinear with the normal vector; it is indeed the required orthogonal projection.

3. *[1 pt]* $OH = \frac{8}{\sqrt{3}}$; for every $P$ in the plane, $OP^2 = OH^2 + HP^2 \geq OH^2$, with equality only for $P = H$.

4. *[1 pt]* The coordinate differences give $AB = AC = BC = 8\sqrt{2}$. The triangle is equilateral. Its height splits a side into two equal parts and equals

$$\sqrt{(8\sqrt{2})^2 - (\frac{8\sqrt{2}}{2})^2} = \frac{8\sqrt{6}}{2}.$$

   Its area is therefore $\frac{1}{2} \times 8\sqrt{2} \times \frac{8\sqrt{6}}{2} = 32\sqrt{3}$ square units.

5. *[1 pt]* Base $ABC$: volume $\frac{(32\sqrt{3})(\frac{8}{\sqrt{3}})}{3} = \frac{256}{3}$. Base $OAB$: area 32 and height 8, the same value.

# Test 4 · B

**PAPER RESERVED FOR ASSESSMENT**

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete attempts will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · A logarithmic function — 5 points

Consider the function $f$ defined on $(0, +\infty)$ by $f(x) = \frac{4+\ln x}{x}$. We denote by $\mathcal{C}_f$ its graph.

1. Determine the limits of $f$ at $0^+$ and at $+\infty$. *(1 pt)*
2. Calculate $f'(x)$, then draw the table of variations of $f$. *(1 pt)*
3. State the value of the maximum of $f$, then solve $f(x) = 0$. *(1 pt)*
4. Study the convexity of $f$ and determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Prove that the equation $f(x) = \frac{1}{2}e^3$ has exactly two solutions in $(0, +\infty)$. *(1 pt)*

## Exercise 2 · Antiderivatives with a given condition — 5 points

The five questions are independent. For each antiderivative calculation, state the expression of the antiderivatives requested and check the result by differentiation.

1. Determine all antiderivatives on $\mathbb{R}$ of $x \mapsto 3x^2 - 6x + 2$. *(1 pt)*
2. Determine the antiderivative $F$ of $x \mapsto e^{3x}$ on $\mathbb{R}$ such that $F(0) = 6$. *(1 pt)*
3. Determine all antiderivatives on $\mathbb{R}$ of $x \mapsto \frac{2x}{x^2+6}$. *(1 pt)*
4. Determine an antiderivative on $\mathbb{R}$ of $x \mapsto 2x(x^2+2)^3$. *(1 pt)*
5. Let $F$ and $G$ be two antiderivatives of the same function on an interval $I$. Prove that there exists a real constant $C$ such that $F(x) - G(x) = C$ for all $x \in I$. *(1 pt)*

Test 4 · B · continued

## Exercise 3 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{2}y + \frac{9}{2}$ on $[0, +\infty)$. In questions 2 to 5, denote by $y$ the solution satisfying $y(0) = 14$.

1. Determine the constant solution of $(E)$. *(1 pt)*
2. Determine the expression of the solution satisfying the given initial condition. *(1 pt)*
3. Determine the limit of $y(t)$ as $t$ tends to $+\infty$. *(1 pt)*
4. Study the variations of $y$, then determine the smallest real number $t \geq 0$ such that $y(t) \leq 10$. *(1 pt)*
5. Apply Euler's method with step size $h > 0$ and denote by $Y_k$ the approximation of $y(kh)$. Give the value of $Y_0$ and express $Y_{k+1}$ in terms of $Y_k$ and $h$. *(1 pt)*

## Exercise 4 · Projection and volume — 5 points

In an orthonormal coordinate system with origin $O$, consider $A(12, 0, 0)$, $B(0, 12, 0)$ and $C(0, 0, 12)$. We are given $\vec{n} = (1, 1, 1)$. Lengths, areas and volumes will be expressed in units of length, area and volume respectively.

1. Prove that $\vec{n}$ is normal to the plane $(ABC)$. *(1 pt)*
2. Let $H$ be the orthogonal projection of $O$ onto $(ABC)$. Determine the coordinates of $H$ using $\overrightarrow{AH} \cdot \vec{n} = 0$. *(1 pt)*
3. Prove that $OH$ is the distance from $O$ to the plane $(ABC)$: for every $P \in (ABC)$, prove that $OP^2 \geq OH^2$. *(1 pt)*
4. Calculate the area of triangle $ABC$. *(1 pt)*
5. Calculate the volume of the tetrahedron $OABC$ in two ways, by choosing two different bases. *(1 pt)*

# Solutions · Test 4 · B

## Exercise 1 · A logarithmic function

1. *[1 pt]* At $0^+$, the numerator tends to $-\infty$ and the denominator to $0^+$: limit $-\infty$; at $+\infty$, limit 0 by comparison of growth rates.

2. *[1 pt]* By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (4 + \ln x)}{x^2} = \frac{-3 - \ln x}{x^2}.$$

The denominator is positive. The numerator vanishes at $s = e^{-3}$, is positive for $0 < x < s$ and negative for $x > s$. Hence $f$ is strictly increasing on $]0, s]$, then strictly decreasing on $[s, +\infty[$.

3. *[1 pt]* The maximum is attained only at $s = e^{-3}$ and equals

$$f(s) = \frac{4 - 3}{e^{-3}} = e^3.$$

Since $x > 0$, $f(x) = 0$ is equivalent to $4 + \ln x = 0$, that is $x = e^{-4}$.

4. *[1 pt]* We differentiate $(-3 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-3 - \ln x)x^{-3} = \frac{2\ln x + 5}{x^3}.$$

The numerator vanishes at $t = e^{-\frac{5}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at $t$ and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. *[1 pt]* Let $m = \frac{e^3}{2}$ and $s = e^{-3}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at 0 and value $2m$ at $s$: the corollary of the intermediate value theorem (IVT) gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from $2m$ to $0 < m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

## Exercise 2 · Antiderivatives with a given condition

1. *[1 pt]* By linearity, the antiderivatives are $F(x) = x^3 - 3x^2 + 2x + C$, $C \in \mathbb{R}$. Indeed, $F'(x) = 3x^2 - 6x + 2$. The derivative does not depend on $C$, and all antiderivatives differ by a constant on $\mathbb{R}$.

2. *[1 pt]* The antiderivatives are $F(x) = \frac{e^{3x}}{3} + C$. The condition $F(0) = 6$ requires $\frac{1}{3} + C = 6$, hence $C = \frac{17}{3}$. The derivative of $\frac{e^{3x}}{3}$ is $e^{3x}$: the function obtained does satisfy both conditions.

3. *[1 pt]* Let $u(x) = x^2 + 6 > 0$ on $\mathbb{R}$; $u'(x) = 2x$. The antiderivatives are therefore $\ln(x^2 + 6) + C$, since their derivative equals $\frac{u'}{u} = \frac{2x}{x^2+6}$.

4. *[1 pt]* An antiderivative is $F(x) = \frac{(x^2+2)^4}{4}$. Indeed, by differentiating a composite function,

$$F'(x) = \frac{4}{4}\,2x(x^2 + 2)^3 = 2x(x^2 + 2)^3.$$

5. *[1 pt]* The derivative of their difference is zero on the interval; the difference is therefore constant. The fact that the domain is an interval is used in this result.

## Exercise 3 · Differential equation and evolution

1. *[1 pt]* $0 = -\frac{1}{2}y + \frac{9}{2}$ gives $y = 9$.

2. *[1 pt]* The constant solution is 9. Let $z = y - 9$; the equation becomes $z' = -\frac{z}{2}$. Its solutions are $z(t) = Ce^{-\frac{t}{2}}$, so all the solutions of the original equation are $y(t) = 9 + Ce^{-\frac{t}{2}}$. The condition $y(0) = 14$ gives $C = 5$. For the function obtained,

$$y'(t) = -\frac{5}{2}e^{-\frac{t}{2}} = -\frac{y(t)}{2} + \frac{9}{2}, \qquad y(0) = 14.$$

   The solution of the initial value problem is therefore $y(t) = 9 + 5e^{-\frac{t}{2}}$.

3. *[1 pt]* $y' = -\frac{5}{2}e^{-\frac{t}{2}} < 0$; limit 9.

4. *[1 pt]* Since $e^{-\frac{t}{2}} > 0$, the derivative $y'(t) = -\frac{5}{2}e^{-\frac{t}{2}}$ is negative: $y$ is strictly decreasing. We solve

$$\begin{aligned} y(t) \leq 10 &\iff 5e^{-\frac{t}{2}} \leq 1 \\ &\iff -\frac{t}{2} \leq -\ln 5 \\ &\iff t \geq 2\ln 5. \end{aligned}$$

   The last step reverses the inequality because we multiply by $-2 < 0$. The first time is $2\ln 5 \simeq 3.2189$.

5. *[1 pt]* $Y_0 = 14$, $Y_{j+1} = Y_j + h(-\frac{Y_j}{2} + \frac{9}{2})$, $t_j = jh$.

## Exercise 4 · Projection and volume

1. *[1 pt]* $\overrightarrow{AB} = (-12, 12, 0)$ and $\overrightarrow{AC} = (-12, 0, 12)$ are non-collinear and their dot products with $\vec{n}$ are zero.

2. *[1 pt]* The line normal to the plane passing through $O$ has direction vector $\vec{n} = (1, 1, 1)$. Its point $H$ is therefore written $(t, t, t)$. For $H$ to lie in $(ABC)$, it is necessary and sufficient that

$$\overrightarrow{AH} \cdot \vec{n} = (t - 12) + t + t = 3t - 12 = 0.$$

   Hence $t = 4$ and $H = (4, 4, 4)$. This point lies in the plane and $\overrightarrow{OH}$ is collinear with the normal vector; it is indeed the required orthogonal projection.

3. *[1 pt]* $OH = \frac{12}{\sqrt{3}}$; for every $P$ in the plane, $OP^2 = OH^2 + HP^2 \geq OH^2$, with equality only for $P = H$.

4. *[1 pt]* The coordinate differences give $AB = AC = BC = 12\sqrt{2}$. The triangle is equilateral. Its height splits a side into two equal parts and equals

$$\sqrt{(12\sqrt{2})^2 - (\frac{12\sqrt{2}}{2})^2} = \frac{12\sqrt{6}}{2}.$$

   Its area is therefore $\frac{1}{2} \times 12\sqrt{2} \times \frac{12\sqrt{6}}{2} = 72\sqrt{3}$ square units.

5. *[1 pt]* Base $ABC$: volume $\frac{(72\sqrt{3})(\frac{12}{\sqrt{3}})}{3} = 288$. Base $OAB$: area 72 and height 12, the same value.

# Test 5 · Practice 1

**PRACTICE PAPER · TRAINING**

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete attempts will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · Integral and sequence — 5 points

For every natural number $n$, we set $I_n = \displaystyle\int_0^1 x^n e^{4x}\,\mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^{4x}$.

1. Calculate $I_0$. *(1 pt)*
2. Using integration by parts, calculate $I_1$. *(1 pt)*
3. Prove, by comparing the integrands, that the sequence $(I_n)$ is positive and decreasing. *(1 pt)*
4. Show that $I_n \leq \frac{e^4}{n+1}$ for all $n \in \mathbb{N}$. Deduce the limit of $(I_n)$. *(1 pt)*
5. Using integration by parts, express $I_{n+1}$ in terms of $I_n$ and $n$. *(1 pt)*

## Exercise 2 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{3}y + \frac{8}{3}$ on $[0, +\infty)$. In questions 2 to 5, denote by $y$ the solution satisfying $y(0) = 12$.

1. Determine the constant solution of $(E)$. *(1 pt)*
2. Determine the expression of the solution satisfying the given initial condition. *(1 pt)*
3. Determine the limit of $y(t)$ as $t$ tends to $+\infty$. *(1 pt)*
4. Study the variations of $y$, then determine the smallest real number $t \geq 0$ such that $y(t) \leq 9$. *(1 pt)*
5. Apply Euler's method with step size $h > 0$ and denote by $Y_k$ the approximation of $y(kh)$. Give the value of $Y_0$ and express $Y_{k+1}$ in terms of $Y_k$ and $h$. *(1 pt)*

Test 5 · Practice 1 · continued

## Exercise 3 · Projection and volume — 5 points

In an orthonormal coordinate system with origin $O$, consider $A(9,0,0)$, $B(0,9,0)$ and $C(0,0,9)$. We are given $\vec{n} = (1,1,1)$. Lengths, areas and volumes will be expressed in units of length, area and volume respectively.

1. Prove that $\vec{n}$ is normal to the plane $(ABC)$. *(1 pt)*
2. Let $H$ be the orthogonal projection of $O$ onto $(ABC)$. Determine the coordinates of $H$ using $\overrightarrow{AH} \cdot \vec{n} = 0$. *(1 pt)*
3. Prove that $OH$ is the distance from $O$ to the plane $(ABC)$: for every $P \in (ABC)$, prove that $OP^2 \geq OH^2$. *(1 pt)*
4. Calculate the area of triangle $ABC$. *(1 pt)*
5. Calculate the volume of the tetrahedron $OABC$ in two ways, by choosing two different bases. *(1 pt)*

## Exercise 4 · Sums of random variables — 5 points

We set $n = 17$. Consider $n$ independent random variables $X_1, \ldots, X_{17}$, which all follow the Bernoulli distribution with parameter $p = \frac{1}{5}$. We set $S = \sum_{i=1}^{17} X_i$ and $G = 9S - 17$.

1. Justify the distribution followed by $S$ and calculate $E(S)$. *(1 pt)*
2. Using the variance of a Bernoulli random variable and independence, calculate $V(S)$. *(1 pt)*
3. Calculate $E(G)$ and $V(G)$. *(1 pt)*
4. We set $M = \frac{S}{17}$. Calculate $E(M)$ and $V(M)$. *(1 pt)*
5. For this question only, we replace the 17 variables by 17 variables all equal to $X_1$. Then calculate the variance of their mean and compare it with that of question 4. Explain why the results differ. *(1 pt)*

# Solutions · Test 5 · Practice 1

## Exercise 1 · Integral and sequence

1. *[1 pt]* The function $x \mapsto \frac{e^{4x}}{4}$ is an antiderivative of $e^{4x}$. Hence $I_0 = [\frac{e^{4x}}{4}]_0^1 = \frac{e^4-1}{4}$.

2. *[1 pt]* Let us choose $u(x) = x$ and $v(x) = \frac{e^{4x}}{4}$, two differentiable functions with continuous derivatives on $[0,1]$. Then $u' = 1$ and $v' = e^{4x}$. Integration by parts gives

$$\begin{aligned} I_1 &= \left[\frac{xe^{4x}}{4}\right]_0^1 - \frac{1}{4}\int_0^1 e^{4x}\,\mathrm{d}x \\ &= \frac{e^4}{4} - \frac{e^4-1}{16} = \frac{3e^4+1}{16}. \end{aligned}$$

3. *[1 pt]* For $x \in [0,1]$, $0 \le x^{n+1} \le x^n$. Since $e^{4x} > 0$, integrating over increasing bounds gives $0 \le I_{n+1} \le I_n$. The sequence is therefore positive and weakly decreasing. The inequalities are even strict: the integrands of $I_n$ and $I_n - I_{n+1}$ are continuous and strictly positive on $]0,1[$.

4. *[1 pt]* $e^{4x} \le e^4$; hence $0 \le I_n \le e^4 \int_0^1 x^n\,\mathrm{d}x = \frac{e^4}{n+1} \to 0$.

5. *[1 pt]* Apply integration by parts to $u(x) = x^{n+1}$ and $v(x) = \frac{e^{4x}}{4}$. Since $n+1 \ge 1$, the term at $x = 0$ is zero. Hence

$$\begin{aligned} I_{n+1} &= \left[\frac{x^{n+1}e^{4x}}{4}\right]_0^1 - \frac{n+1}{4}\int_0^1 x^n e^{4x}\,\mathrm{d}x \\ &= \frac{e^4}{4} - \frac{n+1}{4} I_n. \end{aligned}$$

## Exercise 2 · Differential equation and evolution

1. *[1 pt]* $0 = -\frac{1}{3}y + \frac{8}{3}$ gives $y = 8$.

2. *[1 pt]* The constant solution is 8. Let $z = y - 8$; the equation becomes $z' = -\frac{z}{3}$. Its solutions are $z(t) = Ce^{-\frac{t}{3}}$, so all the solutions of the original equation are $y(t) = 8 + Ce^{-\frac{t}{3}}$. The condition $y(0) = 12$ gives $C = 4$. For the function obtained,

$$y'(t) = -\frac{4}{3}e^{-\frac{t}{3}} = -\frac{y(t)}{3} + \frac{8}{3}, \qquad y(0) = 12.$$

   The solution of the initial value problem is therefore $y(t) = 8 + 4e^{-\frac{t}{3}}$.

3. *[1 pt]* $y' = -\frac{4}{3}e^{-\frac{t}{3}} < 0$; limit 8.

4. *[1 pt]* Since $e^{-\frac{t}{3}} > 0$, the derivative $y'(t) = -\frac{4}{3}e^{-\frac{t}{3}}$ is negative: $y$ is strictly decreasing. We solve

$$\begin{aligned} y(t) \le 9 &\iff 4e^{-\frac{t}{3}} \le 1 \\ &\iff -\frac{t}{3} \le -\ln 4 \\ &\iff t \ge 3\ln 4. \end{aligned}$$

   The last step reverses the inequality because we multiply by $-3 < 0$. The first time is $3\ln 4 \simeq 4.1589$.

5. *[1 pt]* $Y_0 = 12$, $Y_{j+1} = Y_j + h(-\frac{Y_j}{3} + \frac{8}{3})$, $t_j = jh$.

## Exercise 3 · Projection and volume

1. *[1 pt]* $\overrightarrow{AB} = (-9,9,0)$ and $\overrightarrow{AC} = (-9,0,9)$ are non-collinear and their dot products with $\vec{n}$ are zero.

2. *[1 pt]* The line normal to the plane passing through $O$ has direction vector $\vec{n} = (1,1,1)$. Its point $H$ is therefore written $(t,t,t)$. For $H$ to lie in $(ABC)$, it is necessary and sufficient that

$$\overrightarrow{AH} \cdot \vec{n} = (t-9) + t + t = 3t - 9 = 0.$$

   Hence $t = 3$ and $H = (3,3,3)$. This point lies in the plane and $\overrightarrow{OH}$ is collinear with the normal vector; it is indeed the required orthogonal projection.

3. *[1 pt]* $OH = \frac{9}{\sqrt{3}}$; for every $P$ in the plane, $OP^2 = OH^2 + HP^2 \geq OH^2$, with equality only for $P = H$.

4. *[1 pt]* The coordinate differences give $AB = AC = BC = 9\sqrt{2}$. The triangle is equilateral. Its height splits a side into two equal parts and equals

$$\sqrt{(9\sqrt{2})^2 - (\frac{9\sqrt{2}}{2})^2} = \frac{9\sqrt{6}}{2}.$$

   Its area is therefore $\frac{1}{2} \times 9\sqrt{2} \times \frac{9\sqrt{6}}{2} = \frac{81}{2}\sqrt{3}$ square units.

5. *[1 pt]* Base $ABC$: volume $\frac{(\frac{81}{2}\sqrt{3})(\frac{9}{\sqrt{3}})}{3} = \frac{243}{2}$. Base $OAB$: area $\frac{81}{2}$ and height 9, the same value.

## Exercise 4 · Sums of random variables

1. *[1 pt]* $S \sim \mathcal{B}(17, \frac{1}{5})$; $E(S) = \frac{17}{5}$.

2. *[1 pt]* Each $X_i$ equals 0 or 1, so $X_i^2 = X_i$. Hence $E(X_i) = p = \frac{1}{5}$ and

$$V(X_i) = E(X_i^2) - E(X_i)^2 = p - p^2 = p(1-p) = \frac{4}{25}.$$

   The variables are independent; additivity of the variance gives

$$V(S) = \sum_{i=1}^{17} V(X_i) = 17p(1-p) = \frac{68}{25}.$$

3. *[1 pt]* By linearity of the expected value and by $V(\lambda S + \mu) = \lambda^2 V(S)$,

$$E(G) = 9E(S) - 17 = \frac{68}{5}, \qquad V(G) = 81V(S) = \frac{5508}{25}.$$

   The constant $-17$ does not change the variance.

4. *[1 pt]* $E(M) = \frac{1}{5}$ and $V(M) = \frac{4}{425}$; the variance factor is $\frac{1}{n^2}$.

5. *[1 pt]* The mean of 17 variables all equal to $X_1$ is $X_1$ itself: its variance is $\frac{4}{25}$, that is 17 times the variance $\frac{4}{425}$ found in question 4. The results differ because the additivity of variance used in question 4 requires independence, which no longer holds here.

# Test 5 · Practice 2

**PRACTICE PAPER · TRAINING**

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete attempts will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · Integral and sequence — 5 points

For every natural number $n$, we set $I_n = \displaystyle\int_0^1 x^n e^{3x}\,\mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^{3x}$.

1. Calculate $I_0$. *(1 pt)*
2. Using integration by parts, calculate $I_1$. *(1 pt)*
3. Prove, by comparing the integrands, that the sequence $(I_n)$ is positive and decreasing. *(1 pt)*
4. Show that $I_n \leq \frac{e^3}{n+1}$ for all $n \in \mathbb{N}$. Deduce the limit of $(I_n)$. *(1 pt)*
5. Using integration by parts, express $I_{n+1}$ in terms of $I_n$ and $n$. *(1 pt)*

## Exercise 2 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{2}y + 5$ on $[0, +\infty)$. In questions 2 to 5, denote by $y$ the solution satisfying $y(0) = 15$.

1. Determine the constant solution of $(E)$. *(1 pt)*
2. Determine the expression of the solution satisfying the given initial condition. *(1 pt)*
3. Determine the limit of $y(t)$ as $t$ tends to $+\infty$. *(1 pt)*
4. Study the variations of $y$, then determine the smallest real number $t \geq 0$ such that $y(t) \leq 11$. *(1 pt)*
5. Apply Euler's method with step size $h > 0$ and denote by $Y_k$ the approximation of $y(kh)$. Give the value of $Y_0$ and express $Y_{k+1}$ in terms of $Y_k$ and $h$. *(1 pt)*

Test 5 · Practice 2 · continued

## Exercise 3 · Projection and volume — 5 points

In an orthonormal coordinate system with origin $O$, consider $A(13,0,0)$, $B(0,13,0)$ and $C(0,0,13)$. We are given $\vec{n} = (1,1,1)$. Lengths, areas and volumes will be expressed in units of length, area and volume respectively.

1. Prove that $\vec{n}$ is normal to the plane $(ABC)$. *(1 pt)*
2. Let $H$ be the orthogonal projection of $O$ onto $(ABC)$. Determine the coordinates of $H$ using $\overrightarrow{AH} \cdot \vec{n} = 0$. *(1 pt)*
3. Prove that $OH$ is the distance from $O$ to the plane $(ABC)$: for every $P \in (ABC)$, prove that $OP^2 \geq OH^2$. *(1 pt)*
4. Calculate the area of triangle $ABC$. *(1 pt)*
5. Calculate the volume of the tetrahedron $OABC$ in two ways, by choosing two different bases. *(1 pt)*

## Exercise 4 · Sums of random variables — 5 points

We set $n = 14$. Consider $n$ independent random variables $X_1, \ldots, X_{14}$, which all follow the Bernoulli distribution with parameter $p = \frac{1}{4}$. We set $S = \sum_{i=1}^{14} X_i$ and $G = 6S - 14$.

1. Justify the distribution followed by $S$ and calculate $E(S)$. *(1 pt)*
2. Using the variance of a Bernoulli random variable and independence, calculate $V(S)$. *(1 pt)*
3. Calculate $E(G)$ and $V(G)$. *(1 pt)*
4. We set $M = \frac{S}{14}$. Calculate $E(M)$ and $V(M)$. *(1 pt)*
5. For this question only, we replace the 14 variables by 14 variables all equal to $X_1$. Then calculate the variance of their mean and compare it with that of question 4. Explain why the results differ. *(1 pt)*

# Solutions · Test 5 · Practice 2

## Exercise 1 · Integral and sequence

1. *[1 pt]* The function $x \mapsto \frac{e^{3x}}{3}$ is an antiderivative of $e^{3x}$. Hence $I_0 = [\frac{e^{3x}}{3}]_0^1 = \frac{e^3-1}{3}$.

2. *[1 pt]* Let us choose $u(x) = x$ and $v(x) = \frac{e^{3x}}{3}$, two differentiable functions with continuous derivatives on $[0,1]$. Then $u' = 1$ and $v' = e^{3x}$. Integration by parts gives

$$\begin{aligned} I_1 &= \left[\frac{xe^{3x}}{3}\right]_0^1 - \frac{1}{3}\int_0^1 e^{3x}\,\mathrm{d}x \\ &= \frac{e^3}{3} - \frac{e^3-1}{9} = \frac{2e^3+1}{9}. \end{aligned}$$

3. *[1 pt]* For $x \in [0,1]$, $0 \leq x^{n+1} \leq x^n$. Since $e^{3x} > 0$, integrating over increasing bounds gives $0 \leq I_{n+1} \leq I_n$. The sequence is therefore positive and weakly decreasing. The inequalities are even strict: the integrands of $I_n$ and $I_n - I_{n+1}$ are continuous and strictly positive on $]0,1[$.

4. *[1 pt]* $e^{3x} \leq e^3$; hence $0 \leq I_n \leq e^3 \int_0^1 x^n\,\mathrm{d}x = \frac{e^3}{n+1} \to 0$.

5. *[1 pt]* Apply integration by parts to $u(x) = x^{n+1}$ and $v(x) = \frac{e^{3x}}{3}$. Since $n+1 \geq 1$, the term at $x = 0$ is zero. Hence

$$\begin{aligned} I_{n+1} &= \left[\frac{x^{n+1}e^{3x}}{3}\right]_0^1 - \frac{n+1}{3}\int_0^1 x^n e^{3x}\,\mathrm{d}x \\ &= \frac{e^3}{3} - \frac{n+1}{3}I_n. \end{aligned}$$

## Exercise 2 · Differential equation and evolution

1. *[1 pt]* $0 = -\frac{1}{2}y + 5$ gives $y = 10$.

2. *[1 pt]* The constant solution is 10. Let $z = y - 10$; the equation becomes $z' = -\frac{z}{2}$. Its solutions are $z(t) = Ce^{-\frac{t}{2}}$, so all the solutions of the original equation are $y(t) = 10 + Ce^{-\frac{t}{2}}$. The condition $y(0) = 15$ gives $C = 5$. For the function obtained,

$$y'(t) = -\frac{5}{2}e^{-\frac{t}{2}} = -\frac{y(t)}{2} + 5, \qquad y(0) = 15.$$

   The solution of the initial value problem is therefore $y(t) = 10 + 5e^{-\frac{t}{2}}$.

3. *[1 pt]* $y' = -\frac{5}{2}e^{-\frac{t}{2}} < 0$; limit 10.

4. *[1 pt]* Since $e^{-\frac{t}{2}} > 0$, the derivative $y'(t) = -\frac{5}{2}e^{-\frac{t}{2}}$ is negative: $y$ is strictly decreasing. We solve

$$\begin{aligned} y(t) \leq 11 &\iff 5e^{-\frac{t}{2}} \leq 1 \\ &\iff -\frac{t}{2} \leq -\ln 5 \\ &\iff t \geq 2\ln 5. \end{aligned}$$

   The last step reverses the inequality because we multiply by $-2 < 0$. The first time is $2\ln 5 \simeq 3.2189$.

5. *[1 pt]* $Y_0 = 15$, $Y_{j+1} = Y_j + h(-\frac{Y_j}{2} + 5)$, $t_j = jh$.

## Exercise 3 · Projection and volume

1. *[1 pt]* $\overrightarrow{AB} = (-13, 13, 0)$ and $\overrightarrow{AC} = (-13, 0, 13)$ are non-collinear and their dot products with $\vec{n}$ are zero.

2. *[1 pt]* The line normal to the plane passing through $O$ has direction vector $\vec{n} = (1, 1, 1)$. Its point $H$ is therefore written $(t, t, t)$. For $H$ to lie in $(ABC)$, it is necessary and sufficient that

$$\overrightarrow{AH} \cdot \vec{n} = (t - 13) + t + t = 3t - 13 = 0.$$

   Hence $t = \frac{13}{3}$ and $H = (\frac{13}{3}, \frac{13}{3}, \frac{13}{3})$. This point lies in the plane and $\overrightarrow{OH}$ is collinear with the normal vector; it is indeed the required orthogonal projection.

3. *[1 pt]* $OH = \frac{13}{\sqrt{3}}$; for every $P$ in the plane, $OP^2 = OH^2 + HP^2 \geq OH^2$, with equality only for $P = H$.

4. *[1 pt]* The coordinate differences give $AB = AC = BC = 13\sqrt{2}$. The triangle is equilateral. Its height splits a side into two equal parts and equals

$$\sqrt{(13\sqrt{2})^2 - (\frac{13\sqrt{2}}{2})^2} = \frac{13\sqrt{6}}{2}.$$

   Its area is therefore $\frac{1}{2} \times 13\sqrt{2} \times \frac{13\sqrt{6}}{2} = \frac{169}{2}\sqrt{3}$ square units.

5. *[1 pt]* Base $ABC$: volume $\frac{(\frac{169}{2}\sqrt{3})(\frac{13}{\sqrt{3}})}{3} = \frac{2197}{6}$. Base $OAB$: area $\frac{169}{2}$ and height 13, the same value.

## Exercise 4 · Sums of random variables

1. *[1 pt]* $S \sim \mathcal{B}(14, \frac{1}{4})$; $E(S) = \frac{7}{2}$.

2. *[1 pt]* Each $X_i$ equals 0 or 1, so $X_i^2 = X_i$. Hence $E(X_i) = p = \frac{1}{4}$ and

$$V(X_i) = E(X_i^2) - E(X_i)^2 = p - p^2 = p(1 - p) = \frac{3}{16}.$$

   The variables are independent; additivity of the variance gives

$$V(S) = \sum_{i=1}^{14} V(X_i) = 14p(1 - p) = \frac{21}{8}.$$

3. *[1 pt]* By linearity of the expected value and by $V(\lambda S + \mu) = \lambda^2 V(S)$,

$$E(G) = 6E(S) - 14 = 7, \qquad V(G) = 36V(S) = \frac{189}{2}.$$

   The constant $-14$ does not change the variance.

4. *[1 pt]* $E(M) = \frac{1}{4}$ and $V(M) = \frac{3}{224}$; the variance factor is $\frac{1}{n^2}$.

5. *[1 pt]* The mean of 14 variables all equal to $X_1$ is $X_1$ itself: its variance is $\frac{3}{16}$, that is 14 times the variance $\frac{3}{224}$ found in question 4. The results differ because the additivity of variance used in question 4 requires independence, which no longer holds here.

# Test 5 · A

**PAPER RESERVED FOR ASSESSMENT**

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete attempts will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · Integral and sequence — 5 points

For every natural number $n$, we set $I_n = \displaystyle\int_0^1 x^n e^{2x}\,\mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^{2x}$.

1. Calculate $I_0$. *(1 pt)*
2. Using integration by parts, calculate $I_1$. *(1 pt)*
3. Prove, by comparing the integrands, that the sequence $(I_n)$ is positive and decreasing. *(1 pt)*
4. Show that $I_n \leq \frac{e^2}{n+1}$ for all $n \in \mathbb{N}$. Deduce the limit of $(I_n)$. *(1 pt)*
5. Using integration by parts, express $I_{n+1}$ in terms of $I_n$ and $n$. *(1 pt)*

## Exercise 2 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{5}y + \frac{6}{5}$ on $[0, +\infty)$. In questions 2 to 5, denote by $y$ the solution satisfying $y(0) = 12$.

1. Determine the constant solution of $(E)$. *(1 pt)*
2. Determine the expression of the solution satisfying the given initial condition. *(1 pt)*
3. Determine the limit of $y(t)$ as $t$ tends to $+\infty$. *(1 pt)*
4. Study the variations of $y$, then determine the smallest real number $t \geq 0$ such that $y(t) \leq 7$. *(1 pt)*
5. Apply Euler's method with step size $h > 0$ and denote by $Y_k$ the approximation of $y(kh)$. Give the value of $Y_0$ and express $Y_{k+1}$ in terms of $Y_k$ and $h$. *(1 pt)*

Test 5 · A · continued

## Exercise 3 · Projection and volume — 5 points

In an orthonormal coordinate system with origin $O$, consider $A(10,0,0)$, $B(0,10,0)$ and $C(0,0,10)$. We are given $\vec{n} = (1,1,1)$. Lengths, areas and volumes will be expressed in units of length, area and volume respectively.

1. Prove that $\vec{n}$ is normal to the plane $(ABC)$. *(1 pt)*
2. Let $H$ be the orthogonal projection of $O$ onto $(ABC)$. Determine the coordinates of $H$ using $\overrightarrow{AH} \cdot \vec{n} = 0$. *(1 pt)*
3. Prove that $OH$ is the distance from $O$ to the plane $(ABC)$: for every $P \in (ABC)$, prove that $OP^2 \geq OH^2$. *(1 pt)*
4. Calculate the area of triangle $ABC$. *(1 pt)*
5. Calculate the volume of the tetrahedron $OABC$ in two ways, by choosing two different bases. *(1 pt)*

## Exercise 4 · Sums of random variables — 5 points

We set $n = 16$. Consider $n$ independent random variables $X_1, \ldots, X_{16}$, which all follow the Bernoulli distribution with parameter $p = \frac{1}{3}$. We set $S = \sum_{i=1}^{16} X_i$ and $G = 8S - 16$.

1. Justify the distribution followed by $S$ and calculate $E(S)$. *(1 pt)*
2. Using the variance of a Bernoulli random variable and independence, calculate $V(S)$. *(1 pt)*
3. Calculate $E(G)$ and $V(G)$. *(1 pt)*
4. We set $M = \frac{S}{16}$. Calculate $E(M)$ and $V(M)$. *(1 pt)*
5. For this question only, we replace the 16 variables by 16 variables all equal to $X_1$. Then calculate the variance of their mean and compare it with that of question 4. Explain why the results differ. *(1 pt)*

# Solutions · Test 5 · A

## Exercise 1 · Integral and sequence

1. *[1 pt]* The function $x \mapsto \frac{e^{2x}}{2}$ is an antiderivative of $e^{2x}$. Hence $I_0 = [\frac{e^{2x}}{2}]_0^1 = \frac{e^2-1}{2}$.

2. *[1 pt]* Let us choose $u(x) = x$ and $v(x) = \frac{e^{2x}}{2}$, two differentiable functions with continuous derivatives on $[0,1]$. Then $u' = 1$ and $v' = e^{2x}$. Integration by parts gives

$$\begin{aligned} I_1 &= \left[\frac{xe^{2x}}{2}\right]_0^1 - \frac{1}{2}\int_0^1 e^{2x}\,\mathrm{d}x \\ &= \frac{e^2}{2} - \frac{e^2-1}{4} = \frac{e^2+1}{4}. \end{aligned}$$

3. *[1 pt]* For $x \in [0,1]$, $0 \leq x^{n+1} \leq x^n$. Since $e^{2x} > 0$, integrating over increasing bounds gives $0 \leq I_{n+1} \leq I_n$. The sequence is therefore positive and weakly decreasing. The inequalities are even strict: the integrands of $I_n$ and $I_n - I_{n+1}$ are continuous and strictly positive on $]0,1[$.

4. *[1 pt]* $e^{2x} \leq e^2$; hence $0 \leq I_n \leq e^2 \int_0^1 x^n\,\mathrm{d}x = \frac{e^2}{n+1} \to 0$.

5. *[1 pt]* Apply integration by parts to $u(x) = x^{n+1}$ and $v(x) = \frac{e^{2x}}{2}$. Since $n+1 \geq 1$, the term at $x = 0$ is zero. Hence

$$\begin{aligned} I_{n+1} &= \left[\frac{x^{n+1}e^{2x}}{2}\right]_0^1 - \frac{n+1}{2}\int_0^1 x^n e^{2x}\,\mathrm{d}x \\ &= \frac{e^2}{2} - \frac{n+1}{2} I_n. \end{aligned}$$

## Exercise 2 · Differential equation and evolution

1. *[1 pt]* $0 = -\frac{1}{5}y + \frac{6}{5}$ gives $y = 6$.

2. *[1 pt]* The constant solution is 6. Let $z = y - 6$; the equation becomes $z' = -\frac{z}{5}$. Its solutions are $z(t) = Ce^{-\frac{t}{5}}$, so all the solutions of the original equation are $y(t) = 6 + Ce^{-\frac{t}{5}}$. The condition $y(0) = 12$ gives $C = 6$. For the function obtained,

$$y'(t) = -\frac{6}{5}e^{-\frac{t}{5}} = -\frac{y(t)}{5} + \frac{6}{5}, \qquad y(0) = 12.$$

   The solution of the initial value problem is therefore $y(t) = 6 + 6e^{-\frac{t}{5}}$.

3. *[1 pt]* $y' = -\frac{6}{5}e^{-\frac{t}{5}} < 0$; limit 6.

4. *[1 pt]* Since $e^{-\frac{t}{5}} > 0$, the derivative $y'(t) = -\frac{6}{5}e^{-\frac{t}{5}}$ is negative: $y$ is strictly decreasing. We solve

$$\begin{aligned} y(t) \leq 7 &\iff 6e^{-\frac{t}{5}} \leq 1 \\ &\iff -\frac{t}{5} \leq -\ln 6 \\ &\iff t \geq 5\ln 6. \end{aligned}$$

   The last step reverses the inequality because we multiply by $-5 < 0$. The first time is $5\ln 6 \simeq 8.9588$.

5. *[1 pt]* $Y_0 = 12$, $Y_{j+1} = Y_j + h(-\frac{Y_j}{5} + \frac{6}{5})$, $t_j = jh$.

## Exercise 3 · Projection and volume

1. *[1 pt]* $\overrightarrow{AB} = (-10, 10, 0)$ and $\overrightarrow{AC} = (-10, 0, 10)$ are non-collinear and their dot products with $\vec{n}$ are zero.

2. *[1 pt]* The line normal to the plane passing through $O$ has direction vector $\vec{n} = (1, 1, 1)$. Its point $H$ is therefore written $(t, t, t)$. For $H$ to lie in $(ABC)$, it is necessary and sufficient that

$$\overrightarrow{AH} \cdot \vec{n} = (t - 10) + t + t = 3t - 10 = 0.$$

   Hence $t = \frac{10}{3}$ and $H = (\frac{10}{3}, \frac{10}{3}, \frac{10}{3})$. This point lies in the plane and $\overrightarrow{OH}$ is collinear with the normal vector; it is indeed the required orthogonal projection.

3. *[1 pt]* $OH = \frac{10}{\sqrt{3}}$; for every $P$ in the plane, $OP^2 = OH^2 + HP^2 \geq OH^2$, with equality only for $P = H$.

4. *[1 pt]* The coordinate differences give $AB = AC = BC = 10\sqrt{2}$. The triangle is equilateral. Its height splits a side into two equal parts and equals

$$\sqrt{(10\sqrt{2})^2 - (\frac{10\sqrt{2}}{2})^2} = \frac{10\sqrt{6}}{2}.$$

   Its area is therefore $\frac{1}{2} \times 10\sqrt{2} \times \frac{10\sqrt{6}}{2} = 50\sqrt{3}$ square units.

5. *[1 pt]* Base $ABC$: volume $\frac{(50\sqrt{3})(\frac{10}{\sqrt{3}})}{3} = \frac{500}{3}$. Base $OAB$: area 50 and height 10, the same value.

## Exercise 4 · Sums of random variables

1. *[1 pt]* $S \sim \mathcal{B}(16, \frac{1}{3})$; $E(S) = \frac{16}{3}$.

2. *[1 pt]* Each $X_i$ equals 0 or 1, so $X_i^2 = X_i$. Hence $E(X_i) = p = \frac{1}{3}$ and

$$V(X_i) = E(X_i^2) - E(X_i)^2 = p - p^2 = p(1 - p) = \frac{2}{9}.$$

   The variables are independent; additivity of the variance gives

$$V(S) = \sum_{i=1}^{16} V(X_i) = 16p(1 - p) = \frac{32}{9}.$$

3. *[1 pt]* By linearity of the expected value and by $V(\lambda S + \mu) = \lambda^2 V(S)$,

$$E(G) = 8E(S) - 16 = \frac{80}{3}, \qquad V(G) = 64V(S) = \frac{2048}{9}.$$

   The constant $-16$ does not change the variance.

4. *[1 pt]* $E(M) = \frac{1}{3}$ and $V(M) = \frac{1}{72}$; the variance factor is $\frac{1}{n^2}$.

5. *[1 pt]* The mean of 16 variables all equal to $X_1$ is $X_1$ itself: its variance is $\frac{2}{9}$, that is 16 times the variance $\frac{1}{72}$ found in question 4. The results differ because the additivity of variance used in question 4 requires independence, which no longer holds here.

# Test 5 · B

**PAPER RESERVED FOR ASSESSMENT**

**Duration: 2 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete attempts will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · Integral and sequence — 5 points

For every natural number $n$, we set $I_n = \displaystyle\int_0^1 x^n e^{5x}\,\mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^{5x}$.

1. Calculate $I_0$. *(1 pt)*
2. Using integration by parts, calculate $I_1$. *(1 pt)*
3. Prove, by comparing the integrands, that the sequence $(I_n)$ is positive and decreasing. *(1 pt)*
4. Show that $I_n \leq \frac{e^5}{n+1}$ for all $n \in \mathbb{N}$. Deduce the limit of $(I_n)$. *(1 pt)*
5. Using integration by parts, express $I_{n+1}$ in terms of $I_n$ and $n$. *(1 pt)*

## Exercise 2 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{4}y + 2$ on $[0, +\infty)$. In questions 2 to 5, denote by $y$ the solution satisfying $y(0) = 12$.

1. Determine the constant solution of $(E)$. *(1 pt)*
2. Determine the expression of the solution satisfying the given initial condition. *(1 pt)*
3. Determine the limit of $y(t)$ as $t$ tends to $+\infty$. *(1 pt)*
4. Study the variations of $y$, then determine the smallest real number $t \geq 0$ such that $y(t) \leq 9$. *(1 pt)*
5. Apply Euler's method with step size $h > 0$ and denote by $Y_k$ the approximation of $y(kh)$. Give the value of $Y_0$ and express $Y_{k+1}$ in terms of $Y_k$ and $h$. *(1 pt)*

Test 5 · B · continued

## Exercise 3 · Projection and volume — 5 points

In an orthonormal coordinate system with origin $O$, consider $A(6,0,0)$, $B(0,6,0)$ and $C(0,0,6)$. We are given $\vec{n} = (1,1,1)$. Lengths, areas and volumes will be expressed in units of length, area and volume respectively.

1. Prove that $\vec{n}$ is normal to the plane $(ABC)$. *(1 pt)*
2. Let $H$ be the orthogonal projection of $O$ onto $(ABC)$. Determine the coordinates of $H$ using $\overrightarrow{AH} \cdot \vec{n} = 0$. *(1 pt)*
3. Prove that $OH$ is the distance from $O$ to the plane $(ABC)$: for every $P \in (ABC)$, prove that $OP^2 \geq OH^2$. *(1 pt)*
4. Calculate the area of triangle $ABC$. *(1 pt)*
5. Calculate the volume of the tetrahedron $OABC$ in two ways, by choosing two different bases. *(1 pt)*

## Exercise 4 · Sums of random variables — 5 points

We set $n = 18$. Consider $n$ independent random variables $X_1, \ldots, X_{18}$, which all follow the Bernoulli distribution with parameter $p = \frac{1}{2}$. We set $S = \sum_{i=1}^{18} X_i$ and $G = 10S - 18$.

1. Justify the distribution followed by $S$ and calculate $E(S)$. *(1 pt)*
2. Using the variance of a Bernoulli random variable and independence, calculate $V(S)$. *(1 pt)*
3. Calculate $E(G)$ and $V(G)$. *(1 pt)*
4. We set $M = \frac{S}{18}$. Calculate $E(M)$ and $V(M)$. *(1 pt)*
5. For this question only, we replace the 18 variables by 18 variables all equal to $X_1$. Then calculate the variance of their mean and compare it with that of question 4. Explain why the results differ. *(1 pt)*

# Solutions · Test 5 · B

## Exercise 1 · Integral and sequence

1. *[1 pt]* The function $x \mapsto \frac{e^{5x}}{5}$ is an antiderivative of $e^{5x}$. Hence $I_0 = [\frac{e^{5x}}{5}]_0^1 = \frac{e^5-1}{5}$.

2. *[1 pt]* Let us choose $u(x) = x$ and $v(x) = \frac{e^{5x}}{5}$, two differentiable functions with continuous derivatives on $[0,1]$. Then $u' = 1$ and $v' = e^{5x}$. Integration by parts gives

$$\begin{aligned} I_1 &= \left[\frac{xe^{5x}}{5}\right]_0^1 - \frac{1}{5}\int_0^1 e^{5x}\,\mathrm{d}x \\ &= \frac{e^5}{5} - \frac{e^5-1}{25} = \frac{4e^5+1}{25}. \end{aligned}$$

3. *[1 pt]* For $x \in [0,1]$, $0 \leq x^{n+1} \leq x^n$. Since $e^{5x} > 0$, integrating over increasing bounds gives $0 \leq I_{n+1} \leq I_n$. The sequence is therefore positive and weakly decreasing. The inequalities are even strict: the integrands of $I_n$ and $I_n - I_{n+1}$ are continuous and strictly positive on $]0,1[$.

4. *[1 pt]* $e^{5x} \leq e^5$; hence $0 \leq I_n \leq e^5 \int_0^1 x^n\,\mathrm{d}x = \frac{e^5}{n+1} \to 0$.

5. *[1 pt]* Apply integration by parts to $u(x) = x^{n+1}$ and $v(x) = \frac{e^{5x}}{5}$. Since $n+1 \geq 1$, the term at $x = 0$ is zero. Hence

$$\begin{aligned} I_{n+1} &= \left[\frac{x^{n+1}e^{5x}}{5}\right]_0^1 - \frac{n+1}{5}\int_0^1 x^n e^{5x}\,\mathrm{d}x \\ &= \frac{e^5}{5} - \frac{n+1}{5} I_n. \end{aligned}$$

## Exercise 2 · Differential equation and evolution

1. *[1 pt]* $0 = -\frac{1}{4}y + 2$ gives $y = 8$.

2. *[1 pt]* The constant solution is 8. Let $z = y - 8$; the equation becomes $z' = -\frac{z}{4}$. Its solutions are $z(t) = Ce^{-\frac{t}{4}}$, so all the solutions of the original equation are $y(t) = 8 + Ce^{-\frac{t}{4}}$. The condition $y(0) = 12$ gives $C = 4$. For the function obtained,

$$y'(t) = -e^{-\frac{t}{4}} = -\frac{y(t)}{4} + 2, \qquad y(0) = 12.$$

   The solution of the initial value problem is therefore $y(t) = 8 + 4e^{-\frac{t}{4}}$.

3. *[1 pt]* $y' = -e^{-\frac{t}{4}} < 0$; limit 8.

4. *[1 pt]* Since $e^{-\frac{t}{4}} > 0$, the derivative $y'(t) = -e^{-\frac{t}{4}}$ is negative: $y$ is strictly decreasing. We solve

$$\begin{aligned} y(t) \leq 9 &\iff 4e^{-\frac{t}{4}} \leq 1 \\ &\iff -\frac{t}{4} \leq -\ln 4 \\ &\iff t \geq 4\ln 4. \end{aligned}$$

   The last step reverses the inequality because we multiply by $-4 < 0$. The first time is $4\ln 4 \simeq 5.5452$.

5. *[1 pt]* $Y_0 = 12$, $Y_{j+1} = Y_j + h(-\frac{Y_j}{4} + 2)$, $t_j = jh$.

## Exercise 3 · Projection and volume

1. *[1 pt]* $\overrightarrow{AB} = (-6,6,0)$ and $\overrightarrow{AC} = (-6,0,6)$ are non-collinear and their dot products with $\vec{n}$ are zero.

2. *[1 pt]* The line normal to the plane passing through $O$ has direction vector $\vec{n} = (1,1,1)$. Its point $H$ is therefore written $(t,t,t)$. For $H$ to lie in $(ABC)$, it is necessary and sufficient that

$$\overrightarrow{AH} \cdot \vec{n} = (t-6) + t + t = 3t - 6 = 0.$$

   Hence $t = 2$ and $H = (2,2,2)$. This point lies in the plane and $\overrightarrow{OH}$ is collinear with the normal vector; it is indeed the required orthogonal projection.

3. *[1 pt]* $OH = \frac{6}{\sqrt{3}}$; for every $P$ in the plane, $OP^2 = OH^2 + HP^2 \geq OH^2$, with equality only for $P = H$.

4. *[1 pt]* The coordinate differences give $AB = AC = BC = 6\sqrt{2}$. The triangle is equilateral. Its height splits a side into two equal parts and equals

$$\sqrt{(6\sqrt{2})^2 - (\frac{6\sqrt{2}}{2})^2} = \frac{6\sqrt{6}}{2}.$$

   Its area is therefore $\frac{1}{2} \times 6\sqrt{2} \times \frac{6\sqrt{6}}{2} = 18\sqrt{3}$ square units.

5. *[1 pt]* Base $ABC$: volume $\frac{(18\sqrt{3})(\frac{6}{\sqrt{3}})}{3} = 36$. Base $OAB$: area 18 and height 6, the same value.

## Exercise 4 · Sums of random variables

1. *[1 pt]* $S \sim \mathcal{B}(18, \frac{1}{2})$; $E(S) = 9$.

2. *[1 pt]* Each $X_i$ equals 0 or 1, so $X_i^2 = X_i$. Hence $E(X_i) = p = \frac{1}{2}$ and

$$V(X_i) = E(X_i^2) - E(X_i)^2 = p - p^2 = p(1-p) = \frac{1}{4}.$$

   The variables are independent; additivity of the variance gives

$$V(S) = \sum_{i=1}^{18} V(X_i) = 18p(1-p) = \frac{9}{2}.$$

3. *[1 pt]* By linearity of the expected value and by $V(\lambda S + \mu) = \lambda^2 V(S)$,

$$E(G) = 10E(S) - 18 = 72, \qquad V(G) = 100V(S) = 450.$$

   The constant $-18$ does not change the variance.

4. *[1 pt]* $E(M) = \frac{1}{2}$ and $V(M) = \frac{1}{72}$; the variance factor is $\frac{1}{n^2}$.

5. *[1 pt]* The mean of 18 variables all equal to $X_1$ is $X_1$ itself: its variance is $\frac{1}{4}$, that is 18 times the variance $\frac{1}{72}$ found in question 4. The results differ because the additivity of variance used in question 4 requires independence, which no longer holds here.

# Mock 2 · Practice 1

**PRACTICE PAPER · TRAINING**

**Duration: 4 hours · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete approaches will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · Integral and sequence — 5 points

For every natural number $n$, let $I_n = \displaystyle\int_0^1 x^n e^{5x}\,\mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^{5x}$.

1. Calculate $I_0$. *(0.5 pt)*
2. Using integration by parts, calculate $I_1$. *(0.5 pt)*
3. Prove, by comparing the integrands, that $(I_n)$ is positive and decreasing. *(0.5 pt)*
4. Show that $I_n \leq \frac{e^5}{n+1}$ for all $n \in \mathbb{N}$. Deduce the limit of $(I_n)$. *(1 pt)*
5. Using integration by parts, express $I_{n+1}$ in terms of $I_n$ and $n$. *(0.5 pt)*
6. Show that $I_n \geq \frac{1}{n+1}$ for all $n \in \mathbb{N}$. *(0.5 pt)*
7. Split $[0,1]$ into $N$ intervals of equal length, with $N \geq 1$. Write the sums $L_N$ and $U_N$ of the areas of the rectangles built using the values of $e^{5x}$ at the left and right endpoints. Justify $L_N \leq I_0 \leq U_N$, then calculate $U_N - L_N$. *(0.5 pt)*
8. Find the smallest integer $N$ guaranteeing that this bracketing has a width at most equal to $10^{-2}$. Write a Python function that returns $N$, $L_N$ and $U_N$. *(1 pt)*

## Exercise 2 · A logarithmic function — 5 points

Consider the function $f$ defined on $(0, +\infty)$ by $f(x) = \frac{4+\ln x}{x}$. On note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of $f$ at $0^+$ and at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw up the table of variations of $f$. *(0.5 pt)*
3. State the value of the maximum of $f$, then solve $f(x) = 0$. *(0.5 pt)*
4. Study the convexity of $f$ and determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Prove that the equation $f(x) = \frac{1}{2}e^3$ has exactly two solutions in $(0, +\infty)$. *(0.5 pt)*
6. Determine an antiderivative of $f$ on $(0, +\infty)$. *(0.5 pt)*
7. Calculate the mean value of $f$ on $[1, e]$. *(0.5 pt)*
8. Independently of the previous questions, solve $\ln(x-3) + \ln(x+3) = \ln(27)$, after stating the set on which this equation is defined. *(1 pt)*

Mock 2 · Practice 1 · continued

## Exercise 3 · Line, plane and orthogonal projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x + 2y + 2z = 10$ and the point $M(3, 0, 0)$. The line $d$ passes through $M$ and has direction vector $\vec{n} = (1, 2, 2)$.

1. Give a parametric representation of $d$. *(0.5 pt)*
2. Prove that $d$ meets $\mathcal{P}$ at a point $H$, then calculate the coordinates of $H$. *(0.5 pt)*
3. Justify that $H$ is the orthogonal projection of $M$ onto $\mathcal{P}$, then calculate the distance from $M$ to $\mathcal{P}$. *(0.5 pt)*
4. Consider the plane $\mathcal{Q}$ with equation $y = z$. Prove that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it. *(1 pt)*
5. Check that $H \in \mathcal{Q}$. Deduce the distance from $M$ to the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through $M$. *(0.5 pt)*
7. Determine the angle between $d$ and the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
8. Define the point $N$ by $\overrightarrow{MN} = \vec{n}$. Calculate the distances from $N$ to the plane $\mathcal{P}$ and to the line $\mathcal{P} \cap \mathcal{Q}$, then compare them. *(1 pt)*

## Exercise 4 · Repeated trials and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 20 times. The trials are independent and the probability of success is $p = \frac{3}{10}$ on each trial. Let $X$ denote the number of successes obtained.

1. Justify that $X$ follows a binomial distribution and state its parameters. *(0.5 pt)*
2. Calculate the probability of obtaining exactly two successes. *(0.5 pt)*
3. Using complementary events, express and then calculate the probability of obtaining at least two successes. *(0.5 pt)*
4. Calculate the expected value and the variance of $X$. *(1 pt)*
5. We now repeat the same trial $N \geq 1$ times, again independently. Find the smallest integer $N$ for which the probability of obtaining at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. *(0.5 pt)*
6. For an integer $j \in \{0, \ldots, 20\}$, prove the formula for $P(X = j)$ by counting the possible positions of the successes. *(0.5 pt)*
7. A series of 20 trials is accepted if its number of successes is between 1 and 3, inclusive. Calculate the probability that a series is accepted. *(0.5 pt)*
8. Write a Python function with no argument that returns the smallest integer $j$ such that $P(X \leq j) \geq 0.95$. Run the function and check the minimality of the result. *(1 pt)*

# Solutions · Mock 2 · Practice 1

## Exercise 1 · Integral and sequence

1. *[0.5 pt]* La fonction $x \mapsto \frac{e^{5x}}{5}$ is an antiderivative of $e^{5x}$. Thus $I_0 = [\frac{e^{5x}}{5}]_0^1 = \frac{e^5-1}{5}$.

2. *[0.5 pt]* Take $u(x) = x$ and $v(x) = \frac{e^{5x}}{5}$, two differentiable functions with continuous derivatives on $[0,1]$. Then $u' = 1$ and $v' = e^{5x}$. Integration by parts gives

$$\begin{aligned} I_1 &= \left[\frac{xe^{5x}}{5}\right]_0^1 - \frac{1}{5}\int_0^1 e^{5x}\,\mathrm{d}x \\ &= \frac{e^5}{5} - \frac{e^5-1}{25} = \frac{4e^5+1}{25}. \end{aligned}$$

3. *[0.5 pt]* For $x \in [0,1]$, $0 \le x^{n+1} \le x^n$. Since $e^{5x} > 0$, integrating over increasing bounds gives $0 \le I_{n+1} \le I_n$. The sequence is therefore positive and decreasing (in the weak sense). The inequalities are even strict: the integrands of $I_n$ and of $I_n - I_{n+1}$ are continuous and strictly positive on $]0,1[$.

4. *[1 pt]* $e^{5x} \le e^5$; so $0 \le I_n \le e^5 \int_0^1 x^n\,\mathrm{d}x = \frac{e^5}{n+1} \to 0$.

5. *[0.5 pt]* Apply integration by parts with $u(x) = x^{n+1}$ and $v(x) = \frac{e^{5x}}{5}$. Since $n+1 \ge 1$, the term at $x = 0$ is zero. Hence

$$\begin{aligned} I_{n+1} &= \left[\frac{x^{n+1}e^{5x}}{5}\right]_0^1 - \frac{n+1}{5}\int_0^1 x^n e^{5x}\,\mathrm{d}x \\ &= \frac{e^5}{5} - \frac{n+1}{5} I_n. \end{aligned}$$

6. *[0.5 pt]* For $0 \le x \le 1$, $e^{5x} \ge 1$ and $x^n \ge 0$. So $x^n \le x^n e^{5x}$; integrating,

$$I_n \ge \int_0^1 x^n\,\mathrm{d}x = \left[\frac{x^{n+1}}{n+1}\right]_0^1 = \frac{1}{n+1}.$$

7. *[0.5 pt]* The function $x \mapsto e^{5x}$ is increasing. On each interval $[\frac{j}{N}, \frac{j+1}{N}]$, the values at the endpoints bound the function. Integrating and then summing,

$$L_N = \frac{1}{N}\sum_{j=0}^{N-1} e^{\frac{5j}{N}} \le I_0 \le U_N = \frac{1}{N}\sum_{j=1}^{N} e^{\frac{5j}{N}}.$$

   The interior terms cancel in the difference, so $U_N - L_N = \frac{e^5-1}{N}$.

8. *[1 pt]* The width is at most $10^{-2}$ if $N \geq 100(e^5 - 1)$. The first integer satisfying this bound is $N = 14742$. Here is the computation of the bounds:

```python
from math import exp, ceil

def bornes():
    N = ceil(100*(exp(5)-1))
    bas = sum(exp(5*j/N) for j in range(N))/N
    haut = bas+(exp(5)-1)/N
    return N, bas, haut
```

## Exercise 2 · A logarithmic function

1. *[0.5 pt]* At $0^+$, the numerator tends to $-\infty$ and the denominator to $0^+$: limit $-\infty$; at $+\infty$, limit 0 by comparison of growth rates.

2. *[0.5 pt]* By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (4 + \ln x)}{x^2} = \frac{-3 - \ln x}{x^2}.$$

   The denominator is positive. The numerator vanishes at $s = e^{-3}$, is positive for $0 < x < s$ and negative for $x > s$. Thus $f$ is strictly increasing on $]0, s]$, then strictly decreasing on $[s, +\infty[$.

3. *[0.5 pt]* The maximum is attained only at $s = e^{-3}$ and equals

$$f(s) = \frac{4 - 3}{e^{-3}} = e^3.$$

   Since $x > 0$, $f(x) = 0$ is equivalent to $4 + \ln x = 0$, that is $x = e^{-4}$.

4. *[1 pt]* Differentiating $(-3 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-3 - \ln x)x^{-3} = \frac{2\ln x + 5}{x^3}.$$

   The numerator vanishes at $t = e^{-\frac{5}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at $t$ and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. *[0.5 pt]* Let $m = \frac{e^3}{2}$ and $s = e^{-3}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at 0 and value $2m$ at $s$: the corollary of the intermediate value theorem (IVT) gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from $2m$ to $0 < m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

6. *[0.5 pt]* Let $F(x) = 4\ln x + \frac{1}{2}(\ln x)^2$, differentiable for $x > 0$. Then $F'(x) = \frac{4}{x} + \frac{\ln x}{x} = f(x)$; $F$ is an antiderivative on $]0, +\infty[$.

7. *[0.5 pt]* With this antiderivative, $F(e) = \frac{9}{2}$ and $F(1) = 0$. The integral is therefore $\frac{9}{2}$, and the mean value is

$$\mu = \frac{1}{e - 1}\int_1^e f(x)\ \mathrm{d}x = \frac{9}{2(e - 1)}.$$

8. *[1 pt]* Domain $x > 3$; $x^2 - 9 = 27$, so $x = 6$, the negative root being excluded.

## Exercise 3 · Line, plane and orthogonal projection

1. *[0.5 pt]* $(x, y, z) = (3 + t, 2t, 2t)$, $t \in \mathbb{R}$.
2. *[0.5 pt]* $3 + 9t = 10$ gives $t = \frac{7}{9}$; $H = (\frac{34}{9}, \frac{14}{9}, \frac{14}{9})$.
3. *[0.5 pt]* The vector $\vec{n} = (1, 2, 2)$ is normal to $\mathcal{P}$. Since $H \in \mathcal{P}$ and $\overrightarrow{MH} = \frac{7}{9}\vec{n}$, $H$ is the orthogonal projection of $M$ onto the plane. The coordinate system being orthonormal, $\|\vec{n}\| = \sqrt{1+4+4} = 3$. Therefore
$$d(M, \mathcal{P}) = MH = |\frac{7}{9}|\, 3 = \frac{7}{3}.$$
4. *[1 pt]* Set $y = z = s$; $x = 10 - 4s$. So $(x, y, z) = (10, 0, 0) + s(-4, 1, 1)$.
5. *[0.5 pt]* Yes, its last two coordinates are equal. $H$ lies on the line and $\overrightarrow{MH}$ is orthogonal to every direction of the plane, hence to this line. Distance $\frac{7}{3}$.
6. *[0.5 pt]* $x + 2y + 2z = 3$; same normal vector, passing through $M$.
7. *[0.5 pt]* $\vec{n} \cdot (-4, 1, 1) = -4 + 2 + 2 = 0$: angle $\frac{\pi}{2}$.
8. *[1 pt]* $N = M + \vec{n}$ lies on the same normal line as $M$; its orthogonal projection onto $\mathcal{P}$ is therefore also $H$. Now $H \in \mathcal{P} \cap \mathcal{Q}$. For every point $P$ of this line, $NP^2 = NH^2 + HP^2$, so its minimum distance to $N$ is $NH$. Finally,
$$d(N, \mathcal{P}) = d(N, \mathcal{P} \cap \mathcal{Q}) = \|\overrightarrow{NH}\| = |\frac{7}{9} - 1|\, \|\vec{n}\| = \frac{2}{3}.$$

## Exercise 4 · Repeated trials and estimation

1. *[0.5 pt]* Repetition of a fixed number 20 of identical independent trials with two outcomes; $X \sim \mathcal{B}(20, \frac{3}{10})$.
2. *[0.5 pt]* $P(X = 2) = \binom{20}{2}(\frac{3}{10})^2(\frac{7}{10})^{18} \simeq 0.027846$.
3. *[0.5 pt]* $1 - P(X = 0) - P(X = 1) = 1 - (\frac{7}{10})^{20} - 20(\frac{3}{10})(\frac{7}{10})^{19} \simeq 0.992363$.
4. *[1 pt]* $E(X) = 6$, $V(X) = \frac{21}{5}$.
5. *[0.5 pt]* For $N$ independent trials, the probability of no success is $(1-p)^N$. The required probability is therefore $1 - (\frac{7}{10})^N$. It is strictly increasing with $N$, since $0 < \frac{7}{10} < 1$. The calculations give
$$P_8 \simeq 0.942352 < 0.95, \qquad P_9 \simeq 0.959646 \geq 0.95.$$
Thus 9 is the smallest suitable number of trials.
6. *[0.5 pt]* Let $n = 20$. For $j \in \{0, \ldots, n\}$, an ordered outcome containing $j$ successes and $n - j$ failures has, by independence, probability $p^j(1-p)^{n-j}$. There are $\binom{n}{j}$ choices for the positions of the successes. These outcomes are mutually exclusive, so
$$P(X = j) = \binom{n}{j} p^j (1-p)^{n-j}, \qquad p = \frac{3}{10}.$$
7. *[0.5 pt]* The events $X = 1$, $X = 2$, $X = 3$ are mutually exclusive, so
$$P(1 \leq X \leq 3) = \sum_{j=1}^{3} \binom{20}{j} (\frac{3}{10})^j (\frac{7}{10})^{20-j} \simeq 0.106289.$$

8. *[1 pt]*

```python
from math import comb

def quantile():
    n, p = 20, 3/10
    j, cumul = 0, (1-p)**n
    while cumul < 0.95:
        j += 1
        cumul += comb(n, j)*p**j*(1-p)**(n-j)
    return j
```

The probabilities added are positive and the final cumulative value is 1, so the loop terminates. It returns 9. The cumulative values at ranks 8 and 9 are approximately 0.886669 and 0.952038: this confirms minimality.

# Mock 2 · Practice 2

**PRACTICE PAPER · TRAINING**

**Duration: 4 hours · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete approaches will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · Integral and sequence — 5 points

For every natural number $n$, let $I_n = \displaystyle\int_0^1 x^n e^{4x}\,\mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^{4x}$.

1. Calculate $I_0$. *(0.5 pt)*
2. Using integration by parts, calculate $I_1$. *(0.5 pt)*
3. Prove, by comparing the integrands, that $(I_n)$ is positive and decreasing. *(0.5 pt)*
4. Show that $I_n \leq \frac{e^4}{n+1}$ for all $n \in \mathbb{N}$. Deduce the limit of $(I_n)$. *(1 pt)*
5. Using integration by parts, express $I_{n+1}$ in terms of $I_n$ and $n$. *(0.5 pt)*
6. Show that $I_n \geq \frac{1}{n+1}$ for all $n \in \mathbb{N}$. *(0.5 pt)*
7. Split $[0,1]$ into $N$ intervals of equal length, with $N \geq 1$. Write the sums $L_N$ and $U_N$ of the areas of the rectangles built using the values of $e^{4x}$ at the left and right endpoints. Justify $L_N \leq I_0 \leq U_N$, then calculate $U_N - L_N$. *(0.5 pt)*
8. Find the smallest integer $N$ guaranteeing that this bracketing has a width at most equal to $10^{-2}$. Write a Python function that returns $N$, $L_N$ and $U_N$. *(1 pt)*

## Exercise 2 · A logarithmic function — 5 points

Consider the function $f$ defined on $(0, +\infty)$ by $f(x) = \frac{3+\ln x}{x}$. On note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of $f$ at $0^+$ and at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw up the table of variations of $f$. *(0.5 pt)*
3. State the value of the maximum of $f$, then solve $f(x) = 0$. *(0.5 pt)*
4. Study the convexity of $f$ and determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Prove that the equation $f(x) = \frac{1}{2}e^2$ has exactly two solutions in $(0, +\infty)$. *(0.5 pt)*
6. Determine an antiderivative of $f$ on $(0, +\infty)$. *(0.5 pt)*
7. Calculate the mean value of $f$ on $[1, e]$. *(0.5 pt)*
8. Independently of the previous questions, solve $\ln(x-1) + \ln(x+1) = \ln(63)$, after stating the set on which this equation is defined. *(1 pt)*

Mock 2 · Practice 2 · continued

## Exercise 3 · Line, plane and orthogonal projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x + 2y + 2z = 6$ and the point $M(1, 0, 0)$. The line $d$ passes through $M$ and has direction vector $\vec{n} = (1, 2, 2)$.

1. Give a parametric representation of $d$. *(0.5 pt)*
2. Prove that $d$ meets $\mathcal{P}$ at a point $H$, then calculate the coordinates of $H$. *(0.5 pt)*
3. Justify that $H$ is the orthogonal projection of $M$ onto $\mathcal{P}$, then calculate the distance from $M$ to $\mathcal{P}$. *(0.5 pt)*
4. Consider the plane $\mathcal{Q}$ with equation $y = z$. Prove that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it. *(1 pt)*
5. Check that $H \in \mathcal{Q}$. Deduce the distance from $M$ to the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through $M$. *(0.5 pt)*
7. Determine the angle between $d$ and the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
8. Define the point $N$ by $\overrightarrow{MN} = \vec{n}$. Calculate the distances from $N$ to the plane $\mathcal{P}$ and to the line $\mathcal{P} \cap \mathcal{Q}$, then compare them. *(1 pt)*

## Exercise 4 · Repeated trials and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 17 times. The trials are independent and the probability of success is $p = \frac{3}{10}$ on each trial. Let $X$ denote the number of successes obtained.

1. Justify that $X$ follows a binomial distribution and state its parameters. *(0.5 pt)*
2. Calculate the probability of obtaining exactly two successes. *(0.5 pt)*
3. Using complementary events, express and then calculate the probability of obtaining at least two successes. *(0.5 pt)*
4. Calculate the expected value and the variance of $X$. *(1 pt)*
5. We now repeat the same trial $N \geq 1$ times, again independently. Find the smallest integer $N$ for which the probability of obtaining at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. *(0.5 pt)*
6. For an integer $j \in \{0, \ldots, 17\}$, prove the formula for $P(X = j)$ by counting the possible positions of the successes. *(0.5 pt)*
7. A series of 17 trials is accepted if its number of successes is between 1 and 3, inclusive. Calculate the probability that a series is accepted. *(0.5 pt)*
8. Write a Python function with no argument that returns the smallest integer $j$ such that $P(X \leq j) \geq 0.95$. Run the function and check the minimality of the result. *(1 pt)*

# Solutions · Mock 2 · Practice 2

## Exercise 1 · Integral and sequence

1. *[0.5 pt]* La fonction $x \mapsto \frac{e^{4x}}{4}$ is an antiderivative of $e^{4x}$. Thus $I_0 = [\frac{e^{4x}}{4}]_0^1 = \frac{e^4-1}{4}$.

2. *[0.5 pt]* Take $u(x) = x$ and $v(x) = \frac{e^{4x}}{4}$, two differentiable functions with continuous derivatives on $[0,1]$. Then $u' = 1$ and $v' = e^{4x}$. Integration by parts gives

$$\begin{aligned} I_1 &= \left[\frac{xe^{4x}}{4}\right]_0^1 - \frac{1}{4}\int_0^1 e^{4x}\,\mathrm{d}x \\ &= \frac{e^4}{4} - \frac{e^4-1}{16} = \frac{3e^4+1}{16}. \end{aligned}$$

3. *[0.5 pt]* For $x \in [0,1]$, $0 \le x^{n+1} \le x^n$. Since $e^{4x} > 0$, integrating over increasing bounds gives $0 \le I_{n+1} \le I_n$. The sequence is therefore positive and decreasing (in the weak sense). The inequalities are even strict: the integrands of $I_n$ and of $I_n - I_{n+1}$ are continuous and strictly positive on $]0,1[$.

4. *[1 pt]* $e^{4x} \le e^4$; so $0 \le I_n \le e^4 \int_0^1 x^n\,\mathrm{d}x = \frac{e^4}{n+1} \to 0$.

5. *[0.5 pt]* Apply integration by parts with $u(x) = x^{n+1}$ and $v(x) = \frac{e^{4x}}{4}$. Since $n+1 \ge 1$, the term at $x = 0$ is zero. Hence

$$\begin{aligned} I_{n+1} &= \left[\frac{x^{n+1}e^{4x}}{4}\right]_0^1 - \frac{n+1}{4}\int_0^1 x^n e^{4x}\,\mathrm{d}x \\ &= \frac{e^4}{4} - \frac{n+1}{4} I_n. \end{aligned}$$

6. *[0.5 pt]* For $0 \le x \le 1$, $e^{4x} \ge 1$ and $x^n \ge 0$. So $x^n \le x^n e^{4x}$; integrating,

$$I_n \ge \int_0^1 x^n\,\mathrm{d}x = \left[\frac{x^{n+1}}{n+1}\right]_0^1 = \frac{1}{n+1}.$$

7. *[0.5 pt]* The function $x \mapsto e^{4x}$ is increasing. On each interval $[\frac{j}{N}, \frac{j+1}{N}]$, the values at the endpoints bound the function. Integrating and then summing,

$$L_N = \frac{1}{N}\sum_{j=0}^{N-1} e^{\frac{4j}{N}} \le I_0 \le U_N = \frac{1}{N}\sum_{j=1}^{N} e^{\frac{4j}{N}}.$$

   The interior terms cancel in the difference, so $U_N - L_N = \frac{e^4-1}{N}$.

8. *[1 pt]* The width is at most $10^{-2}$ if $N \geq 100(e^4 - 1)$. The first integer satisfying this bound is $N = 5360$. Here is the computation of the bounds:

```
from math import exp, ceil

def bornes():
    N = ceil(100*(exp(4)-1))
    bas = sum(exp(4*j/N) for j in range(N))/N
    haut = bas+(exp(4)-1)/N
    return N, bas, haut
```

## Exercise 2 · A logarithmic function

1. *[0.5 pt]* At $0^+$, the numerator tends to $-\infty$ and the denominator to $0^+$: limit $-\infty$; at $+\infty$, limit 0 by comparison of growth rates.

2. *[0.5 pt]* By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (3 + \ln x)}{x^2} = \frac{-2 - \ln x}{x^2}.$$

   The denominator is positive. The numerator vanishes at $s = e^{-2}$, is positive for $0 < x < s$ and negative for $x > s$. Thus $f$ is strictly increasing on $]0, s]$, then strictly decreasing on $[s, +\infty[$.

3. *[0.5 pt]* The maximum is attained only at $s = e^{-2}$ and equals

$$f(s) = \frac{3 - 2}{e^{-2}} = e^2.$$

   Since $x > 0$, $f(x) = 0$ is equivalent to $3 + \ln x = 0$, that is $x = e^{-3}$.

4. *[1 pt]* Differentiating $(-2 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-2 - \ln x)x^{-3} = \frac{2\ln x + 3}{x^3}.$$

   The numerator vanishes at $t = e^{-\frac{3}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at $t$ and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. *[0.5 pt]* Let $m = \frac{e^2}{2}$ and $s = e^{-2}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at 0 and value $2m$ at $s$: the corollary of the intermediate value theorem (IVT) gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from $2m$ to $0 < m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

6. *[0.5 pt]* Let $F(x) = 3\ln x + \frac{1}{2}(\ln x)^2$, differentiable for $x > 0$. Then $F'(x) = \frac{3}{x} + \frac{\ln x}{x} = f(x)$; $F$ is an antiderivative on $]0, +\infty[$.

7. *[0.5 pt]* With this antiderivative, $F(e) = \frac{7}{2}$ and $F(1) = 0$. The integral is therefore $\frac{7}{2}$, and the mean value is

$$\mu = \frac{1}{e - 1}\int_1^e f(x)\ \mathrm{d}x = \frac{7}{2(e - 1)}.$$

8. *[1 pt]* Domain $x > 1$; $x^2 - 1 = 63$, so $x = 8$, the negative root being excluded.

## Exercise 3 · Line, plane and orthogonal projection

1. *[0.5 pt]* $(x, y, z) = (1 + t, 2t, 2t)$, $t \in \mathbb{R}$.
2. *[0.5 pt]* $1 + 9t = 6$ gives $t = \frac{5}{9}$; $H = (\frac{14}{9}, \frac{10}{9}, \frac{10}{9})$.
3. *[0.5 pt]* The vector $\vec{n} = (1, 2, 2)$ is normal to $\mathcal{P}$. Since $H \in \mathcal{P}$ and $\overrightarrow{MH} = \frac{5}{9}\vec{n}$, $H$ is the orthogonal projection of $M$ onto the plane. The coordinate system being orthonormal, $\|\vec{n}\| = \sqrt{1+4+4} = 3$. Therefore
$$d(M, \mathcal{P}) = MH = |\frac{5}{9}|\, 3 = \frac{5}{3}.$$
4. *[1 pt]* Set $y = z = s$; $x = 6 - 4s$. So $(x, y, z) = (6, 0, 0) + s(-4, 1, 1)$.
5. *[0.5 pt]* Yes, its last two coordinates are equal. $H$ lies on the line and $\overrightarrow{MH}$ is orthogonal to every direction of the plane, hence to this line. Distance $\frac{5}{3}$.
6. *[0.5 pt]* $x + 2y + 2z = 1$; same normal vector, passing through $M$.
7. *[0.5 pt]* $\vec{n} \cdot (-4, 1, 1) = -4 + 2 + 2 = 0$: angle $\frac{\pi}{2}$.
8. *[1 pt]* $N = M + \vec{n}$ lies on the same normal line as $M$; its orthogonal projection onto $\mathcal{P}$ is therefore also $H$. Now $H \in \mathcal{P} \cap \mathcal{Q}$. For every point $P$ of this line, $NP^2 = NH^2 + HP^2$, so its minimum distance to $N$ is $NH$. Finally,
$$d(N, \mathcal{P}) = d(N, \mathcal{P} \cap \mathcal{Q}) = \|\overrightarrow{NH}\| = |\frac{5}{9} - 1|\, \|\vec{n}\| = \frac{4}{3}.$$

## Exercise 4 · Repeated trials and estimation

1. *[0.5 pt]* Repetition of a fixed number 17 of identical independent trials with two outcomes; $X \sim \mathcal{B}(17, \frac{3}{10})$.
2. *[0.5 pt]* $P(X = 2) = \binom{17}{2}(\frac{3}{10})^2(\frac{7}{10})^{15} \simeq 0.058110$.
3. *[0.5 pt]* $1 - P(X = 0) - P(X = 1) = 1 - (\frac{7}{10})^{17} - 17(\frac{3}{10})(\frac{7}{10})^{16} \simeq 0.980725$.
4. *[1 pt]* $E(X) = \frac{51}{10}$, $V(X) = \frac{357}{100}$.
5. *[0.5 pt]* For $N$ independent trials, the probability of no success is $(1-p)^N$. The required probability is therefore $1 - (\frac{7}{10})^N$. It is strictly increasing with $N$, since $0 < \frac{7}{10} < 1$. The calculations give
$$P_8 \simeq 0.942352 < 0.95, \qquad P_9 \simeq 0.959646 \geq 0.95.$$
Thus 9 is the smallest suitable number of trials.
6. *[0.5 pt]* Let $n = 17$. For $j \in \{0, \ldots, n\}$, an ordered outcome containing $j$ successes and $n - j$ failures has, by independence, probability $p^j(1-p)^{n-j}$. There are $\binom{n}{j}$ choices for the positions of the successes. These outcomes are mutually exclusive, so
$$P(X = j) = \binom{n}{j} p^j (1-p)^{n-j}, \qquad p = \frac{3}{10}.$$
7. *[0.5 pt]* The events $X = 1$, $X = 2$, $X = 3$ are mutually exclusive, so
$$P(1 \leq X \leq 3) = \sum_{j=1}^{3} \binom{17}{j} (\frac{3}{10})^j (\frac{7}{10})^{17-j} \simeq 0.199581.$$

8. *[1 pt]*

```
from math import comb

def quantile():
    n, p = 17, 3/10
    j, cumul = 0, (1-p)**n
    while cumul < 0.95:
        j += 1
        cumul += comb(n, j)*p**j*(1-p)**(n-j)
    return j
```

The probabilities added are positive and the final cumulative value is 1, so the loop terminates. It returns 8. The cumulative values at ranks 7 and 8 are approximately 0.895360 and 0.959723: this confirms minimality.

# Mock 2 · A

**PAPER RESERVED FOR ASSESSMENT**

**Duration: 4 hours · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete approaches will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · Integral and sequence — 5 points

For every natural number $n$, let $I_n = \displaystyle\int_0^1 x^n e^{3x}\,\mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^{3x}$.

1. Calculate $I_0$. *(0.5 pt)*
2. Using integration by parts, calculate $I_1$. *(0.5 pt)*
3. Prove, by comparing the integrands, that $(I_n)$ is positive and decreasing. *(0.5 pt)*
4. Show that $I_n \leq \frac{e^3}{n+1}$ for all $n \in \mathbb{N}$. Deduce the limit of $(I_n)$. *(1 pt)*
5. Using integration by parts, express $I_{n+1}$ in terms of $I_n$ and $n$. *(0.5 pt)*
6. Show that $I_n \geq \frac{1}{n+1}$ for all $n \in \mathbb{N}$. *(0.5 pt)*
7. Split $[0,1]$ into $N$ intervals of equal length, with $N \geq 1$. Write the sums $L_N$ and $U_N$ of the areas of the rectangles built using the values of $e^{3x}$ at the left and right endpoints. Justify $L_N \leq I_0 \leq U_N$, then calculate $U_N - L_N$. *(0.5 pt)*
8. Find the smallest integer $N$ guaranteeing that this bracketing has a width at most equal to $10^{-2}$. Write a Python function that returns $N$, $L_N$ and $U_N$. *(1 pt)*

## Exercise 2 · A logarithmic function — 5 points

Consider the function $f$ defined on $(0, +\infty)$ by $f(x) = \frac{2+\ln x}{x}$. On note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of $f$ at $0^+$ and at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw up the table of variations of $f$. *(0.5 pt)*
3. State the value of the maximum of $f$, then solve $f(x) = 0$. *(0.5 pt)*
4. Study the convexity of $f$ and determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Prove that the equation $f(x) = \frac{1}{2}e^1$ has exactly two solutions in $(0, +\infty)$. *(0.5 pt)*
6. Determine an antiderivative of $f$ on $(0, +\infty)$. *(0.5 pt)*
7. Calculate the mean value of $f$ on $[1, e]$. *(0.5 pt)*
8. Independently of the previous questions, solve $\ln(x-2) + \ln(x+2) = \ln(21)$, after stating the set on which this equation is defined. *(1 pt)*

Mock 2 · A · continued

## Exercise 3 · Line, plane and orthogonal projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x + 2y + 2z = 10$ and the point $M(2, 0, 0)$. The line $d$ passes through $M$ and has direction vector $\vec{n} = (1, 2, 2)$.

1. Give a parametric representation of $d$. *(0.5 pt)*
2. Prove that $d$ meets $\mathcal{P}$ at a point $H$, then calculate the coordinates of $H$. *(0.5 pt)*
3. Justify that $H$ is the orthogonal projection of $M$ onto $\mathcal{P}$, then calculate the distance from $M$ to $\mathcal{P}$. *(0.5 pt)*
4. Consider the plane $\mathcal{Q}$ with equation $y = z$. Prove that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it. *(1 pt)*
5. Check that $H \in \mathcal{Q}$. Deduce the distance from $M$ to the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through $M$. *(0.5 pt)*
7. Determine the angle between $d$ and the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
8. Define the point $N$ by $\overrightarrow{MN} = \vec{n}$. Calculate the distances from $N$ to the plane $\mathcal{P}$ and to the line $\mathcal{P} \cap \mathcal{Q}$, then compare them. *(1 pt)*

## Exercise 4 · Repeated trials and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 18 times. The trials are independent and the probability of success is $p = \frac{1}{5}$ on each trial. Let $X$ denote the number of successes obtained.

1. Justify that $X$ follows a binomial distribution and state its parameters. *(0.5 pt)*
2. Calculate the probability of obtaining exactly two successes. *(0.5 pt)*
3. Using complementary events, express and then calculate the probability of obtaining at least two successes. *(0.5 pt)*
4. Calculate the expected value and the variance of $X$. *(1 pt)*
5. We now repeat the same trial $N \geq 1$ times, again independently. Find the smallest integer $N$ for which the probability of obtaining at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. *(0.5 pt)*
6. For an integer $j \in \{0, \ldots, 18\}$, prove the formula for $P(X = j)$ by counting the possible positions of the successes. *(0.5 pt)*
7. A series of 18 trials is accepted if its number of successes is between 1 and 3, inclusive. Calculate the probability that a series is accepted. *(0.5 pt)*
8. Write a Python function with no argument that returns the smallest integer $j$ such that $P(X \leq j) \geq 0.95$. Run the function and check the minimality of the result. *(1 pt)*

# Solutions · Mock 2 · A

## Exercise 1 · Integral and sequence

1. *[0.5 pt]* La fonction $x \mapsto \frac{e^{3x}}{3}$ is an antiderivative of $e^{3x}$. Thus $I_0 = [\frac{e^{3x}}{3}]_0^1 = \frac{e^3-1}{3}$.

2. *[0.5 pt]* Take $u(x) = x$ and $v(x) = \frac{e^{3x}}{3}$, two differentiable functions with continuous derivatives on $[0,1]$. Then $u' = 1$ and $v' = e^{3x}$. Integration by parts gives

$$\begin{aligned} I_1 &= \left[\frac{xe^{3x}}{3}\right]_0^1 - \frac{1}{3}\int_0^1 e^{3x}\,\mathrm{d}x \\ &= \frac{e^3}{3} - \frac{e^3-1}{9} = \frac{2e^3+1}{9}. \end{aligned}$$

3. *[0.5 pt]* For $x \in [0,1]$, $0 \le x^{n+1} \le x^n$. Since $e^{3x} > 0$, integrating over increasing bounds gives $0 \le I_{n+1} \le I_n$. The sequence is therefore positive and decreasing (in the weak sense). The inequalities are even strict: the integrands of $I_n$ and of $I_n - I_{n+1}$ are continuous and strictly positive on $]0,1[$.

4. *[1 pt]* $e^{3x} \le e^3$; so $0 \le I_n \le e^3 \int_0^1 x^n\,\mathrm{d}x = \frac{e^3}{n+1} \to 0$.

5. *[0.5 pt]* Apply integration by parts with $u(x) = x^{n+1}$ and $v(x) = \frac{e^{3x}}{3}$. Since $n+1 \ge 1$, the term at $x = 0$ is zero. Hence

$$\begin{aligned} I_{n+1} &= \left[\frac{x^{n+1}e^{3x}}{3}\right]_0^1 - \frac{n+1}{3}\int_0^1 x^n e^{3x}\,\mathrm{d}x \\ &= \frac{e^3}{3} - \frac{n+1}{3} I_n. \end{aligned}$$

6. *[0.5 pt]* For $0 \le x \le 1$, $e^{3x} \ge 1$ and $x^n \ge 0$. So $x^n \le x^n e^{3x}$; integrating,

$$I_n \ge \int_0^1 x^n\,\mathrm{d}x = \left[\frac{x^{n+1}}{n+1}\right]_0^1 = \frac{1}{n+1}.$$

7. *[0.5 pt]* The function $x \mapsto e^{3x}$ is increasing. On each interval $[\frac{j}{N}, \frac{j+1}{N}]$, the values at the endpoints bound the function. Integrating and then summing,

$$L_N = \frac{1}{N}\sum_{j=0}^{N-1} e^{\frac{3j}{N}} \le I_0 \le U_N = \frac{1}{N}\sum_{j=1}^{N} e^{\frac{3j}{N}}.$$

   The interior terms cancel in the difference, so $U_N - L_N = \frac{e^3-1}{N}$.

8. *[1 pt]* The width is at most $10^{-2}$ if $N \geq 100(e^3 - 1)$. The first integer satisfying this bound is $N = 1909$. Here is the computation of the bounds:

```python
from math import exp, ceil

def bornes():
    N = ceil(100*(exp(3)-1))
    bas = sum(exp(3*j/N) for j in range(N))/N
    haut = bas+(exp(3)-1)/N
    return N, bas, haut
```

## Exercise 2 · A logarithmic function

1. *[0.5 pt]* At $0^+$, the numerator tends to $-\infty$ and the denominator to $0^+$: limit $-\infty$; at $+\infty$, limit 0 by comparison of growth rates.

2. *[0.5 pt]* By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (2 + \ln x)}{x^2} = \frac{-1 - \ln x}{x^2}.$$

   The denominator is positive. The numerator vanishes at $s = e^{-1}$, is positive for $0 < x < s$ and negative for $x > s$. Thus $f$ is strictly increasing on $]0, s]$, then strictly decreasing on $[s, +\infty[$.

3. *[0.5 pt]* The maximum is attained only at $s = e^{-1}$ and equals

$$f(s) = \frac{2 - 1}{e^{-1}} = e^1.$$

   Since $x > 0$, $f(x) = 0$ is equivalent to $2 + \ln x = 0$, that is $x = e^{-2}$.

4. *[1 pt]* Differentiating $(-1 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-1 - \ln x)x^{-3} = \frac{2 \ln x + 1}{x^3}.$$

   The numerator vanishes at $t = e^{-\frac{1}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at $t$ and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. *[0.5 pt]* Let $m = \frac{e^1}{2}$ and $s = e^{-1}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at 0 and value $2m$ at $s$: the corollary of the intermediate value theorem (IVT) gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from $2m$ to $0 < m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

6. *[0.5 pt]* Let $F(x) = 2 \ln x + \frac{1}{2}(\ln x)^2$, differentiable for $x > 0$. Then $F'(x) = \frac{2}{x} + \frac{\ln x}{x} = f(x)$; $F$ is an antiderivative on $]0, +\infty[$.

7. *[0.5 pt]* With this antiderivative, $F(e) = \frac{5}{2}$ and $F(1) = 0$. The integral is therefore $\frac{5}{2}$, and the mean value is

$$\mu = \frac{1}{e - 1} \int_1^e f(x)\, \mathrm{d}x = \frac{5}{2(e - 1)}.$$

8. *[1 pt]* Domain $x > 2$; $x^2 - 4 = 21$, so $x = 5$, the negative root being excluded.

## Exercise 3 · Line, plane and orthogonal projection

1. *[0.5 pt]* $(x, y, z) = (2 + t, 2t, 2t)$, $t \in \mathbb{R}$.
2. *[0.5 pt]* $2 + 9t = 10$ gives $t = \frac{8}{9}$; $H = (\frac{26}{9}, \frac{16}{9}, \frac{16}{9})$.
3. *[0.5 pt]* The vector $\vec{n} = (1, 2, 2)$ is normal to $\mathcal{P}$. Since $H \in \mathcal{P}$ and $\overrightarrow{MH} = \frac{8}{9}\vec{n}$, $H$ is the orthogonal projection of $M$ onto the plane. The coordinate system being orthonormal, $\|\vec{n}\| = \sqrt{1+4+4} = 3$. Therefore
$$d(M, \mathcal{P}) = MH = |\frac{8}{9}|\, 3 = \frac{8}{3}.$$
4. *[1 pt]* Set $y = z = s$; $x = 10 - 4s$. So $(x, y, z) = (10, 0, 0) + s(-4, 1, 1)$.
5. *[0.5 pt]* Yes, its last two coordinates are equal. $H$ lies on the line and $\overrightarrow{MH}$ is orthogonal to every direction of the plane, hence to this line. Distance $\frac{8}{3}$.
6. *[0.5 pt]* $x + 2y + 2z = 2$; same normal vector, passing through $M$.
7. *[0.5 pt]* $\vec{n} \cdot (-4, 1, 1) = -4 + 2 + 2 = 0$: angle $\frac{\pi}{2}$.
8. *[1 pt]* $N = M + \vec{n}$ lies on the same normal line as $M$; its orthogonal projection onto $\mathcal{P}$ is therefore also $H$. Now $H \in \mathcal{P} \cap \mathcal{Q}$. For every point $P$ of this line, $NP^2 = NH^2 + HP^2$, so its minimum distance to $N$ is $NH$. Finally,
$$d(N, \mathcal{P}) = d(N, \mathcal{P} \cap \mathcal{Q}) = \|\overrightarrow{NH}\| = |\frac{8}{9} - 1|\, \|\vec{n}\| = \frac{1}{3}.$$

## Exercise 4 · Repeated trials and estimation

1. *[0.5 pt]* Repetition of a fixed number 18 of identical independent trials with two outcomes; $X \sim \mathcal{B}(18, \frac{1}{5})$.
2. *[0.5 pt]* $P(X = 2) = \binom{18}{2}(\frac{1}{5})^2(\frac{4}{5})^{16} \simeq 0.172263$.
3. *[0.5 pt]* $1 - P(X = 0) - P(X = 1) = 1 - (\frac{4}{5})^{18} - 18(\frac{1}{5})(\frac{4}{5})^{17} \simeq 0.900921$.
4. *[1 pt]* $E(X) = \frac{18}{5}$, $V(X) = \frac{72}{25}$.
5. *[0.5 pt]* For $N$ independent trials, the probability of no success is $(1-p)^N$. The required probability is therefore $1 - (\frac{4}{5})^N$. It is strictly increasing with $N$, since $0 < \frac{4}{5} < 1$. The calculations give
$$P_{13} \simeq 0.945024 < 0.95, \qquad P_{14} \simeq 0.956020 \geq 0.95.$$
Thus 14 is the smallest suitable number of trials.
6. *[0.5 pt]* Let $n = 18$. For $j \in \{0, \ldots, n\}$, an ordered outcome containing $j$ successes and $n - j$ failures has, by independence, probability $p^j(1-p)^{n-j}$. There are $\binom{n}{j}$ choices for the positions of the successes. These outcomes are mutually exclusive, so
$$P(X = j) = \binom{n}{j} p^j (1-p)^{n-j}, \qquad p = \frac{1}{5}.$$
7. *[0.5 pt]* The events $X = 1$, $X = 2$, $X = 3$ are mutually exclusive, so
$$P(1 \leq X \leq 3) = \sum_{j=1}^{3} \binom{18}{j} (\frac{1}{5})^j (\frac{4}{5})^{18-j} \simeq 0.483011.$$

8. *[1 pt]*

```python
from math import comb

def quantile():
    n, p = 18, 2/10
    j, cumul = 0, (1-p)**n
    while cumul < 0.95:
        j += 1
        cumul += comb(n, j)*p**j*(1-p)**(n-j)
    return j
```

The probabilities added are positive and the final cumulative value is 1, so the loop terminates. It returns 7. The cumulative values at ranks 6 and 7 are approximately 0.948729 and 0.983720: this confirms minimality.

# Mock 2 · B

**PAPER RESERVED FOR ASSESSMENT**

**Duration: 4 hours · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete approaches will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · Integral and sequence — 5 points

For every natural number $n$, let $I_n = \displaystyle\int_0^1 x^n e^{2x}\,\mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^{2x}$.

1. Calculate $I_0$. *(0.5 pt)*
2. Using integration by parts, calculate $I_1$. *(0.5 pt)*
3. Prove, by comparing the integrands, that $(I_n)$ is positive and decreasing. *(0.5 pt)*
4. Show that $I_n \leq \frac{e^2}{n+1}$ for all $n \in \mathbb{N}$. Deduce the limit of $(I_n)$. *(1 pt)*
5. Using integration by parts, express $I_{n+1}$ in terms of $I_n$ and $n$. *(0.5 pt)*
6. Show that $I_n \geq \frac{1}{n+1}$ for all $n \in \mathbb{N}$. *(0.5 pt)*
7. Split $[0,1]$ into $N$ intervals of equal length, with $N \geq 1$. Write the sums $L_N$ and $U_N$ of the areas of the rectangles built using the values of $e^{2x}$ at the left and right endpoints. Justify $L_N \leq I_0 \leq U_N$, then calculate $U_N - L_N$. *(0.5 pt)*
8. Find the smallest integer $N$ guaranteeing that this bracketing has a width at most equal to $10^{-2}$. Write a Python function that returns $N$, $L_N$ and $U_N$. *(1 pt)*

## Exercise 2 · A logarithmic function — 5 points

Consider the function $f$ defined on $(0, +\infty)$ by $f(x) = \frac{5+\ln x}{x}$. On note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of $f$ at $0^+$ and at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw up the table of variations of $f$. *(0.5 pt)*
3. State the value of the maximum of $f$, then solve $f(x) = 0$. *(0.5 pt)*
4. Study the convexity of $f$ and determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Prove that the equation $f(x) = \frac{1}{2}e^4$ has exactly two solutions in $(0, +\infty)$. *(0.5 pt)*
6. Determine an antiderivative of $f$ on $(0, +\infty)$. *(0.5 pt)*
7. Calculate the mean value of $f$ on $[1, e]$. *(0.5 pt)*
8. Independently of the previous questions, solve $\ln(x-3) + \ln(x+3) = \ln(27)$, after stating the set on which this equation is defined. *(1 pt)*

Mock 2 · B · continued

## Exercise 3 · Line, plane and orthogonal projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x + 2y + 2z = 11$ and the point $M(3, 0, 0)$. The line $d$ passes through $M$ and has direction vector $\vec{n} = (1, 2, 2)$.

1. Give a parametric representation of $d$. *(0.5 pt)*
2. Prove that $d$ meets $\mathcal{P}$ at a point $H$, then calculate the coordinates of $H$. *(0.5 pt)*
3. Justify that $H$ is the orthogonal projection of $M$ onto $\mathcal{P}$, then calculate the distance from $M$ to $\mathcal{P}$. *(0.5 pt)*
4. Consider the plane $\mathcal{Q}$ with equation $y = z$. Prove that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it. *(1 pt)*
5. Check that $H \in \mathcal{Q}$. Deduce the distance from $M$ to the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through $M$. *(0.5 pt)*
7. Determine the angle between $d$ and the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
8. Define the point $N$ by $\overrightarrow{MN} = \vec{n}$. Calculate the distances from $N$ to the plane $\mathcal{P}$ and to the line $\mathcal{P} \cap \mathcal{Q}$, then compare them. *(1 pt)*

## Exercise 4 · Repeated trials and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 20 times. The trials are independent and the probability of success is $p = \frac{1}{10}$ on each trial. Let $X$ denote the number of successes obtained.

1. Justify that $X$ follows a binomial distribution and state its parameters. *(0.5 pt)*
2. Calculate the probability of obtaining exactly two successes. *(0.5 pt)*
3. Using complementary events, express and then calculate the probability of obtaining at least two successes. *(0.5 pt)*
4. Calculate the expected value and the variance of $X$. *(1 pt)*
5. We now repeat the same trial $N \geq 1$ times, again independently. Find the smallest integer $N$ for which the probability of obtaining at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. *(0.5 pt)*
6. For an integer $j \in \{0, \ldots, 20\}$, prove the formula for $P(X = j)$ by counting the possible positions of the successes. *(0.5 pt)*
7. A series of 20 trials is accepted if its number of successes is between 1 and 3, inclusive. Calculate the probability that a series is accepted. *(0.5 pt)*
8. Write a Python function with no argument that returns the smallest integer $j$ such that $P(X \leq j) \geq 0.95$. Run the function and check the minimality of the result. *(1 pt)*

# Solutions · Mock 2 · B

## Exercise 1 · Integral and sequence

1. *[0.5 pt]* La fonction $x \mapsto \frac{e^{2x}}{2}$ is an antiderivative of $e^{2x}$. Thus $I_0 = [\frac{e^{2x}}{2}]_0^1 = \frac{e^2-1}{2}$.

2. *[0.5 pt]* Take $u(x) = x$ and $v(x) = \frac{e^{2x}}{2}$, two differentiable functions with continuous derivatives on $[0,1]$. Then $u' = 1$ and $v' = e^{2x}$. Integration by parts gives

$$\begin{aligned} I_1 &= \left[\frac{xe^{2x}}{2}\right]_0^1 - \frac{1}{2}\int_0^1 e^{2x}\,\mathrm{d}x \\ &= \frac{e^2}{2} - \frac{e^2-1}{4} = \frac{e^2+1}{4}. \end{aligned}$$

3. *[0.5 pt]* For $x \in [0,1]$, $0 \leq x^{n+1} \leq x^n$. Since $e^{2x} > 0$, integrating over increasing bounds gives $0 \leq I_{n+1} \leq I_n$. The sequence is therefore positive and decreasing (in the weak sense). The inequalities are even strict: the integrands of $I_n$ and of $I_n - I_{n+1}$ are continuous and strictly positive on $]0,1[$.

4. *[1 pt]* $e^{2x} \leq e^2$; so $0 \leq I_n \leq e^2 \int_0^1 x^n\,\mathrm{d}x = \frac{e^2}{n+1} \to 0$.

5. *[0.5 pt]* Apply integration by parts with $u(x) = x^{n+1}$ and $v(x) = \frac{e^{2x}}{2}$. Since $n+1 \geq 1$, the term at $x = 0$ is zero. Hence

$$\begin{aligned} I_{n+1} &= \left[\frac{x^{n+1}e^{2x}}{2}\right]_0^1 - \frac{n+1}{2}\int_0^1 x^n e^{2x}\,\mathrm{d}x \\ &= \frac{e^2}{2} - \frac{n+1}{2} I_n. \end{aligned}$$

6. *[0.5 pt]* For $0 \leq x \leq 1$, $e^{2x} \geq 1$ and $x^n \geq 0$. So $x^n \leq x^n e^{2x}$; integrating,

$$I_n \geq \int_0^1 x^n\,\mathrm{d}x = \left[\frac{x^{n+1}}{n+1}\right]_0^1 = \frac{1}{n+1}.$$

7. *[0.5 pt]* The function $x \mapsto e^{2x}$ is increasing. On each interval $[\frac{j}{N}, \frac{j+1}{N}]$, the values at the endpoints bound the function. Integrating and then summing,

$$L_N = \frac{1}{N}\sum_{j=0}^{N-1} e^{\frac{2j}{N}} \leq I_0 \leq U_N = \frac{1}{N}\sum_{j=1}^{N} e^{\frac{2j}{N}}.$$

   The interior terms cancel in the difference, so $U_N - L_N = \frac{e^2-1}{N}$.

8. *[1 pt]* The width is at most $10^{-2}$ if $N \geq 100(e^2-1)$. The first integer satisfying this bound is $N = 639$. Here is the computation of the bounds:

```python
from math import exp, ceil

def bornes():
    N = ceil(100*(exp(2)-1))
    bas = sum(exp(2*j/N) for j in range(N))/N
    haut = bas+(exp(2)-1)/N
    return N, bas, haut
```

## Exercise 2 · A logarithmic function

1. *[0.5 pt]* At $0^+$, the numerator tends to $-\infty$ and the denominator to $0^+$: limit $-\infty$; at $+\infty$, limit 0 by comparison of growth rates.

2. *[0.5 pt]* By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (5+\ln x)}{x^2} = \frac{-4-\ln x}{x^2}.$$

   The denominator is positive. The numerator vanishes at $s = e^{-4}$, is positive for $0 < x < s$ and negative for $x > s$. Thus $f$ is strictly increasing on $]0, s]$, then strictly decreasing on $[s, +\infty[$.

3. *[0.5 pt]* The maximum is attained only at $s = e^{-4}$ and equals

$$f(s) = \frac{5-4}{e^{-4}} = e^4.$$

   Since $x > 0$, $f(x) = 0$ is equivalent to $5 + \ln x = 0$, that is $x = e^{-5}$.

4. *[1 pt]* Differentiating $(-4-\ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-4-\ln x)x^{-3} = \frac{2\ln x + 7}{x^3}.$$

   The numerator vanishes at $t = e^{-\frac{7}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at $t$ and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. *[0.5 pt]* Let $m = \frac{e^4}{2}$ and $s = e^{-4}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at 0 and value $2m$ at $s$: the corollary of the intermediate value theorem (IVT) gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from $2m$ to $0 < m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

6. *[0.5 pt]* Let $F(x) = 5\ln x + \frac{1}{2}(\ln x)^2$, differentiable for $x > 0$. Then $F'(x) = \frac{5}{x} + \frac{\ln x}{x} = f(x)$; $F$ is an antiderivative on $]0, +\infty[$.

7. *[0.5 pt]* With this antiderivative, $F(e) = \frac{11}{2}$ and $F(1) = 0$. The integral is therefore $\frac{11}{2}$, and the mean value is

$$\mu = \frac{1}{e-1}\int_1^e f(x)\,\mathrm{d}x = \frac{11}{2(e-1)}.$$

8. *[1 pt]* Domain $x > 3$; $x^2 - 9 = 27$, so $x = 6$, the negative root being excluded.

## Exercise 3 · Line, plane and orthogonal projection

1. *[0.5 pt]* $(x, y, z) = (3 + t, 2t, 2t)$, $t \in \mathbb{R}$.
2. *[0.5 pt]* $3 + 9t = 11$ gives $t = \frac{8}{9}$; $H = (\frac{35}{9}, \frac{16}{9}, \frac{16}{9})$.
3. *[0.5 pt]* The vector $\vec{n} = (1, 2, 2)$ is normal to $\mathcal{P}$. Since $H \in \mathcal{P}$ and $\overrightarrow{MH} = \frac{8}{9}\vec{n}$, $H$ is the orthogonal projection of $M$ onto the plane. The coordinate system being orthonormal, $\|\vec{n}\| = \sqrt{1+4+4} = 3$. Therefore
$$d(M, \mathcal{P}) = MH = |\frac{8}{9}|\, 3 = \frac{8}{3}.$$
4. *[1 pt]* Set $y = z = s$; $x = 11 - 4s$. So $(x, y, z) = (11, 0, 0) + s(-4, 1, 1)$.
5. *[0.5 pt]* Yes, its last two coordinates are equal. $H$ lies on the line and $\overrightarrow{MH}$ is orthogonal to every direction of the plane, hence to this line. Distance $\frac{8}{3}$.
6. *[0.5 pt]* $x + 2y + 2z = 3$; same normal vector, passing through $M$.
7. *[0.5 pt]* $\vec{n} \cdot (-4, 1, 1) = -4 + 2 + 2 = 0$: angle $\frac{\pi}{2}$.
8. *[1 pt]* $N = M + \vec{n}$ lies on the same normal line as $M$; its orthogonal projection onto $\mathcal{P}$ is therefore also $H$. Now $H \in \mathcal{P} \cap \mathcal{Q}$. For every point $P$ of this line, $NP^2 = NH^2 + HP^2$, so its minimum distance to $N$ is $NH$. Finally,
$$d(N, \mathcal{P}) = d(N, \mathcal{P} \cap \mathcal{Q}) = \|\overrightarrow{NH}\| = |\frac{8}{9} - 1|\, \|\vec{n}\| = \frac{1}{3}.$$

## Exercise 4 · Repeated trials and estimation

1. *[0.5 pt]* Repetition of a fixed number 20 of identical independent trials with two outcomes; $X \sim \mathcal{B}(20, \frac{1}{10})$.
2. *[0.5 pt]* $P(X = 2) = \binom{20}{2}(\frac{1}{10})^2(\frac{9}{10})^{18} \simeq 0.285180$.
3. *[0.5 pt]* $1 - P(X = 0) - P(X = 1) = 1 - (\frac{9}{10})^{20} - 20(\frac{1}{10})(\frac{9}{10})^{19} \simeq 0.608253$.
4. *[1 pt]* $E(X) = 2$, $V(X) = \frac{9}{5}$.
5. *[0.5 pt]* For $N$ independent trials, the probability of no success is $(1-p)^N$. The required probability is therefore $1 - (\frac{9}{10})^N$. It is strictly increasing with $N$, since $0 < \frac{9}{10} < 1$. The calculations give
$$P_{28} \simeq 0.947665 < 0.95, \qquad P_{29} \simeq 0.952899 \geq 0.95.$$
Thus 29 is the smallest suitable number of trials.
6. *[0.5 pt]* Let $n = 20$. For $j \in \{0, \ldots, n\}$, an ordered outcome containing $j$ successes and $n - j$ failures has, by independence, probability $p^j(1-p)^{n-j}$. There are $\binom{n}{j}$ choices for the positions of the successes. These outcomes are mutually exclusive, so
$$P(X = j) = \binom{n}{j} p^j (1-p)^{n-j}, \qquad p = \frac{1}{10}.$$
7. *[0.5 pt]* The events $X = 1$, $X = 2$, $X = 3$ are mutually exclusive, so
$$P(1 \leq X \leq 3) = \sum_{j=1}^{3} \binom{20}{j} (\frac{1}{10})^j (\frac{9}{10})^{20-j} \simeq 0.745470.$$

8. *[1 pt]*

```python
from math import comb

def quantile():
    n, p = 20, 1/10
    j, cumul = 0, (1-p)**n
    while cumul < 0.95:
        j += 1
        cumul += comb(n, j)*p**j*(1-p)**(n-j)
    return j
```

The probabilities added are positive and the final cumulative value is 1, so the loop terminates. It returns 4. The cumulative values at ranks 3 and 4 are approximately 0.867047 and 0.956826: this confirms minimality.

# Final mock · Practice 1

**PRACTICE PAPER · TRAINING**

**Duration: 4 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · A logarithmic function — 5 points

Consider the function $f$ defined on $(0, +\infty)$ by $f(x) = \frac{2+\ln x}{x}$. Let $\mathcal{C}_f$ denote its graph.

1. Determine the limits of $f$ at $0^+$ and at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw up the table of variations of $f$. *(0.5 pt)*
3. State the maximum value of $f$, then solve $f(x) = 0$. *(0.5 pt)*
4. Study the convexity of $f$ and determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Prove that the equation $f(x) = \frac{1}{2}e^1$ has exactly two solutions in $(0, +\infty)$. *(0.5 pt)*
6. Find an antiderivative of $f$ on $(0, +\infty)$. *(0.5 pt)*
7. Calculate the mean value of $f$ on $[1, e]$. *(0.5 pt)*
8. Independently of the previous questions, solve $\ln(x-2) + \ln(x+2) = \ln(32)$, after stating the set on which this equation is defined. *(1 pt)*

## Exercise 2 · Periodic phenomenon — 5 points

A point moves on a wheel turning at constant speed in a vertical plane. Its height above the ground is modelled by $h(t) = 9 - 3\cos\left(\frac{\pi t}{5}\right)$, where $t$ is expressed in seconds and $h(t)$ in metres. We study one complete revolution, for $t \in [0, 10]$.

1. Give a period of the model extended to $\mathbb{R}$, then the minimum and maximum heights over one revolution. *(0.5 pt)*
2. Calculate $h'(t)$ and draw up the table of variations of $h$ on $[0, 10]$. *(0.5 pt)*
3. Find all times $t \in [0, 10]$ such that $h(t) = \frac{21}{2}$. *(0.5 pt)*
4. Calculate the mean value of $h$ on $[0, 10]$. *(1 pt)*
5. Study the convexity of $h$ on the interval of study and determine the coordinates of the points of inflection of its graph. *(0.5 pt)*
6. Determine the total time during which the height is at least equal to the level in question 3. *(0.5 pt)*
7. The vertical speed is given by $h'(t)$. Determine its maximum and minimum values, in metres per second, and the times at which they are reached. *(0.5 pt)*
8. Calculate the vertical displacement $h(10) - h(0)$, then the sum of the height travelled on the way up and that travelled on the way down. Explain why these two results are different. *(1 pt)*

Final mock · Practice 1 · continued

## Exercise 3 · Line, plane and projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x + 2y + 2z = 11$ and the point $M(2, 0, 0)$. The line $d$ passes through $M$ and has direction vector $\vec{n} = (1, 2, 2)$.

1. Give a parametric representation of $d$. *(0.5 pt)*
2. Prove that $d$ meets $\mathcal{P}$ at a point $H$, then calculate the coordinates of $H$. *(0.5 pt)*
3. Justify that $H$ is the orthogonal projection of $M$ onto $\mathcal{P}$, then calculate the distance from $M$ to $\mathcal{P}$. *(0.5 pt)*
4. Consider the plane $\mathcal{Q}$ with equation $y = z$. Prove that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it. *(1 pt)*
5. Check that $H \in \mathcal{Q}$. Deduce the distance from $M$ to the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through $M$. *(0.5 pt)*
7. Determine the angle between $d$ and the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
8. Define the point $N$ by $\overrightarrow{MN} = \vec{n}$. Calculate the distances from $N$ to the plane $\mathcal{P}$ and to the line $\mathcal{P} \cap \mathcal{Q}$, then compare them. *(1 pt)*

## Exercise 4 · Repetitions and estimation — 5 points

We repeat 20 times a trial with two outcomes, called success and failure. The trials are independent and the probability of success is $p = \frac{1}{5}$ on each trial. Let $X$ denote the number of successes obtained.

1. Justify that $X$ follows a binomial distribution and state its parameters. *(0.5 pt)*
2. Calculate the probability of obtaining exactly two successes. *(0.5 pt)*
3. Using complementary events, express and then calculate the probability of obtaining at least two successes. *(0.5 pt)*
4. Calculate the expected value and the variance of $X$. *(1 pt)*
5. We now repeat the same trial a number $N \geq 1$ of times, still independently. Determine the smallest integer $N$ for which the probability of obtaining at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. *(0.5 pt)*
6. For an integer $j \in \{0, \ldots, 20\}$, prove the formula for $P(X = j)$ by counting the possible positions of the successes. *(0.5 pt)*
7. For $N$ independent trials with the same parameter $p$, let $F_N$ denote the frequency of successes. Prove the expressions for $E(F_N)$ and $V(F_N)$ in terms of $N$ and $p$. *(0.5 pt)*
8. Using the Bienaymé–Chebyshev inequality, determine a sufficient size $N$ so that $P(|F_N - p| < 0.02) \geq 0.95$. Is the size obtained necessarily the smallest possible for this probability? Justify. *(1 pt)*

# Solutions · Final mock · Practice 1

## Exercise 1 · A logarithmic function

1. *[0.5 pt]* At $0^+$, the numerator tends to $-\infty$ and the denominator to $0^+$: limit $-\infty$; at $+\infty$, limit $0$ by comparison of growth rates.

2. *[0.5 pt]* By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (2+\ln x)}{x^2} = \frac{-1-\ln x}{x^2}.$$

The denominator is positive. The numerator vanishes at $s = e^{-1}$, is positive for $0 < x < s$ and negative for $x > s$. Hence $f$ is strictly increasing on $]0, s]$, then strictly decreasing on $[s, +\infty[$.

3. *[0.5 pt]* The maximum is attained only at $s = e^{-1}$ and equals

$$f(s) = \frac{2-1}{e^{-1}} = e^1.$$

Since $x > 0$, $f(x) = 0$ is equivalent to $2 + \ln x = 0$, that is $x = e^{-2}$.

4. *[1 pt]* Differentiating $(-1-\ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-1-\ln x)x^{-3} = \frac{2\ln x + 1}{x^3}.$$

The numerator vanishes at $t = e^{-\frac{1}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at $t$ and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. *[0.5 pt]* Let $m = \frac{e^1}{2}$ and $s = e^{-1}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at $0$ and value $2m$ at $s$: the corollary of the IVT gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from $2m$ to $0 < m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

6. *[0.5 pt]* Let $F(x) = 2\ln x + \frac{1}{2}(\ln x)^2$, differentiable for $x > 0$. Then $F'(x) = \frac{2}{x} + \frac{\ln x}{x} = f(x)$; $F$ is an antiderivative on $]0, +\infty[$.

7. *[0.5 pt]* With this antiderivative, $F(e) = \frac{5}{2}$ and $F(1) = 0$. The integral therefore equals $\frac{5}{2}$, and the mean value is

$$\mu = \frac{1}{e-1}\int_1^e f(x)\,\mathrm{d}x = \frac{5}{2(e-1)}.$$

8. *[1 pt]* Domain $x > 2$; $x^2 - 4 = 32$, so $x = 6$, the negative root being excluded.

## Exercise 2 · Periodic phenomenon

1. *[0.5 pt]* Period 10 s; minimum 6 m, maximum 12 m.

2. *[0.5 pt]* $h'(t) = (\frac{3}{5}\pi)\sin(\frac{\pi t}{5})$; increasing on $[0, 5]$, decreasing on $[5, 10]$.

3. *[0.5 pt]* The equation is equivalent to $\cos(\frac{\pi t}{5}) = -\frac{1}{2}$. Over one revolution, $\frac{\pi t}{5} \in [0, 2\pi]$; the suitable angles are $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$. Hence $t = \frac{10}{3}$ s or $t = \frac{20}{3}$ s.

4. *[1 pt]* An antiderivative is $F(t) = 9t - \frac{15}{\pi}\sin(\frac{\pi t}{5})$. Since $\sin 0 = \sin(2\pi) = 0$,

$$\overline{h} = \frac{1}{10}\int_0^{10} h(t)\,\mathrm{d}t = \frac{F(10) - F(0)}{10} = \frac{90}{10} = 9 \text{ m}.$$

5. *[0.5 pt]* The second derivative is $h''(t) = \frac{3\pi^2}{25}\cos(\frac{\pi t}{5})$. Its sign is that of the cosine: positive on $[0, \frac{5}{2}[$ and $]\frac{15}{2}, 10]$, negative between these two intervals. The function is convex on $[0, \frac{5}{2}]$ and $[\frac{15}{2}, 10]$, concave on $[\frac{5}{2}, \frac{15}{2}]$. The two changes of convexity give the points of inflection $(\frac{5}{2}, 9)$ and $(\frac{15}{2}, 9)$.

6. *[0.5 pt]* Between $\frac{10}{3}$ and $\frac{20}{3}$; duration $\frac{10}{3}$ s.

7. *[0.5 pt]* $h'$ reaches $\frac{3}{5}\pi$ m s$^{-1}$ at $t = \frac{5}{2}$ and its opposite at $t = \frac{15}{2}$.

8. *[1 pt]* Displacement $h(10) - h(0) = 0$. Rise of 6 m then fall of 6 m: total distance 12 m, equal to the integral of $|h'|$.

## Exercise 3 · Line, plane and projection

1. *[0.5 pt]* $(x, y, z) = (2 + t, 2t, 2t)$, $t \in \mathbb{R}$.

2. *[0.5 pt]* $2 + 9t = 11$ gives $t = 1$; $H = (3, 2, 2)$.

3. *[0.5 pt]* The vector $\vec{n} = (1, 2, 2)$ is normal to $\mathcal{P}$. Since $H \in \mathcal{P}$ and $\overrightarrow{MH} = \vec{n}$, $H$ is the orthogonal projection of $M$ onto the plane. The coordinate system being orthonormal, $\|\vec{n}\| = \sqrt{1 + 4 + 4} = 3$. Hence
$$d(M, \mathcal{P}) = MH = |1|\,3 = 3.$$

4. *[1 pt]* Set $y = z = s$; $x = 11 - 4s$. Hence $(x, y, z) = (11, 0, 0) + s(-4, 1, 1)$.

5. *[0.5 pt]* Yes, its last two coordinates are equal. $H$ lies on the line and $\overrightarrow{MH}$ is orthogonal to every direction of the plane, hence to this line. Distance 3.

6. *[0.5 pt]* $x + 2y + 2z = 2$; same normal vector and passing through $M$.

7. *[0.5 pt]* $\vec{n} \cdot (-4, 1, 1) = -4 + 2 + 2 = 0$: angle $\frac{\pi}{2}$.

8. *[1 pt]* $N = M + \vec{n}$ lies on the same normal line as $M$; its projection onto $\mathcal{P}$ is therefore also $H$. Now $H \in \mathcal{P} \cap \mathcal{Q}$. For every $P$ on this line, $NP^2 = NH^2 + HP^2$, so its minimal distance to $N$ is $NH$. Finally,
$$d(N, \mathcal{P}) = d(N, \mathcal{P} \cap \mathcal{Q}) = \|\overrightarrow{NH}\| = |1 - 1|\,\|\vec{n}\| = 0.$$

## Exercise 4 · Repetitions and estimation

1. *[0.5 pt]* Repetition of a fixed number 20 of identical independent trials with two outcomes; $X \sim \mathcal{B}(20, \frac{1}{5})$.

2. *[0.5 pt]* $P(X = 2) = \binom{20}{2}(\frac{1}{5})^2(\frac{4}{5})^{18} \simeq 0.136909$.

3. *[0.5 pt]* $1 - P(X = 0) - P(X = 1) = 1 - (\frac{4}{5})^{20} - 20(\frac{1}{5})(\frac{4}{5})^{19} \simeq 0.930825$.

4. *[1 pt]* $E(X) = 4$, $V(X) = \frac{16}{5}$.

5. *[0.5 pt]* For $N$ independent trials, the probability of no success is $(1 - p)^N$. The required probability is therefore $1 - (\frac{4}{5})^N$. It is strictly increasing in $N$, since $0 < \frac{4}{5} < 1$. The calculations give
$$P_{13} \simeq 0.945024 < 0.95, \qquad P_{14} \simeq 0.956020 \geq 0.95.$$
Thus 14 is the smallest suitable number of trials.

6. *[0.5 pt]* Let $n = 20$. For $j \in \{0, \dots, n\}$, an ordered outcome containing $j$ successes and $n - j$ failures has, by independence, probability $p^j(1-p)^{n-j}$. There are $\binom{n}{j}$ choices for the positions of the successes. These outcomes are mutually exclusive, so

$$P(X = j) = \binom{n}{j} p^j (1-p)^{n-j}, \qquad p = \frac{1}{5}.$$

7. *[0.5 pt]* If $B_1, \dots, B_N$ are the independent indicator variables of the successes, $F_N = \frac{B_1 + \cdots + B_N}{N}$. With $E(B_i) = p$ and $V(B_i) = p(1-p)$,

$$E(F_N) = \frac{Np}{N} = \frac{1}{5}, \qquad V(F_N) = \frac{Np(1-p)}{N^2} = \frac{\frac{4}{25}}{N}.$$

   Independence is required for adding variances, not for the linearity of expected value.

8. *[1 pt]* The Bienaymé–Chebyshev inequality applied to $F_N$ gives

$$P(|F_N - p| \geq 0.02) \leq \frac{p(1-p)}{N(0.02)^2}.$$

   To guarantee that this probability is at most 0.05, it suffices that

$$N \geq \frac{\frac{4}{25}}{0.05 \times (0.02)^2} = 8000.$$

   Taking the complement, we then have $P(|F_N - p| < 0.02) \geq 0.95$. The size 8000 is sufficient; the inequality being an upper bound, it does not necessarily give the minimum size for the exact probability.

# Final mock · Practice 2

**PRACTICE PAPER · TRAINING**

**Duration: 4 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · A logarithmic function — 5 points

Consider the function $f$ defined on $(0, +\infty)$ by $f(x) = \frac{5+\ln x}{x}$. Let $\mathcal{C}_f$ denote its graph.

1. Determine the limits of $f$ at $0^+$ and at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw up the table of variations of $f$. *(0.5 pt)*
3. State the maximum value of $f$, then solve $f(x) = 0$. *(0.5 pt)*
4. Study the convexity of $f$ and determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Prove that the equation $f(x) = \frac{1}{2}e^4$ has exactly two solutions in $(0, +\infty)$. *(0.5 pt)*
6. Find an antiderivative of $f$ on $(0, +\infty)$. *(0.5 pt)*
7. Calculate the mean value of $f$ on $[1, e]$. *(0.5 pt)*
8. Independently of the previous questions, solve $\ln(x-3) + \ln(x+3) = \ln(40)$, after stating the set on which this equation is defined. *(1 pt)*

## Exercise 2 · Periodic phenomenon — 5 points

A point moves on a wheel turning at constant speed in a vertical plane. Its height above the ground is modelled by $h(t) = 12 - 4\cos\left(\frac{\pi t}{4}\right)$, where $t$ is expressed in seconds and $h(t)$ in metres. We study one complete revolution, for $t \in [0, 8]$.

1. Give a period of the model extended to $\mathbb{R}$, then the minimum and maximum heights over one revolution. *(0.5 pt)*
2. Calculate $h'(t)$ and draw up the table of variations of $h$ on $[0, 8]$. *(0.5 pt)*
3. Find all times $t \in [0, 8]$ such that $h(t) = 14$. *(0.5 pt)*
4. Calculate the mean value of $h$ on $[0, 8]$. *(1 pt)*
5. Study the convexity of $h$ on the interval of study and determine the coordinates of the points of inflection of its graph. *(0.5 pt)*
6. Determine the total time during which the height is at least equal to the level in question 3. *(0.5 pt)*
7. The vertical speed is given by $h'(t)$. Determine its maximum and minimum values, in metres per second, and the times at which they are reached. *(0.5 pt)*
8. Calculate the vertical displacement $h(8) - h(0)$, then the sum of the height travelled on the way up and that travelled on the way down. Explain why these two results are different. *(1 pt)*

Final mock · Practice 2 · continued

## Exercise 3 · Line, plane and projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x + 2y + 2z = 7$ and the point $M(3, 0, 0)$. The line $d$ passes through $M$ and has direction vector $\vec{n} = (1, 2, 2)$.

1. Give a parametric representation of $d$. *(0.5 pt)*
2. Prove that $d$ meets $\mathcal{P}$ at a point $H$, then calculate the coordinates of $H$. *(0.5 pt)*
3. Justify that $H$ is the orthogonal projection of $M$ onto $\mathcal{P}$, then calculate the distance from $M$ to $\mathcal{P}$. *(0.5 pt)*
4. Consider the plane $\mathcal{Q}$ with equation $y = z$. Prove that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it. *(1 pt)*
5. Check that $H \in \mathcal{Q}$. Deduce the distance from $M$ to the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through $M$. *(0.5 pt)*
7. Determine the angle between $d$ and the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
8. Define the point $N$ by $\overrightarrow{MN} = \vec{n}$. Calculate the distances from $N$ to the plane $\mathcal{P}$ and to the line $\mathcal{P} \cap \mathcal{Q}$, then compare them. *(1 pt)*

## Exercise 4 · Repetitions and estimation — 5 points

We repeat 17 times a trial with two outcomes, called success and failure. The trials are independent and the probability of success is $p = \frac{3}{10}$ on each trial. Let $X$ denote the number of successes obtained.

1. Justify that $X$ follows a binomial distribution and state its parameters. *(0.5 pt)*
2. Calculate the probability of obtaining exactly two successes. *(0.5 pt)*
3. Using complementary events, express and then calculate the probability of obtaining at least two successes. *(0.5 pt)*
4. Calculate the expected value and the variance of $X$. *(1 pt)*
5. We now repeat the same trial a number $N \geq 1$ of times, still independently. Determine the smallest integer $N$ for which the probability of obtaining at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. *(0.5 pt)*
6. For an integer $j \in \{0, \ldots, 17\}$, prove the formula for $P(X = j)$ by counting the possible positions of the successes. *(0.5 pt)*
7. For $N$ independent trials with the same parameter $p$, let $F_N$ denote the frequency of successes. Prove the expressions for $E(F_N)$ and $V(F_N)$ in terms of $N$ and $p$. *(0.5 pt)*
8. Using the Bienaymé–Chebyshev inequality, determine a sufficient size $N$ so that $P(|F_N - p| < 0.02) \geq 0.95$. Is the size obtained necessarily the smallest possible for this probability? Justify. *(1 pt)*

# Solutions · Final mock · Practice 2

## Exercise 1 · A logarithmic function

1. *[0.5 pt]* At $0^+$, the numerator tends to $-\infty$ and the denominator to $0^+$: limit $-\infty$; at $+\infty$, limit 0 by comparison of growth rates.

2. *[0.5 pt]* By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (5 + \ln x)}{x^2} = \frac{-4 - \ln x}{x^2}.$$

The denominator is positive. The numerator vanishes at $s = e^{-4}$, is positive for $0 < x < s$ and negative for $x > s$. Hence $f$ is strictly increasing on $]0, s]$, then strictly decreasing on $[s, +\infty[$.

3. *[0.5 pt]* The maximum is attained only at $s = e^{-4}$ and equals

$$f(s) = \frac{5 - 4}{e^{-4}} = e^4.$$

Since $x > 0$, $f(x) = 0$ is equivalent to $5 + \ln x = 0$, that is $x = e^{-5}$.

4. *[1 pt]* Differentiating $(-4 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-4 - \ln x)x^{-3} = \frac{2\ln x + 7}{x^3}.$$

The numerator vanishes at $t = e^{-\frac{7}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at $t$ and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. *[0.5 pt]* Let $m = \frac{e^4}{2}$ and $s = e^{-4}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at 0 and value $2m$ at $s$: the corollary of the IVT gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from $2m$ to $0 < m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

6. *[0.5 pt]* Let $F(x) = 5\ln x + \frac{1}{2}(\ln x)^2$, differentiable for $x > 0$. Then $F'(x) = \frac{5}{x} + \frac{\ln x}{x} = f(x)$; $F$ is an antiderivative on $]0, +\infty[$.

7. *[0.5 pt]* With this antiderivative, $F(e) = \frac{11}{2}$ and $F(1) = 0$. The integral therefore equals $\frac{11}{2}$, and the mean value is

$$\mu = \frac{1}{e - 1}\int_1^e f(x)\ \mathrm{d}x = \frac{11}{2(e - 1)}.$$

8. *[1 pt]* Domain $x > 3$; $x^2 - 9 = 40$, so $x = 7$, the negative root being excluded.

## Exercise 2 · Periodic phenomenon

1. *[0.5 pt]* Period 8 s; minimum 8 m, maximum 16 m.

2. *[0.5 pt]* $h'(t) = \pi \sin(\frac{\pi t}{4})$; increasing on $[0, 4]$, decreasing on $[4, 8]$.

3. *[0.5 pt]* The equation is equivalent to $\cos(\frac{\pi t}{4}) = -\frac{1}{2}$. Over one revolution, $\frac{\pi t}{4} \in [0, 2\pi]$; the suitable angles are $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$. Hence $t = \frac{8}{3}$ s or $t = \frac{16}{3}$ s.

4. *[1 pt]* An antiderivative is $F(t) = 12t - \frac{16}{\pi}\sin(\frac{\pi t}{4})$. Since $\sin 0 = \sin(2\pi) = 0$,

$$\overline{h} = \frac{1}{8}\int_0^8 h(t)\ \mathrm{d}t = \frac{F(8) - F(0)}{8} = \frac{96}{8} = 12 \text{ m}.$$

5. *[0.5 pt]* The second derivative is $h''(t) = \frac{4\pi^2}{16}\cos(\frac{\pi t}{4})$. Its sign is that of the cosine: positive on $[0,2[$ and $]6,8]$, negative between these two intervals. The function is convex on $[0,2]$ and $[6,8]$, concave on $[2,6]$. The two changes of convexity give the points of inflection $(2,12)$ and $(6,12)$.

6. *[0.5 pt]* Between $\frac{8}{3}$ and $\frac{16}{3}$; duration $\frac{8}{3}$ s.

7. *[0.5 pt]* $h'$ reaches $1\pi$ m s$^{-1}$ at $t = 2$ and its opposite at $t = 6$.

8. *[1 pt]* Displacement $h(8) - h(0) = 0$. Rise of 8 m then fall of 8 m: total distance 16 m, equal to the integral of $|h'|$.

## Exercise 3 · Line, plane and projection

1. *[0.5 pt]* $(x,y,z) = (3+t, 2t, 2t)$, $t \in \mathbb{R}$.

2. *[0.5 pt]* $3 + 9t = 7$ gives $t = \frac{4}{9}$; $H = (\frac{31}{9}, \frac{8}{9}, \frac{8}{9})$.

3. *[0.5 pt]* The vector $\vec{n} = (1,2,2)$ is normal to $\mathcal{P}$. Since $H \in \mathcal{P}$ and $\overrightarrow{MH} = \frac{4}{9}\vec{n}$, $H$ is the orthogonal projection of $M$ onto the plane. The coordinate system being orthonormal, $\|\vec{n}\| = \sqrt{1+4+4} = 3$. Hence
$$d(M,\mathcal{P}) = MH = |\frac{4}{9}|\,3 = \frac{4}{3}.$$

4. *[1 pt]* Set $y = z = s$; $x = 7 - 4s$. Hence $(x,y,z) = (7,0,0) + s(-4,1,1)$.

5. *[0.5 pt]* Yes, its last two coordinates are equal. $H$ lies on the line and $\overrightarrow{MH}$ is orthogonal to every direction of the plane, hence to this line. Distance $\frac{4}{3}$.

6. *[0.5 pt]* $x + 2y + 2z = 3$; same normal vector and passing through $M$.

7. *[0.5 pt]* $\vec{n} \cdot (-4,1,1) = -4 + 2 + 2 = 0$: angle $\frac{\pi}{2}$.

8. *[1 pt]* $N = M + \vec{n}$ lies on the same normal line as $M$; its projection onto $\mathcal{P}$ is therefore also $H$. Now $H \in \mathcal{P} \cap \mathcal{Q}$. For every $P$ on this line, $NP^2 = NH^2 + HP^2$, so its minimal distance to $N$ is $NH$. Finally,
$$d(N,\mathcal{P}) = d(N,\mathcal{P}\cap\mathcal{Q}) = \|\overrightarrow{NH}\| = |\frac{4}{9} - 1|\,\|\vec{n}\| = \frac{5}{3}.$$

## Exercise 4 · Repetitions and estimation

1. *[0.5 pt]* Repetition of a fixed number 17 of identical independent trials with two outcomes; $X \sim \mathcal{B}(17, \frac{3}{10})$.

2. *[0.5 pt]* $P(X=2) = \binom{17}{2}(\frac{3}{10})^2(\frac{7}{10})^{15} \simeq 0.058110$.

3. *[0.5 pt]* $1 - P(X=0) - P(X=1) = 1 - (\frac{7}{10})^{17} - 17(\frac{3}{10})(\frac{7}{10})^{16} \simeq 0.980725$.

4. *[1 pt]* $E(X) = \frac{51}{10}$, $V(X) = \frac{357}{100}$.

5. *[0.5 pt]* For $N$ independent trials, the probability of no success is $(1-p)^N$. The required probability is therefore $1 - (\frac{7}{10})^N$. It is strictly increasing in $N$, since $0 < \frac{7}{10} < 1$. The calculations give
$$P_8 \simeq 0.942352 < 0.95, \qquad P_9 \simeq 0.959646 \geq 0.95.$$
Thus 9 is the smallest suitable number of trials.

6. *[0.5 pt]* Let $n = 17$. For $j \in \{0, \dots, n\}$, an ordered outcome containing $j$ successes and $n - j$ failures has, by independence, probability $p^j(1-p)^{n-j}$. There are $\binom{n}{j}$ choices for the positions of the successes. These outcomes are mutually exclusive, so

$$P(X = j) = \binom{n}{j} p^j (1-p)^{n-j}, \qquad p = \frac{3}{10}.$$

7. *[0.5 pt]* If $B_1, \dots, B_N$ are the independent indicator variables of the successes, $F_N = \frac{B_1 + \cdots + B_N}{N}$. With $E(B_i) = p$ and $V(B_i) = p(1-p)$,

$$E(F_N) = \frac{Np}{N} = \frac{3}{10}, \qquad V(F_N) = \frac{Np(1-p)}{N^2} = \frac{\frac{21}{100}}{N}.$$

   Independence is required for adding variances, not for the linearity of expected value.

8. *[1 pt]* The Bienaymé–Chebyshev inequality applied to $F_N$ gives

$$P(|F_N - p| \geq 0.02) \leq \frac{p(1-p)}{N(0.02)^2}.$$

   To guarantee that this probability is at most 0.05, it suffices that

$$N \geq \frac{\frac{21}{100}}{0.05 \times (0.02)^2} = 10500.$$

   Taking the complement, we then have $P(|F_N - p| < 0.02) \geq 0.95$. The size 10500 is sufficient; the inequality being an upper bound, it does not necessarily give the minimum size for the exact probability.

# Final mock · A

**PAPER RESERVED FOR ASSESSMENT**

**Duration: 4 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · A logarithmic function — 5 points

Consider the function $f$ defined on $(0, +\infty)$ by $f(x) = \frac{4+\ln x}{x}$. Let $\mathcal{C}_f$ denote its graph.

1. Determine the limits of $f$ at $0^+$ and at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw up the table of variations of $f$. *(0.5 pt)*
3. State the maximum value of $f$, then solve $f(x) = 0$. *(0.5 pt)*
4. Study the convexity of $f$ and determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Prove that the equation $f(x) = \frac{1}{2}e^3$ has exactly two solutions in $(0, +\infty)$. *(0.5 pt)*
6. Find an antiderivative of $f$ on $(0, +\infty)$. *(0.5 pt)*
7. Calculate the mean value of $f$ on $[1, e]$. *(0.5 pt)*
8. Independently of the previous questions, solve $\ln(x-1) + \ln(x+1) = \ln(15)$, after stating the set on which this equation is defined. *(1 pt)*

## Exercise 2 · Periodic phenomenon — 5 points

A point moves on a wheel turning at constant speed in a vertical plane. Its height above the ground is modelled by $h(t) = 7 - 2\cos\left(\frac{\pi t}{3}\right)$, where $t$ is expressed in seconds and $h(t)$ in metres. We study one complete revolution, for $t \in [0, 6]$.

1. Give a period of the model extended to $\mathbb{R}$, then the minimum and maximum heights over one revolution. *(0.5 pt)*
2. Calculate $h'(t)$ and draw up the table of variations of $h$ on $[0, 6]$. *(0.5 pt)*
3. Find all times $t \in [0, 6]$ such that $h(t) = 8$. *(0.5 pt)*
4. Calculate the mean value of $h$ on $[0, 6]$. *(1 pt)*
5. Study the convexity of $h$ on the interval of study and determine the coordinates of the points of inflection of its graph. *(0.5 pt)*
6. Determine the total time during which the height is at least equal to the level in question 3. *(0.5 pt)*
7. The vertical speed is given by $h'(t)$. Determine its maximum and minimum values, in metres per second, and the times at which they are reached. *(0.5 pt)*
8. Calculate the vertical displacement $h(6) - h(0)$, then the sum of the height travelled on the way up and that travelled on the way down. Explain why these two results are different. *(1 pt)*

Final mock · A · continued

## Exercise 3 · Line, plane and projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x + 2y + 2z = 8$ and the point $M(1, 0, 0)$. The line $d$ passes through $M$ and has direction vector $\vec{n} = (1, 2, 2)$.

1. Give a parametric representation of $d$. *(0.5 pt)*
2. Prove that $d$ meets $\mathcal{P}$ at a point $H$, then calculate the coordinates of $H$. *(0.5 pt)*
3. Justify that $H$ is the orthogonal projection of $M$ onto $\mathcal{P}$, then calculate the distance from $M$ to $\mathcal{P}$. *(0.5 pt)*
4. Consider the plane $\mathcal{Q}$ with equation $y = z$. Prove that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it. *(1 pt)*
5. Check that $H \in \mathcal{Q}$. Deduce the distance from $M$ to the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through $M$. *(0.5 pt)*
7. Determine the angle between $d$ and the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
8. Define the point $N$ by $\overrightarrow{MN} = \vec{n}$. Calculate the distances from $N$ to the plane $\mathcal{P}$ and to the line $\mathcal{P} \cap \mathcal{Q}$, then compare them. *(1 pt)*

## Exercise 4 · Repetitions and estimation — 5 points

We repeat 18 times a trial with two outcomes, called success and failure. The trials are independent and the probability of success is $p = \frac{1}{10}$ on each trial. Let $X$ denote the number of successes obtained.

1. Justify that $X$ follows a binomial distribution and state its parameters. *(0.5 pt)*
2. Calculate the probability of obtaining exactly two successes. *(0.5 pt)*
3. Using complementary events, express and then calculate the probability of obtaining at least two successes. *(0.5 pt)*
4. Calculate the expected value and the variance of $X$. *(1 pt)*
5. We now repeat the same trial a number $N \geq 1$ of times, still independently. Determine the smallest integer $N$ for which the probability of obtaining at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. *(0.5 pt)*
6. For an integer $j \in \{0, \dots, 18\}$, prove the formula for $P(X = j)$ by counting the possible positions of the successes. *(0.5 pt)*
7. For $N$ independent trials with the same parameter $p$, let $F_N$ denote the frequency of successes. Prove the expressions for $E(F_N)$ and $V(F_N)$ in terms of $N$ and $p$. *(0.5 pt)*
8. Using the Bienaymé–Chebyshev inequality, determine a sufficient size $N$ so that $P(|F_N - p| < 0.02) \geq 0.95$. Is the size obtained necessarily the smallest possible for this probability? Justify. *(1 pt)*

# Solutions · Final mock · A

## Exercise 1 · A logarithmic function

1. *[0.5 pt]* At $0^+$, the numerator tends to $-\infty$ and the denominator to $0^+$: limit $-\infty$; at $+\infty$, limit $0$ by comparison of growth rates.

2. *[0.5 pt]* By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (4 + \ln x)}{x^2} = \frac{-3 - \ln x}{x^2}.$$

   The denominator is positive. The numerator vanishes at $s = e^{-3}$, is positive for $0 < x < s$ and negative for $x > s$. Hence $f$ is strictly increasing on $]0, s]$, then strictly decreasing on $[s, +\infty[$.

3. *[0.5 pt]* The maximum is attained only at $s = e^{-3}$ and equals

$$f(s) = \frac{4 - 3}{e^{-3}} = e^3.$$

   Since $x > 0$, $f(x) = 0$ is equivalent to $4 + \ln x = 0$, that is $x = e^{-4}$.

4. *[1 pt]* Differentiating $(-3 - \ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-3 - \ln x)x^{-3} = \frac{2\ln x + 5}{x^3}.$$

   The numerator vanishes at $t = e^{-\frac{5}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at $t$ and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. *[0.5 pt]* Let $m = \frac{e^3}{2}$ and $s = e^{-3}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at $0$ and value $2m$ at $s$: the corollary of the IVT gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from $2m$ to $0 < m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

6. *[0.5 pt]* Let $F(x) = 4\ln x + \frac{1}{2}(\ln x)^2$, differentiable for $x > 0$. Then $F'(x) = \frac{4}{x} + \frac{\ln x}{x} = f(x)$; $F$ is an antiderivative on $]0, +\infty[$.

7. *[0.5 pt]* With this antiderivative, $F(e) = \frac{9}{2}$ and $F(1) = 0$. The integral therefore equals $\frac{9}{2}$, and the mean value is

$$\mu = \frac{1}{e - 1}\int_1^e f(x)\,\mathrm{d}x = \frac{9}{2(e-1)}.$$

8. *[1 pt]* Domain $x > 1$; $x^2 - 1 = 15$, so $x = 4$, the negative root being excluded.

## Exercise 2 · Periodic phenomenon

1. *[0.5 pt]* Period 6 s; minimum 5 m, maximum 9 m.

2. *[0.5 pt]* $h'(t) = (\frac{2}{3}\pi)\sin(\frac{\pi t}{3})$; increasing on $[0, 3]$, decreasing on $[3, 6]$.

3. *[0.5 pt]* The equation is equivalent to $\cos(\frac{\pi t}{3}) = -\frac{1}{2}$. Over one revolution, $\frac{\pi t}{3} \in [0, 2\pi]$; the suitable angles are $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$. Hence $t = 2$ s or $t = 4$ s.

4. *[1 pt]* An antiderivative is $F(t) = 7t - \frac{6}{\pi}\sin(\frac{\pi t}{3})$. Since $\sin 0 = \sin(2\pi) = 0$,

$$\overline{h} = \frac{1}{6}\int_0^6 h(t)\,\mathrm{d}t = \frac{F(6) - F(0)}{6} = \frac{42}{6} = 7 \text{ m}.$$

5. *[0.5 pt]* The second derivative is $h''(t) = \frac{2\pi^2}{9}\cos(\frac{\pi t}{3})$. Its sign is that of the cosine: positive on $[0, \frac{3}{2}[$ and $]\frac{9}{2}, 6]$, negative between these two intervals. The function is convex on $[0, \frac{3}{2}]$ and $[\frac{9}{2}, 6]$, concave on $[\frac{3}{2}, \frac{9}{2}]$. The two changes of convexity give the points of inflection $(\frac{3}{2}, 7)$ and $(\frac{9}{2}, 7)$.

6. *[0.5 pt]* Between 2 and 4; duration 2 s.

7. *[0.5 pt]* $h'$ reaches $\frac{2}{3}\pi$ m s$^{-1}$ at $t = \frac{3}{2}$ and its opposite at $t = \frac{9}{2}$.

8. *[1 pt]* Displacement $h(6) - h(0) = 0$. Rise of 4 m then fall of 4 m: total distance 8 m, equal to the integral of $|h'|$.

## Exercise 3 · Line, plane and projection

1. *[0.5 pt]* $(x, y, z) = (1 + t, 2t, 2t)$, $t \in \mathbb{R}$.

2. *[0.5 pt]* $1 + 9t = 8$ gives $t = \frac{7}{9}$; $H = (\frac{16}{9}, \frac{14}{9}, \frac{14}{9})$.

3. *[0.5 pt]* The vector $\vec{n} = (1, 2, 2)$ is normal to $\mathcal{P}$. Since $H \in \mathcal{P}$ and $\overrightarrow{MH} = \frac{7}{9}\vec{n}$, $H$ is the orthogonal projection of $M$ onto the plane. The coordinate system being orthonormal, $\|\vec{n}\| = \sqrt{1+4+4} = 3$. Hence
$$d(M, \mathcal{P}) = MH = |\frac{7}{9}|\,3 = \frac{7}{3}.$$

4. *[1 pt]* Set $y = z = s$; $x = 8 - 4s$. Hence $(x, y, z) = (8, 0, 0) + s(-4, 1, 1)$.

5. *[0.5 pt]* Yes, its last two coordinates are equal. $H$ lies on the line and $\overrightarrow{MH}$ is orthogonal to every direction of the plane, hence to this line. Distance $\frac{7}{3}$.

6. *[0.5 pt]* $x + 2y + 2z = 1$; same normal vector and passing through $M$.

7. *[0.5 pt]* $\vec{n} \cdot (-4, 1, 1) = -4 + 2 + 2 = 0$: angle $\frac{\pi}{2}$.

8. *[1 pt]* $N = M + \vec{n}$ lies on the same normal line as $M$; its projection onto $\mathcal{P}$ is therefore also $H$. Now $H \in \mathcal{P} \cap \mathcal{Q}$. For every $P$ on this line, $NP^2 = NH^2 + HP^2$, so its minimal distance to $N$ is $NH$. Finally,
$$d(N, \mathcal{P}) = d(N, \mathcal{P} \cap \mathcal{Q}) = \|\overrightarrow{NH}\| = |\frac{7}{9} - 1|\,\|\vec{n}\| = \frac{2}{3}.$$

## Exercise 4 · Repetitions and estimation

1. *[0.5 pt]* Repetition of a fixed number 18 of identical independent trials with two outcomes; $X \sim \mathcal{B}(18, \frac{1}{10})$.

2. *[0.5 pt]* $P(X = 2) = \binom{18}{2}(\frac{1}{10})^2(\frac{9}{10})^{16} \simeq 0.283512$.

3. *[0.5 pt]* $1 - P(X = 0) - P(X = 1) = 1 - (\frac{9}{10})^{18} - 18(\frac{1}{10})(\frac{9}{10})^{17} \simeq 0.549716$.

4. *[1 pt]* $E(X) = \frac{9}{5}$, $V(X) = \frac{81}{50}$.

5. *[0.5 pt]* For $N$ independent trials, the probability of no success is $(1 - p)^N$. The required probability is therefore $1 - (\frac{9}{10})^N$. It is strictly increasing in $N$, since $0 < \frac{9}{10} < 1$. The calculations give
$$P_{28} \simeq 0.947665 < 0.95, \qquad P_{29} \simeq 0.952899 \geq 0.95.$$
Thus 29 is the smallest suitable number of trials.

6. *[0.5 pt]* Let $n = 18$. For $j \in \{0, \ldots, n\}$, an ordered outcome containing $j$ successes and $n - j$ failures has, by independence, probability $p^j(1-p)^{n-j}$. There are $\binom{n}{j}$ choices for the positions of the successes. These outcomes are mutually exclusive, so

$$P(X = j) = \binom{n}{j} p^j (1-p)^{n-j}, \qquad p = \frac{1}{10}.$$

7. *[0.5 pt]* If $B_1, \ldots, B_N$ are the independent indicator variables of the successes, $F_N = \frac{B_1 + \cdots + B_N}{N}$. With $E(B_i) = p$ and $V(B_i) = p(1-p)$,

$$E(F_N) = \frac{Np}{N} = \frac{1}{10}, \qquad V(F_N) = \frac{Np(1-p)}{N^2} = \frac{\frac{9}{100}}{N}.$$

   Independence is required for adding variances, not for the linearity of expected value.

8. *[1 pt]* The Bienaymé–Chebyshev inequality applied to $F_N$ gives

$$P(|F_N - p| \geq 0.02) \leq \frac{p(1-p)}{N(0.02)^2}.$$

   To guarantee that this probability is at most 0.05, it suffices that

$$N \geq \frac{\frac{9}{100}}{0.05 \times (0.02)^2} = 4500.$$

   Taking the complement, we then have $P(|F_N - p| < 0.02) \geq 0.95$. The size 4500 is sufficient; the inequality being an upper bound, it does not necessarily give the minimum size for the exact probability.

# Final mock · B

**PAPER RESERVED FOR ASSESSMENT**

**Duration: 4 h · Marked out of 20 · Four independent exercises.**
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to $10^{-6}$ and other approximations to $10^{-4}$, unless stated otherwise.

**Name:** ..... **Class:** ..... **Date:** .....

## Exercise 1 · A logarithmic function — 5 points

Consider the function $f$ defined on $(0, +\infty)$ by $f(x) = \frac{3+\ln x}{x}$. Let $\mathcal{C}_f$ denote its graph.

1. Determine the limits of $f$ at $0^+$ and at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw up the table of variations of $f$. *(0.5 pt)*
3. State the maximum value of $f$, then solve $f(x) = 0$. *(0.5 pt)*
4. Study the convexity of $f$ and determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Prove that the equation $f(x) = \frac{1}{2}e^2$ has exactly two solutions in $(0, +\infty)$. *(0.5 pt)*
6. Find an antiderivative of $f$ on $(0, +\infty)$. *(0.5 pt)*
7. Calculate the mean value of $f$ on $[1, e]$. *(0.5 pt)*
8. Independently of the previous questions, solve $\ln(x-2) + \ln(x+2) = \ln(32)$, after stating the set on which this equation is defined. *(1 pt)*

## Exercise 2 · Periodic phenomenon — 5 points

A point moves on a wheel turning at constant speed in a vertical plane. Its height above the ground is modelled by $h(t) = 10 - 3\cos\left(\frac{\pi t}{2}\right)$, where $t$ is expressed in seconds and $h(t)$ in metres. We study one complete revolution, for $t \in [0, 4]$.

1. Give a period of the model extended to $\mathbb{R}$, then the minimum and maximum heights over one revolution. *(0.5 pt)*
2. Calculate $h'(t)$ and draw up the table of variations of $h$ on $[0, 4]$. *(0.5 pt)*
3. Find all times $t \in [0, 4]$ such that $h(t) = \frac{23}{2}$. *(0.5 pt)*
4. Calculate the mean value of $h$ on $[0, 4]$. *(1 pt)*
5. Study the convexity of $h$ on the interval of study and determine the coordinates of the points of inflection of its graph. *(0.5 pt)*
6. Determine the total time during which the height is at least equal to the level in question 3. *(0.5 pt)*
7. The vertical speed is given by $h'(t)$. Determine its maximum and minimum values, in metres per second, and the times at which they are reached. *(0.5 pt)*
8. Calculate the vertical displacement $h(4) - h(0)$, then the sum of the height travelled on the way up and that travelled on the way down. Explain why these two results are different. *(1 pt)*

Final mock · B · continued

## Exercise 3 · Line, plane and projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x + 2y + 2z = 12$ and the point $M(2, 0, 0)$. The line $d$ passes through $M$ and has direction vector $\vec{n} = (1, 2, 2)$.

1. Give a parametric representation of $d$. *(0.5 pt)*
2. Prove that $d$ meets $\mathcal{P}$ at a point $H$, then calculate the coordinates of $H$. *(0.5 pt)*
3. Justify that $H$ is the orthogonal projection of $M$ onto $\mathcal{P}$, then calculate the distance from $M$ to $\mathcal{P}$. *(0.5 pt)*
4. Consider the plane $\mathcal{Q}$ with equation $y = z$. Prove that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it. *(1 pt)*
5. Check that $H \in \mathcal{Q}$. Deduce the distance from $M$ to the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through $M$. *(0.5 pt)*
7. Determine the angle between $d$ and the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
8. Define the point $N$ by $\overrightarrow{MN} = \vec{n}$. Calculate the distances from $N$ to the plane $\mathcal{P}$ and to the line $\mathcal{P} \cap \mathcal{Q}$, then compare them. *(1 pt)*

## Exercise 4 · Repetitions and estimation — 5 points

We repeat 20 times a trial with two outcomes, called success and failure. The trials are independent and the probability of success is $p = \frac{3}{10}$ on each trial. Let $X$ denote the number of successes obtained.

1. Justify that $X$ follows a binomial distribution and state its parameters. *(0.5 pt)*
2. Calculate the probability of obtaining exactly two successes. *(0.5 pt)*
3. Using complementary events, express and then calculate the probability of obtaining at least two successes. *(0.5 pt)*
4. Calculate the expected value and the variance of $X$. *(1 pt)*
5. We now repeat the same trial a number $N \geq 1$ of times, still independently. Determine the smallest integer $N$ for which the probability of obtaining at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. *(0.5 pt)*
6. For an integer $j \in \{0, \ldots, 20\}$, prove the formula for $P(X = j)$ by counting the possible positions of the successes. *(0.5 pt)*
7. For $N$ independent trials with the same parameter $p$, let $F_N$ denote the frequency of successes. Prove the expressions for $E(F_N)$ and $V(F_N)$ in terms of $N$ and $p$. *(0.5 pt)*
8. Using the Bienaymé–Chebyshev inequality, determine a sufficient size $N$ so that $P(|F_N - p| < 0.02) \geq 0.95$. Is the size obtained necessarily the smallest possible for this probability? Justify. *(1 pt)*

# Solutions · Final mock · B

## Exercise 1 · A logarithmic function

1. *[0.5 pt]* At $0^+$, the numerator tends to $-\infty$ and the denominator to $0^+$: limit $-\infty$; at $+\infty$, limit 0 by comparison of growth rates.

2. *[0.5 pt]* By the quotient rule,

$$f'(x) = \frac{(\frac{1}{x})x - (3+\ln x)}{x^2} = \frac{-2-\ln x}{x^2}.$$

The denominator is positive. The numerator vanishes at $s = e^{-2}$, is positive for $0 < x < s$ and negative for $x > s$. Hence $f$ is strictly increasing on $]0, s]$, then strictly decreasing on $[s, +\infty[$.

3. *[0.5 pt]* The maximum is attained only at $s = e^{-2}$ and equals

$$f(s) = \frac{3-2}{e^{-2}} = e^2.$$

Since $x > 0$, $f(x) = 0$ is equivalent to $3 + \ln x = 0$, that is $x = e^{-3}$.

4. *[1 pt]* Differentiating $(-2-\ln x)x^{-2}$:

$$f''(x) = -x^{-3} - 2(-2-\ln x)x^{-3} = \frac{2\ln x + 3}{x^3}.$$

The numerator vanishes at $t = e^{-\frac{3}{2}}$ and changes from negative to positive. The function is concave on $]0, t]$, convex on $[t, +\infty[$. It changes convexity at $t$ and $f(t) = \frac{3}{2t}$: the point of inflection is $I(t, \frac{3}{2t})$.

5. *[0.5 pt]* Let $m = \frac{e^2}{2}$ and $s = e^{-2}$. The function is continuous on $]0, +\infty[$. On $]0, s]$, it is strictly increasing, with limit $-\infty$ at 0 and value $2m$ at $s$: the corollary of the IVT gives exactly one solution in $]0, s[$. On $[s, +\infty[$, it is strictly decreasing from $2m$ to $0 < m$: exactly one solution in $]s, +\infty[$. These intervals cover the domain and $f(s) \neq m$: there are exactly two solutions.

6. *[0.5 pt]* Let $F(x) = 3\ln x + \frac{1}{2}(\ln x)^2$, differentiable for $x > 0$. Then $F'(x) = \frac{3}{x} + \frac{\ln x}{x} = f(x)$; $F$ is an antiderivative on $]0, +\infty[$.

7. *[0.5 pt]* With this antiderivative, $F(e) = \frac{7}{2}$ and $F(1) = 0$. The integral therefore equals $\frac{7}{2}$, and the mean value is

$$\mu = \frac{1}{e-1}\int_1^e f(x)\,\mathrm{d}x = \frac{7}{2(e-1)}.$$

8. *[1 pt]* Domain $x > 2$; $x^2 - 4 = 32$, so $x = 6$, the negative root being excluded.

## Exercise 2 · Periodic phenomenon

1. *[0.5 pt]* Period 4 s; minimum 7 m, maximum 13 m.

2. *[0.5 pt]* $h'(t) = (\frac{3}{2}\pi)\sin(\frac{\pi t}{2})$; increasing on $[0, 2]$, decreasing on $[2, 4]$.

3. *[0.5 pt]* The equation is equivalent to $\cos(\frac{\pi t}{2}) = -\frac{1}{2}$. Over one revolution, $\frac{\pi t}{2} \in [0, 2\pi]$; the suitable angles are $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$. Hence $t = \frac{4}{3}$ s or $t = \frac{8}{3}$ s.

4. *[1 pt]* An antiderivative is $F(t) = 10t - \frac{6}{\pi}\sin(\frac{\pi t}{2})$. Since $\sin 0 = \sin(2\pi) = 0$,

$$\overline{h} = \frac{1}{4}\int_0^4 h(t)\,\mathrm{d}t = \frac{F(4)-F(0)}{4} = \frac{40}{4} = 10 \text{ m}.$$

5. *[0.5 pt]* The second derivative is $h''(t) = \frac{3\pi^2}{4}\cos(\frac{\pi t}{2})$. Its sign is that of the cosine: positive on $[0,1[$ and $]3,4]$, negative between these two intervals. The function is convex on $[0,1]$ and $[3,4]$, concave on $[1,3]$. The two changes of convexity give the points of inflection $(1,10)$ and $(3,10)$.

6. *[0.5 pt]* Between $\frac{4}{3}$ and $\frac{8}{3}$; duration $\frac{4}{3}$ s.

7. *[0.5 pt]* $h'$ reaches $\frac{3}{2}\pi$ m s$^{-1}$ at $t = 1$ and its opposite at $t = 3$.

8. *[1 pt]* Displacement $h(4) - h(0) = 0$. Rise of 6 m then fall of 6 m: total distance 12 m, equal to the integral of $|h'|$.

## Exercise 3 · Line, plane and projection

1. *[0.5 pt]* $(x,y,z) = (2+t, 2t, 2t)$, $t \in \mathbb{R}$.

2. *[0.5 pt]* $2 + 9t = 12$ gives $t = \frac{10}{9}$; $H = (\frac{28}{9}, \frac{20}{9}, \frac{20}{9})$.

3. *[0.5 pt]* The vector $\vec{n} = (1,2,2)$ is normal to $\mathcal{P}$. Since $H \in \mathcal{P}$ and $\overrightarrow{MH} = \frac{10}{9}\vec{n}$, $H$ is the orthogonal projection of $M$ onto the plane. The coordinate system being orthonormal, $\|\vec{n}\| = \sqrt{1+4+4} = 3$. Hence
$$d(M,\mathcal{P}) = MH = |\frac{10}{9}|\,3 = \frac{10}{3}.$$

4. *[1 pt]* Set $y = z = s$; $x = 12 - 4s$. Hence $(x,y,z) = (12,0,0) + s(-4,1,1)$.

5. *[0.5 pt]* Yes, its last two coordinates are equal. $H$ lies on the line and $\overrightarrow{MH}$ is orthogonal to every direction of the plane, hence to this line. Distance $\frac{10}{3}$.

6. *[0.5 pt]* $x + 2y + 2z = 2$; same normal vector and passing through $M$.

7. *[0.5 pt]* $\vec{n} \cdot (-4,1,1) = -4 + 2 + 2 = 0$: angle $\frac{\pi}{2}$.

8. *[1 pt]* $N = M + \vec{n}$ lies on the same normal line as $M$; its projection onto $\mathcal{P}$ is therefore also $H$. Now $H \in \mathcal{P} \cap \mathcal{Q}$. For every $P$ on this line, $NP^2 = NH^2 + HP^2$, so its minimal distance to $N$ is $NH$. Finally,
$$d(N,\mathcal{P}) = d(N,\mathcal{P}\cap\mathcal{Q}) = \|\overrightarrow{NH}\| = |\frac{10}{9} - 1|\,\|\vec{n}\| = \frac{1}{3}.$$

## Exercise 4 · Repetitions and estimation

1. *[0.5 pt]* Repetition of a fixed number 20 of identical independent trials with two outcomes; $X \sim \mathcal{B}(20, \frac{3}{10})$.

2. *[0.5 pt]* $P(X=2) = \binom{20}{2}(\frac{3}{10})^2(\frac{7}{10})^{18} \simeq 0.027846$.

3. *[0.5 pt]* $1 - P(X=0) - P(X=1) = 1 - (\frac{7}{10})^{20} - 20(\frac{3}{10})(\frac{7}{10})^{19} \simeq 0.992363$.

4. *[1 pt]* $E(X) = 6$, $V(X) = \frac{21}{5}$.

5. *[0.5 pt]* For $N$ independent trials, the probability of no success is $(1-p)^N$. The required probability is therefore $1 - (\frac{7}{10})^N$. It is strictly increasing in $N$, since $0 < \frac{7}{10} < 1$. The calculations give
$$P_8 \simeq 0.942352 < 0.95, \qquad P_9 \simeq 0.959646 \geq 0.95.$$
Thus 9 is the smallest suitable number of trials.

6. *[0.5 pt]* Let $n = 20$. For $j \in \{0, \ldots, n\}$, an ordered outcome containing $j$ successes and $n - j$ failures has, by independence, probability $p^j(1-p)^{n-j}$. There are $\binom{n}{j}$ choices for the positions of the successes. These outcomes are mutually exclusive, so

$$P(X = j) = \binom{n}{j} p^j (1-p)^{n-j}, \qquad p = \frac{3}{10}.$$

7. *[0.5 pt]* If $B_1, \ldots, B_N$ are the independent indicator variables of the successes, $F_N = \frac{B_1 + \cdots + B_N}{N}$. With $E(B_i) = p$ and $V(B_i) = p(1-p)$,

$$E(F_N) = \frac{Np}{N} = \frac{3}{10}, \qquad V(F_N) = \frac{Np(1-p)}{N^2} = \frac{\frac{21}{100}}{N}.$$

   Independence is required for adding variances, not for the linearity of expected value.

8. *[1 pt]* The Bienaymé–Chebyshev inequality applied to $F_N$ gives

$$P(|F_N - p| \geq 0.02) \leq \frac{p(1-p)}{N(0.02)^2}.$$

   To guarantee that this probability is at most 0.05, it suffices that

$$N \geq \frac{\frac{21}{100}}{0.05 \times (0.02)^2} = 10500.$$

   Taking the complement, we then have $P(|F_N - p| < 0.02) \geq 0.95$. The size 10500 is sufficient; the inequality being an upper bound, it does not necessarily give the minimum size for the exact probability.